

Warehouse Metrics Repo

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WAREHOUSE SKIPPED LINE REPORT — UAE & KSA WAREHOUSES

Purpose : tracks skipped, ops-marked-missing, and picked item counts per warehouse category and business date, combining both legacy (GOMS) and new (CW Picking) picking systems into a single unified view

Output : one row per date × warehouse category with skipped_items, ops_marked_missing, ops_marked_damage, ops_marked_expiry, and picked_items

Author : noon Minutes Ops

DEFINITIONS

date : operational business date derived from the picking job timestamp, offset by -6h30m (UAE) or -6h (KSA) to align with the warehouse business day start.
e.g. a job accepted at 06:30 Dubai maps to the same business date as one accepted at 07:00

wh_code : partner_warehouse_code — human-readable WH identifier
e.g. DXBID01, RUHID01

warehouse : logical category resolved from the WH code mapping table (noonbinimops.fulfillment.wh_code_wh_category_mapping). Falls back to 'Other' if code is unmapped.

skipped_items : distinct picking lines/items skipped due to Damaged, Missing, or Expired reasons

ops_marked_missing : subset of skipped items where the item is in mp_status=31 (unfulfillable), representing items flagged by ops as missing/damaged/expired

ops_marked_damage : ops_marked_missing filtered to Damaged reason

ops_marked_expiry : ops_marked_missing filtered to Expired reason

picked_items : distinct items/lines successfully picked (not skipped, not cancelled)

SYSTEMS COVERED

old_base : legacy GOMS picking system
source: noonwh.scgoms_goms (picking_job, picking_job_line, item, warehouse, reason)

skip reasons identified by name_en IN
('Damaged Items', 'Missing Items', 'Expired Items')
skip flag: pjl.id_status = 11
picked flag: pjl.id_status NOT IN (11, 51)

new_base : new CW Picking system
source: noonwh.scgoms_cwpicking (picking_job, picklist_line,
reason, warehouse, picking_container_line)
skip reasons identified by r.code IN
('missing', 'damaged', 'expired')
skip flag: pllst.code = 'skipped'
picked flag: picking_container_line join is non-null
NOTE: item-level data joined via transfer_request_line
using transfer_nr + catalog_sku as composite key

FILTERS

Rolling 31-day window on job created/accepted timestamp
Only closed jobs (id_status=2 in GOMS / pjst.code='closed' in CW)
GOMS: id_mp = 9 (noon marketplace)
Final output limited to last 30 business days

*/

WITH old_base AS (

--
-- LEGACY GOMS: Aggregate skipped and picked items per warehouse
-- and business date from the old picking system.
-- Business date is shifted back 6h30m (UAE) or 6h (KSA) from
-- job accepted_at to align with warehouse operational day.
--

--
SELECT
src_w.partner_warehouse_code AS wh_code,
name_en, -- skip reason label (e.g. 'Damaged Items')
CASE
WHEN src_w.id_country = 1 THEN date(DATETIME(pj.accepted_at, 'Asia/Dubai') - interval
'6' hour - interval '30' minute)
WHEN src_w.id_country = 2 THEN date(DATETIME(pj.accepted_at, 'Asia/Riyadh') -
interval '6' hour)

```

END AS date,

-- Items skipped for Damaged / Missing / Expired reasons
COUNT(DISTINCT case when pjl.id_status = 11 AND name_en IN ('Damaged Items',
'Missing Items', 'Expired Items') then i.id_item end) AS skipped_items,

-- Subset of skipped items that ops flagged as unfulfillable (missing/ops reason)
COUNT(
  DISTINCT CASE
    WHEN pjl.id_status = 11 AND name_en IN ('Damaged Items', 'Missing Items',
'Expired Items') and i.id_mp_status = 31
      AND JSON_EXTRACT_SCALAR(i.misc, '$.unfulfillable_reason') in
('ops_marked_missing','ops')
    THEN i.id_item
  END
) AS ops_marked_missing,

-- Items successfully picked (not skipped, not in a terminal skip/cancel state)
COUNT(DISTINCT case when pjl.id_status <> 11 AND pjl.id_status <> 51 then i.id_item
end) AS picked_items
FROM
  noondwh.scgoms_goms.picking_job pj
  LEFT JOIN noondwh.scgoms_goms.picking_job_line pjl USING (id_picking_job)
  LEFT JOIN noondwh.scgoms_goms.item i ON pjl.id_item = i.id_item
  LEFT JOIN noondwh.scgoms_goms.user u ON pj.id_user = u.id_user
  LEFT JOIN noondwh.scgoms_goms.warehouse src_w ON src_w.id_warehouse =
pj.id_warehouse
  LEFT JOIN noondwh.scgoms_goms.reason r ON r.id_reason = pjl.id_skip_reason
WHERE
  DATE(pj.accepted_at, 'Asia/Dubai') >= CURRENT_DATE - 31 -- rolling 31-day window
  AND pj.id_status = 2 -- closed jobs only
  AND pj.id_mp = 9 -- noon marketplace only
GROUP BY
  ALL
),

```

```

new_base AS (

```

```

  --

```

```

--

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```

-- NEW CW PICKING: Aggregate skipped and picked lines per
-- warehouse and business date from the new picking system.
-- Business date offset same as old_base (6h30m UAE / 6h KSA).
-- Item-level data linked via transfer_request + transfer_nr.

```

```

--
--
SELECT
  src_w.partner_warehouse_code AS wh_code,
  r.code as name_en,          -- skip reason code (e.g. 'missing', 'damaged')
  CASE
    WHEN src_w.id_country = 1 THEN date(DATETIME(pj.created_at, 'Asia/Dubai') - interval
'6' hour - interval '30' minute)
    WHEN src_w.id_country = 2 THEN date(DATETIME(pj.created_at, 'Asia/Riyadh') - interval
'6' hour)
  END AS date,

  -- Picklist lines skipped for missing / damaged / expired reasons
  count(distinct case when pllst.code = 'skipped' AND r.code IN ('missing', 'damaged',
'expired') then pll.id_picklist_line end) AS skipped_items,

  -- Subset of skipped items where item is unfulfillable (mp_status=31),
  -- representing ops-confirmed missing/damaged/expired inventory
  COUNT(
    DISTINCT CASE
      WHEN pllst.code = 'skipped' AND r.code IN ('missing', 'damaged', 'expired') and
i.id_mp_status = 31
      -- AND JSON_EXTRACT_SCALAR(i.misc, '$.unfulfillable_reason') in
('ops_marked_missing','ops')
      THEN i.id_item
    END
  ) AS ops_marked_missing,

  -- Picklist lines successfully picked (confirmed by presence in picking container)
  count(distinct case when plc.id_picking_container_line is not null then pll.id_picklist_line
end) AS picked_items
FROM noondwh.scgoms_cwpicking.picking_job pj
LEFT JOIN noondwh.scgoms_cwpicking.status pjst ON pj.id_status = pjst.id_status
LEFT JOIN noondwh.scgoms_cwpicking.picklist_line pll USING (id_picking_job)
-- Transfer request lines linked within a 32-day window to bound partition scans
left join noondwh.scgoms_cwpicking.transfer_request_line t on t.id_transfer_request_line =
pll.id_transfer_request_line and DATE(t.created_at, 'Asia/Dubai') >= CURRENT_DATE - 32
left join noondwh.scgoms_cwpicking.transfer_request tr on tr.id_transfer_request =
t.id_transfer_request and DATE(tr.created_at, 'Asia/Dubai') >= CURRENT_DATE - 32
-- Item resolved via composite key: transfer_nr + catalog_sku
LEFT JOIN noondwh.scgoms_goms.item i ON tr.transfer_nr = i.transfer_nr and i.catalog_sku =
t.sku and DATE(i.created_at, 'Asia/Dubai') >= CURRENT_DATE - 32
LEFT JOIN noondwh.scgoms_cwpicking.status pllst ON pll.id_status = pllst.id_status

```

```

LEFT JOIN noondwh.scgoms_cwpicking.warehouse src_w ON pll.id_warehouse =
src_w.id_warehouse
LEFT JOIN noondwh.scgoms_cwpicking.reason r ON r.id_reason = pll.id_skip_reason
LEFT JOIN noondwh.scgoms_cwpicking.warehouse_zone wz ON pll.id_warehouse_zone =
wz.id_warehouse_zone
LEFT JOIN noondwh.scgoms_cwpicking.warehouse dst_w ON pll.id_destination =
dst_w.id_warehouse
LEFT JOIN noondwh.scgoms_cwpicking.picking_container_line plc USING (id_picklist_line)
LEFT JOIN noondwh.scgoms_cwpicking.picking_container pcr ON pcr.id_picking_container =
plc.id_picking_container
LEFT JOIN noondwh.scgoms_cwpicking.cluster cl ON (cl.id_cluster = pll.id_cluster)
WHERE
  -- Business date filter: offset by 6h30m to align with WH operational day
  DATE(DATETIME(pj.created_at, 'Asia/Dubai') - INTERVAL '6' HOUR - interval '30' minute) >=
CURRENT_DATE - 31
  and pjst.code = 'closed' -- closed jobs only
group by all
),

```

final_base as

```

(
--

```

```

-- UNION: Merge new CW Picking and legacy GOMS results, resolve
-- warehouse category from the WH code mapping table, and
-- aggregate by date × warehouse. Last 30 days only.
--

```

```

-- New CW Picking system aggregated by warehouse category
SELECT
  date,
  COALESCE(c.warehouse_catrgory, 'Other') AS warehouse,
  SUM(skipped_items) AS skipped_items,
  SUM(ops_marked_missing) AS ops_marked_missing,
  sum(case when name_en in ('Damaged Items') then ops_marked_missing else 0 end ) as
ops_marked_damage,
  sum(case when name_en in ('Expired Items') then ops_marked_missing else 0 end ) as
ops_marked_expiry,
  sum(picked_items) as picked_items
FROM
  new_base b

```

```
LEFT JOIN noonbinimops.fulfillment.wh_code_wh_category_mapping c ON b.wh_code =  
c.warehouse_code  
where date >= CURRENT_DATE - 30  
GROUP BY ALL
```

```
union all
```

```
-- Legacy GOMS system aggregated by warehouse category  
SELECT  
    date,  
    COALESCE(c.warehouse_catrgory, 'Other') AS warehouse,  
    SUM(skipped_items) AS skipped_items,  
    SUM(ops_marked_missing) AS ops_marked_missing,  
    sum(case when name_en in ('Damaged Items') then ops_marked_missing else 0 end ) as  
ops_marked_damage,  
    sum(case when name_en in ('Expired Items') then ops_marked_missing else 0 end ) as  
ops_marked_expiry,  
    sum(picked_items) as picked_items  
FROM  
    old_base b  
    LEFT JOIN noonbinimops.fulfillment.wh_code_wh_category_mapping c ON b.wh_code =  
c.warehouse_code  
    where date >= CURRENT_DATE - 30  
GROUP BY ALL  
)
```

```
--
```

```
-- FINAL OUTPUT: Collapse any duplicates across the two systems  
-- by summing all metrics at date × warehouse grain.  
--
```

```
select  
    date,  
    warehouse,  
    sum(skipped_items) as skipped_items,  
    sum(ops_marked_missing) as ops_marked_missing,  
    sum(ops_marked_damage) as ops_marked_damage,  
    sum(ops_marked_expiry) as ops_marked_expiry,  
    sum(picked_items) as picked_items  
from final_base  
group by all
```

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INCIDENT STOCKTAKE REPORT — UAE & KSA WAREHOUSES

Purpose : tracks job completion metrics for incident stocktake processes per warehouse category and business date

Output : one row per created_date × warehouse
with total_jobs, pending_jobs, pending_confirmation_jobs, skipped_jobs, and completed_jobs

Author : noon Minutes Ops

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DEFINITIONS

created_date : local calendar date the stocktake job was created (Dubai TZ for UAE, Riyadh TZ for KSA)

grouping_key : job grouping identifier — filtered to 'incident_stocktake' (ad-hoc counts)

wh_code : partner_warehouse_code — human-readable WH identifier (e.g. DXBID01, RUHID01)

warehouse : logical category resolved from the WH code mapping table. Falls back to 'Other' if code is unmapped.

skipped_jobs : jobs in 'force_closed' state — stocktake jobs closed without completion

pending_jobs : jobs still in 'pending' state

pending_confirmation_jobs : jobs in 'pending_confirmation' state — awaiting supervisor/ops sign-off

completed_jobs : jobs in 'closed' state — fully counted and confirmed

total_jobs : all jobs regardless of state

completion_percentage : $\text{completed_jobs} / \text{total_jobs} \times 100$

SOURCE TABLES (noondwh.mxdcss_dcscs)

job : stocktake job header with created_at and status

status : lookup for job state codes

warehouse : warehouse master with country and partner_warehouse_code

process : process definition linked to the job

process_type : process type lookup — filtered to 'stock_take'

FILTERS

process_type.code = 'stock_take'
grouping_key = 'incident_stocktake'
Rolling 31-day window on job created_at
Final output limited to last 30 business days

=
*/

with base as

```
(
  SELECT
    -- Localise job creation date to warehouse country timezone
    CASE
      WHEN w.id_country = 1 THEN DATE(j.created_at, 'Asia/Dubai')
      WHEN w.id_country = 2 THEN DATE(j.created_at, "Asia/Riyadh")
    END AS created_date,

    j.grouping_key,
    w.partner_warehouse_code as wh_code,
    COUNT(CASE WHEN s.code = 'force_closed' THEN j.job_nr END) as skipped_jobs,      -- jobs
    closed without completion
    COUNT(CASE WHEN s.code = 'pending' THEN j.job_nr END) AS pending_jobs,        -- not yet
    started
    COUNT(CASE WHEN s.code = 'pending_confirmation' THEN j.job_nr END) AS
    pending_confirmation_jobs, -- awaiting sign-off
    COUNT(CASE WHEN s.code = 'closed' THEN j.job_nr END) AS completed_jobs,      -- fully
    completed
    COUNT(j.job_nr) AS total_jobs,
    COUNT(CASE WHEN s.code = 'closed' THEN j.job_nr END)* 100/COUNT(j.job_nr) AS
    completion_percentage -- % complete
```

```
FROM noondwh.mxdcss_dcsc.job j
JOIN noondwh.mxdcss_dcsc.status s USING (id_status)
JOIN noondwh.mxdcss_dcsc.warehouse w using (id_warehouse)
JOIN noondwh.mxdcss_dcsc.process p USING (id_process)
JOIN noondwh.mxdcss_dcsc.process_type pt USING (id_process_type)
WHERE
  pt.code = 'stock_take' and j.grouping_key = 'incident_stocktake' -- incident stocktakes only
  AND j.created_at >= TIMESTAMP(DATE_SUB(CURRENT_DATE(), INTERVAL 31 DAY)) -- rolling
  31-day window
  --and s.code != 'force_closed'
```

GROUP BY all

)

-- Final output: resolve warehouse category from WH code mapping,
-- aggregate job counts at date × warehouse grain.

-- Last 30 days only.

```
select
  created_date,
  coalesce(c.warehouse_catrgory,'Other') AS warehouse, -- falls back to 'Other' if wh_code unmapped
  sum(total_jobs) total_jobs,
  sum(pending_jobs) as pending_jobs,
  sum(pending_confirmation_jobs) as pending_confirmation_jobs,
  sum(skipped_jobs) as skipped_jobs,
  sum(completed_jobs) as completed_jobs
from base b left join noonbinimops.fulfillment.wh_code_wh_category_mapping c on b.wh_code =
c.warehouse_code
where created_date >= CURRENT_DATE - 30
group by 1,2
```

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MP PRODUCTIVITY REPORT — UAE CENTRAL WAREHOUSES

Purpose : measures per-user throughput and active time for
6 warehouse functions across all UAE central WHs
for a single business day

Output : one row per user × wh_code × biz_date
with units, time (seconds), and total throughput

Author : noon Minutes

DEFINITIONS

biz_date : operational business date, defined as the 24h window
starting 06:00 Dubai time (02:00 UTC).
e.g. biz_date 2026-04-06 = UTC 2026-04-06 02:00 → 2026-04-07 02:00
computed as: DATE(DATETIME(ts,'Asia/Dubai') - INTERVAL 6 HOUR)
so a shift starting at 06:00 Dubai = midnight on biz_date

wh_code : partner_warehouse_code — the human-readable WH identifier
e.g. DXBID01, AUHID01, DXSID03

warehouse_group : logical grouping of WH codes
TP = DXBID01-08 (Technopark, Dubai)
Kezad = AUHID01-02 (Khalifa Economic Zone, Abu Dhabi)
FZ Kezad = AUHFKZ01-03 (Free Zone Kezad)
Dubai South = DXSID01-07 + DXSLQD01
F&V = DXBFNV01 (Fruit & Veg dedicated WH)

id_user : internal numeric user ID across goms/dcss/inbound systems
username : resolved from scgoms_cwpicking.user + scgoms_goms.user.
prefers ops.idp.noon.team usernames (local ops accounts)
over noon.com SSO usernames.
falls back to raw id_user string if not found in either table.
NOTE: dcss-only users (stocktake/movement) may not resolve

FUNCTIONS COVERED

pick : units = distinct picklist lines completed in a closed picking job

time = job duration (job_start → job_end), but if first scan occurs >600s after job start, time starts from first scan (strips idle/setup time at job open)
source: noondwh.scgoms_cwpicking

put : putaway jobs only (id_job_subtype IN 4,5,6)
units = distinct completed job lines (id_status=3)
time = MIN(job.created_at) → MAX(job_line.updated_at)
source: noondwh.mxdcss_dcss.job + job_line

receive : inbound box receipt (id_status IN 13=received, 14=putaway)
units = distinct inbound boxes
time = MIN(created_at) → MAX(updated_at) per user/wh/day
source: noondwh.scinbound_inbound.inbound_box
NOTE: joins on partner_code (not partner_warehouse_code)

sort : goms sort scan events (id_audit_action=21,
to_state='pending_putaway')
units = scan count
time = sum of inter-scan gaps, capped at 300s per gap
(prevents inflating time when sorter is idle)
source: noondwh.scgoms_goms.audit_log

stocktake : dcss stocktake jobs (id_job_subtype=10, process_type LIKE '%stock%take%')
only closed or force_closed-with-pending-confirmation jobs
units = distinct job lines
time = opened_at → pending_confirmation_at (or closed_at if no pending)
source: noondwh.mxdcss_dcss.job + audit_log

movement : dcss internal_movement or migration jobs on deep locations only
(is_deep='1' on source location)
units = SUM(transaction_line.qty)
time = MIN(tl.created_at) → MAX(tl.created_at) per user/wh/day
id_user sourced from job table (not transaction_line)
source: noondwh.mxdcss_dcss.transaction_line + job + process_type

total_throughput : pick + sort + put + recv + stocktake + movement units
(movement_units excluded from time calculations but included in throughput count)

KEY DESIGN DECISIONS

1. biz_date shift: 06:00 Dubai start aligns with operational day.
the -6h offset converts Dubai time to a "logical midnight" so

DATE() returns the correct biz_date for night-shift workers.

2. pick time: the >600s first-scan rule removes cases where a job was opened but the picker wasn't active yet (e.g. job pre-assigned before shift start). uses first actual scan as clock start.
3. sort gap cap: LEAST(..., 300) caps each inter-scan gap at 5 min. without this, a sorter's idle break inflates sort_sec massively. 300s is a conservative max reasonable scan cycle time.
4. stocktake pre-aggregation: audit log timestamps are aggregated into stocktake_job_times CTE first, then SUM'd — avoids BQ multi-level aggregation error (SUM(MIN(...)) not allowed).
5. movement user: transaction_line has no id_user; pulled from parent job table via tl.id_job = j.id_job.
6. movement time: MIN→MAX across all tl rows for user/wh/day — not per-job. understates time for users with multiple movement jobs.

KNOWN LIMITATIONS

- sort only covers goms Sort WHs (DXBID01 is Non-Sort, uses cwpicking)
 - JLM (skip/skipped_flow) not excluded from pick units — these are failed picks logged in scgoms_goms, not cwpicking, so not in scope here
 - username resolution misses dcss-only users → falls back to id_user int
 - movement time is day-level not job-level (see point 6 above)
-
-

*/

-- — VARIABLES

-- biz_date_start / biz_date_end define the UTC window for the target biz date
-- to change date: update TIMESTAMP('2026-04-06 02:00:00 UTC') → next day 02:00 UTC
-- e.g. for 2026-04-07: use TIMESTAMP('2026-04-07 02:00:00 UTC') and '2026-04-08 02:00:00 UTC'

DECLARE biz_date_start TIMESTAMP DEFAULT TIMESTAMP('2026-04-06 02:00:00 UTC');
DECLARE biz_date_end TIMESTAMP DEFAULT TIMESTAMP('2026-04-07 02:00:00 UTC');

WITH

-- — WH SCOPE

-- explicit allowlist of UAE central WH codes to include.

-- add/remove codes here to adjust scope.

-- excludes NIM dark stores and KSA WHs by design.

wh_scope AS (

SELECT wh_code FROM UNNEST([

'AUHID01','AUHID02', -- Kezad

'AUHFKZ01','AUHFKZ02','AUHFKZ03', -- FZ Kezad

'DXBID01','DXBID02','DXBID03','DXBID04','DXBID07', -- TP

'DXSID01','DXSID02','DXSID03','DXSID04','DXSID06','DXSID07', -- Dubai South

'DXSLQD01', -- Dubai South LQ

'DXBFNV01' -- F&V

]) AS wh_code

),

-- — PICK: job_base

-- raw picking data: one row per picking job.

-- qty = distinct picklist lines (= units picked in the job)

-- first_scan = earliest scan time in the job (used for idle-strip logic)

-- only includes jobs with at least one container line (pcl.id_picking_container_line IS NOT NULL)

-- only closed jobs (pjst.code = 'closed')

job_base AS (

SELECT

src_w.partner_warehouse_code AS wh_code,

pj.id_user,

DATE(DATETIME(pj.created_at,'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,

DATETIME(pj.created_at,'Asia/Dubai') AS job_start,

DATETIME(pj.finsihed_picking_at,'Asia/Dubai') AS job_end, -- note: typo in

source table (finsihed), do not correct

COUNT(DISTINCT pll.id_picklist_line) AS qty,

MIN(DATETIME(pcl.created_at,'Asia/Dubai')) AS first_scan

FROM noondwh.scgoms_cwpicking.picking_job pj

JOIN noondwh.scgoms_cwpicking.status pjst

ON pj.id_status = pjst.id_status AND pjst.code = 'closed'

JOIN noondwh.scgoms_cwpicking.picklist_line pll

ON pll.id_picking_job = pj.id_picking_job

JOIN noondwh.scgoms_cwpicking.warehouse src_w

ON pll.id_warehouse = src_w.id_warehouse

JOIN noondwh.scgoms_cwpicking.picking_container_line pcl

ON pcl.id_picklist_line = pll.id_picklist_line

```

JOIN wh_scope ws ON src_w.partner_warehouse_code = ws.wh_code
WHERE pj.created_at >= biz_date_start AND pj.created_at < biz_date_end
  AND pcl.id_picking_container_line IS NOT NULL
GROUP BY 1,2,3,4,5
),

```

```
-- — PICK: aggregated to user/wh/date
```

```

-- applies the >600s idle-strip rule:
-- if first scan is >10min after job open → start clock from first_scan
-- otherwise use full job duration (job_start → job_end)
pick AS (
  SELECT wh_code, id_user, biz_date, 'pick' AS fn,
    SUM(qty) AS units,
    SUM(CASE WHEN DATETIME_DIFF(first_scan,job_start,SECOND) > 600
      THEN DATETIME_DIFF(job_end,first_scan,SECOND)
      ELSE DATETIME_DIFF(job_end,job_start,SECOND) END) AS time_sec
  FROM job_base GROUP BY 1,2,3,4
),

```

```
-- — PUTAWAY
```

```

-- id_job_subtype IN (4,5,6): putaway subtypes only (excludes stocktake, movement, etc.)
-- jl.id_status = 3: completed lines only
-- time = job open → last line update (captures actual putaway duration)
dcss AS (
  SELECT
    w.partner_warehouse_code AS wh_code,
    j.id_user,
    DATE(DATETIME(j.created_at,'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
    'put' AS fn,
    COUNT(DISTINCT jl.id_job_line) AS units,
    DATETIME_DIFF(
      MAX(DATETIME(jl.updated_at,'Asia/Dubai')),
      MIN(DATETIME(j.created_at,'Asia/Dubai')), SECOND) AS time_sec
  FROM noondwh.mxdcss_dcsc.job j
  JOIN noondwh.mxdcss_dcsc.job_line jl ON jl.id_job = j.id_job
  JOIN noondwh.mxdcss_dcsc.warehouse w ON j.id_warehouse = w.id_warehouse
  JOIN wh_scope ws ON w.partner_warehouse_code = ws.wh_code
  WHERE j.id_job_subtype IN (4,5,6)
    AND jl.id_status = 3
    AND j.created_at >= biz_date_start AND j.created_at < biz_date_end
    AND j.id_user IS NOT NULL

```

```

GROUP BY 1,2,3,4
),

-- — RECEIVE

```

```

-- id_status IN (13,14): received or putaway boxes (both count as received)
-- NOTE: inbound.warehouse uses partner_code not partner_warehouse_code
-- time = earliest box created → latest box updated for the user/wh/day session
recv AS (
  SELECT
    w.partner_code                AS wh_code,
    ib.id_user,
    DATE(DATETIME(ib.created_at,'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
    'receive'                     AS fn,
    COUNT(DISTINCT ib.id_inbound_box) AS units,
    DATETIME_DIFF(
      MAX(DATETIME(ib.updated_at,'Asia/Dubai')),
      MIN(DATETIME(ib.created_at,'Asia/Dubai')), SECOND) AS time_sec
  FROM noondwh.scinbound_inbound.inbound_box ib
  JOIN noondwh.scinbound_inbound.warehouse w ON ib.id_warehouse = w.id_warehouse
  JOIN wh_scope ws ON w.partner_code = ws.wh_code
  WHERE ib.id_status IN (13,14)
    AND ib.created_at >= biz_date_start AND ib.created_at < biz_date_end
    AND ib.id_user IS NOT NULL
  GROUP BY 1,2,3,4
),

-- — SORT: raw scan events

```

```

-- id_audit_action = 21: sort scan event in goms
-- to_state = 'pending_putaway': marks a completed sort scan
-- LEAD() gets the next scan time for the same user/wh/date
-- used to compute per-scan gap for time calculation
sort_base AS (
  SELECT
    w.partner_warehouse_code      AS wh_code,
    al.id_user,
    DATE(DATETIME(al.created_at,'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
    DATETIME(al.created_at,'Asia/Dubai') AS scan_at,
    LEAD(DATETIME(al.created_at,'Asia/Dubai')) OVER (
      PARTITION BY al.id_user, w.partner_warehouse_code,
        DATE(DATETIME(al.created_at,'Asia/Dubai') - INTERVAL 6 HOUR)
      ORDER BY al.created_at) AS next_scan

```



```

FROM noondwh.scgoms_goms.audit_log al
JOIN noondwh.scgoms_goms.input_container_line icl
  ON icl.id_input_container_line = al.action_key
JOIN noondwh.scgoms_goms.input_container ic
  ON ic.id_input_container = icl.id_input_container
JOIN noondwh.scgoms_goms.warehouse w ON w.id_warehouse = ic.id_warehouse
JOIN wh_scope ws ON w.partner_warehouse_code = ws.wh_code
WHERE al.id_audit_action = 21
  AND JSON_EXTRACT_SCALAR(al.misc,'$.to') = 'pending_putaway'
  AND al.created_at >= biz_date_start AND al.created_at < biz_date_end
  AND al.id_user IS NOT NULL
),

-- — SORT: aggregated


---


-- COALESCE(gap, 300): last scan of the day has no next_scan → default to 300s
-- LEAST(..., 300): cap each gap at 300s to exclude idle/break time
sort AS (
  SELECT wh_code, id_user, biz_date, 'sort' AS fn,
    COUNT(*) AS units,
    SUM(LEAST(COALESCE(DATETIME_DIFF(next_scan,scan_at,SECOND),300),300)) AS
time_sec
  FROM sort_base GROUP BY 1,2,3,4
),

-- — STOCKTAKE: pre-aggregate audit timestamps per job


---


-- resolves process_type per job via explicit joins (BQ disallows subquery in JOIN)
-- aggregates opened/pending_confirmation/closed timestamps per job in one pass
-- has_pending: count of pending_confirmation events (used to validate force_closed jobs)
-- status_code: closed or force_closed
job_process_type AS (
  SELECT j.id_job, pt.code AS pt_code
  FROM noondwh.mxdcss_dcsc.job j
  JOIN noondwh.mxdcss_dcsc.process p  ON p.id_process = j.id_process
  JOIN noondwh.mxdcss_dcsc.process_type pt ON pt.id_process_type = p.id_process_type
  WHERE j.id_job_subtype = 10
    AND pt.code LIKE '%stock%take%'
    AND j.created_at >= biz_date_start AND j.created_at < biz_date_end
),
stocktake_job_times AS (
  SELECT
    j.id_job,
    j.id_user,

```

```

j.id_warehouse,
MAX(s.code) AS status_code,
MIN(CASE WHEN JSON_EXTRACT_SCALAR(al1.misc,'$.to') = 'opened' THEN
al1.created_at END) AS opened_at,
MIN(CASE WHEN JSON_EXTRACT_SCALAR(al1.misc,'$.to') = 'pending_confirmation'
THEN al1.created_at END) AS pending_at,
MIN(CASE WHEN JSON_EXTRACT_SCALAR(al1.misc,'$.to') = 'closed' THEN
al1.created_at END) AS closed_at,
COUNTIF(JSON_EXTRACT_SCALAR(al1.misc,'$.to') = 'pending_confirmation') AS
has_pending
FROM noondwh.mxdcss_dcss.job j
JOIN noondwh.mxdcss_dcss.status s ON s.id_status = j.id_status
JOIN job_process_type jpt ON jpt.id_job = j.id_job
JOIN noondwh.mxdcss_dcss.audit_log al1
ON al1.id_audit_action = 1
AND al1.action_key = j.id_job
AND al1.created_at >= biz_date_start AND al1.created_at < biz_date_end
WHERE (s.code = 'closed' OR s.code = 'force_closed')
AND j.created_at >= biz_date_start AND j.created_at < biz_date_end
AND j.id_user IS NOT NULL
GROUP BY 1,2,3
),

```

-- — STOCKTAKE: aggregated

```

-- time = opened_at → pending_confirmation_at (preferred)
-- falls back to closed_at if no pending_confirmation event exists
-- excludes force_closed jobs with no pending_confirmation (incomplete counts)
inv_stocktake AS (
SELECT
w.partner_warehouse_code AS wh_code,
jt.id_user,
DATE(DATETIME(jt.opened_at,'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
'stocktake' AS fn,
COUNT(DISTINCT jl.id_job_line) AS units,
SUM(TIMESTAMP_DIFF(
COALESCE(jt.pending_at, jt.closed_at),
jt.opened_at, SECOND)) AS time_sec
FROM stocktake_job_times jt
LEFT JOIN noondwh.mxdcss_dcss.job_line jl ON jl.id_job = jt.id_job
LEFT JOIN noondwh.mxdcss_dcss.warehouse w ON w.id_warehouse = jt.id_warehouse
JOIN wh_scope ws ON w.partner_warehouse_code = ws.wh_code
WHERE jt.opened_at IS NOT NULL
AND (jt.pending_at IS NOT NULL OR jt.closed_at IS NOT NULL)

```

```

    AND NOT (jt.status_code = 'force_closed' AND jt.has_pending = 0)
GROUP BY 1,2,3,4
),

```

-- — INTERNAL MOVEMENT & MIGRATION

```

-- process_type IN ('internal_movement','migration'): stock moves between locations
-- is_deep='1': only deep storage locations (excludes forward pick face moves)
-- units = qty moved (not job lines), can be > 1 per tl row
-- time = MIN→MAX of tl.created_at across all movement lines for user/wh/day
-- NOTE: day-level aggregation, not per-job — understates time if multi-job
-- id_user joined from job table (transaction_line has no id_user column)
inv_movement AS (
SELECT
    w.partner_warehouse_code                AS wh_code,
    j.id_user,
    DATE(DATETIME(tl.created_at,'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
    'movement'                             AS fn,
    SUM(tl.qty)                             AS units,
    DATETIME_DIFF(
        MAX(DATETIME(tl.created_at,'Asia/Dubai')),
        MIN(DATETIME(tl.created_at,'Asia/Dubai')), SECOND) AS time_sec
FROM noondwh.mxdcss_dcss.transaction_line tl
JOIN noondwh.mxdcss_dcss.location l
    ON ('location', l.id_location) = (tl.src_key, CAST(tl.src_value AS INT64))
LEFT JOIN noondwh.mxdcss_dcss.warehouse w    ON w.id_warehouse = l.id_warehouse
JOIN noondwh.mxdcss_dcss.job j              ON j.id_job = tl.id_job
JOIN noondwh.mxdcss_dcss.process p          ON p.id_process = j.id_process
JOIN noondwh.mxdcss_dcss.process_type pt    ON pt.id_process_type = p.id_process_type
JOIN wh_scope ws ON w.partner_warehouse_code = ws.wh_code
WHERE pt.code IN ('internal_movement','migration')
    AND tl.created_at >= biz_date_start AND tl.created_at < biz_date_end
    AND JSON_EXTRACT_SCALAR(l.misc,'$.is_deep') = '1'
    AND j.id_user IS NOT NULL
GROUP BY 1,2,3,4
),

```

-- — UNION ALL 6 FUNCTIONS

```

-- common schema: wh_code, id_user, biz_date, fn (label), units, time_sec
all_fns AS (
SELECT * FROM pick
UNION ALL SELECT * FROM dcss
UNION ALL SELECT * FROM recv

```

```
UNION ALL SELECT * FROM sort
UNION ALL SELECT * FROM inv_stocktake
UNION ALL SELECT * FROM inv_movement
),
```

```
-- — PIVOT
```

```
-- collapses fn rows into columns per user/wh/date
-- CASE WHEN fn=X THEN ... ELSE 0 pattern ensures no NULLs in output
pivoted AS (
  SELECT
    wh_code, id_user, biz_date,
    SUM(CASE WHEN fn='pick'   THEN units   ELSE 0 END) AS pick_units,
    SUM(CASE WHEN fn='pick'   THEN time_sec ELSE 0 END) AS pick_sec,
    SUM(CASE WHEN fn='put'    THEN units   ELSE 0 END) AS put_lines,
    SUM(CASE WHEN fn='put'    THEN time_sec ELSE 0 END) AS put_sec,
    SUM(CASE WHEN fn='receive' THEN units   ELSE 0 END) AS recv_boxes,
    SUM(CASE WHEN fn='receive' THEN time_sec ELSE 0 END) AS recv_sec,
    SUM(CASE WHEN fn='sort'   THEN units   ELSE 0 END) AS sort_items,
    SUM(CASE WHEN fn='sort'   THEN time_sec ELSE 0 END) AS sort_sec,
    SUM(CASE WHEN fn='stocktake' THEN units   ELSE 0 END) AS stocktake_units,
    SUM(CASE WHEN fn='stocktake' THEN time_sec ELSE 0 END) AS stocktake_sec,
    SUM(CASE WHEN fn='movement' THEN units   ELSE 0 END) AS movement_units,
    SUM(CASE WHEN fn='movement' THEN time_sec ELSE 0 END) AS movement_sec
  FROM all_fns
  GROUP BY 1,2,3
),
```

```
-- — USERNAMES
```

```
-- deduplicates users who appear in both cwpicking and goms user tables
-- COALESCE logic: prefer ops.idp.noon.team accounts (local WH logins)
--                over noon.com SSO accounts
-- both sources split on '@' to get just the username prefix
usernames AS (
  SELECT id_user,
    COALESCE(
      MIN(CASE WHEN username LIKE '%ops.idp.noon.team%' THEN
        SPLIT(username, '@')[OFFSET(0)] END),
      MIN(SPLIT(username, '@')[OFFSET(0)])
    ) AS username
  FROM (
```

```

SELECT id_user, username FROM noondwh.scgoms_cwpicking.user
UNION ALL
SELECT id_user, username FROM noondwh.scgoms_goms.user
)
GROUP BY id_user
)

-- — FINAL OUTPUT

```

```

--
-- one row per user × wh_code × biz_date
-- warehouse_group derived from wh_code prefix
-- username falls back to CAST(id_user AS STRING) if not in username table
-- total_throughput = sum of all unit types (movement included in count)
SELECT
  p.biz_date,
  p.wh_code,
  CASE
    WHEN p.wh_code LIKE 'AUHFKZ%' THEN 'FZ Kezad'
    WHEN p.wh_code LIKE 'AUHID%' THEN 'Kezad'
    WHEN p.wh_code LIKE 'DXBID%' THEN 'TP'
    WHEN p.wh_code LIKE 'DXSID%' OR p.wh_code LIKE 'DXSLQ%' THEN 'Dubai South'
    WHEN p.wh_code = 'DXBFNV01' THEN 'F&V'
    ELSE 'Other'
  END
  AS warehouse_group,
  p.id_user,
  COALESCE(u.username, CAST(p.id_user AS STRING)) AS username,
  p.pick_units,
  p.pick_sec,
  p.sort_items,
  p.sort_sec,
  p.put_lines,
  p.put_sec,
  p.recv_boxes,
  p.recv_sec,
  p.stocktake_units,
  p.stocktake_sec,
  p.movement_units,
  p.movement_sec,
  p.pick_units + p.sort_items + p.put_lines
  + p.recv_boxes + p.stocktake_units + p.movement_units AS total_throughput
FROM pivoted p
LEFT JOIN usernames u ON u.id_user = p.id_user
ORDER BY 1,3,2,5

```

/*

WAREHOUSE IPH — WITH AND WITHOUT UTILIZATION

Purpose : Per-user, per-function IPH at two granularities:

- IPH_effective - job time only (no idle; >600s scan-gap rule for pick, 300s gap cap for sort)
- IPH_utilization- total shift window: first_scan → last_scan capped at 12h (43200s); cases exceeding 12h flagged for ops review

Source : Raw tables (~30min refresh) - noondwh.scgoms_cwpicking,
noondwh.mxdcss_dcsc, noondwh.scinbound_inbound,
noondwh.scgoms_goms

Grain : user × wh_code × function × biz_date (detail layer)
wh_code × function × biz_date (summary layer)

Biz day : 06:00 local time = midnight on biz_date
(DATE(DATETIME(ts,'Asia/Dubai') - INTERVAL 6 HOUR))

Date : Change biz_date_start / biz_date_end for a different day.
Current window = yesterday.

DEFINITIONS

effective_time_sec : Active working time per job, excluding idle.

pick → if first scan >600s after job open,
clock starts from first_scan, else full job

sort → sum of per-scan gaps capped at 300s each

put → job open → last completed line

receive → first box created → last box updated

stocktake → job opened → pending_confirmation
(or closed if no pending_conf)

movement → MIN(tl.created_at) → MAX(tl.created_at)

utilization_sec : DATETIME_DIFF(last_event, first_event) per user/wh/day
across ALL functions combined. Capped at 43200s (12h).
Over-12h flag raised when raw window > 43200s.

IPH_effective : units / (effective_time_sec / 3600)
Only active job time in denominator.

IPH_utilization : units / (utilization_sec / 3600)
Total shift window in denominator.
Lower than IPH_effective – penalises idle time.

NOTES

- movement time: day-level MIN-MAX, not per-job (see MP productivity notes)
- sort: only GOMS Sort WHs; DXBID01 is Non-Sort / cwpicking
- receive: scinbound.warehouse uses partner_code not partner_warehouse_code
- stocktake: force_closed with no pending_confirmation excluded (incomplete)
- username: prefers ops.idp.noon.team over noon.com SSO accounts
- Source: directly from raw noonwh.* tables (not wh_picker_metrics_pace derived table) – real-time, reflects jobs completed up to ~30min ago

REF: picker_raw_v2_wh.toml, wh_picker_metrics_pace.toml,
MP productivity query (mp_productivity.sql)

*/

```
-- — DATE WINDOW —  
-- Change these two values to shift the analysis date.  
-- biz_date_start = 02:00 UTC = 06:00 Dubai = business day start  
-- For yesterday: start = DATE_SUB(CURRENT_DATE(), 1) 02:00 UTC  
DECLARE biz_date_start TIMESTAMP DEFAULT TIMESTAMP(  
    CONCAT(FORMAT_DATE('%Y-%m-%d', DATE_SUB(CURRENT_DATE('Asia/Dubai'), INTERVAL 1  
DAY)), ' 02:00:00 UTC')  
);  
DECLARE biz_date_end TIMESTAMP DEFAULT TIMESTAMP(
```

```

        CONCAT(FORMAT_DATE('%Y-%m-%d', CURRENT_DATE('Asia/Dubai')), ' 02:00:00 UTC')
    );

-- -- WH SCOPE -----
-- Explicit allowlist. Mirrors the MP Productivity query scope.
-- Add/remove codes here to adjust coverage.
WITH wh_scope AS (
    SELECT wh_code FROM UNNEST([
        'AUHID01', 'AUHID02',
        'AUHFKZ01', 'AUHFKZ02', 'AUHFKZ03',
        'DXBID01', 'DXBID02', 'DXBID03', 'DXBID04', 'DXBID07',
        'DXSID01', 'DXSID02', 'DXSID03', 'DXSID04', 'DXSID06', 'DXSID07',
        'DXSLQD01',
        'DXBFNV01'
    ]) AS wh_code
),

/*
=====
FUNCTION 1 - PICK
Source : noondwh.scgoms_cwpicking
Units  : distinct picklist lines with a container scan
Time    : job_start→job_end; if first scan >600s after job open,
          clock starts from first_scan (idle-strip rule)
REF     : picker_raw_v2_wh.toml, wh_picker_metrics_pace.toml
=====
*/

pick_jobs AS (
    SELECT
        src_w.partner_warehouse_code AS wh_code,
        pj.id_user,
        DATE(DATETIME(pj.created_at, 'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
        DATETIME(pj.created_at, 'Asia/Dubai') AS job_start,
        DATETIME(pj.finsihed_picking_at, 'Asia/Dubai') AS job_end,
        -- note: 'finsihed_picking_at' is a typo in the source schema - do not correct
        COUNT(DISTINCT pll.id_picklist_line) AS units,

```



```

        MIN(DATETIME(pcl.created_at, 'Asia/Dubai'))                AS first_scan,
        MAX(DATETIME(pcl.created_at, 'Asia/Dubai'))                AS last_scan
FROM noonwh.scgoms_cwpicking.picking_job pj
JOIN noonwh.scgoms_cwpicking.status pjst
    ON pj.id_status = pjst.id_status AND pjst.code = 'closed'
JOIN noonwh.scgoms_cwpicking.picklist_line pll
    ON pll.id_picking_job = pj.id_picking_job
JOIN noonwh.scgoms_cwpicking.warehouse src_w
    ON pll.id_warehouse = src_w.id_warehouse
JOIN noonwh.scgoms_cwpicking.picking_container_line pcl
    ON pcl.id_picklist_line = pll.id_picklist_line
    AND pcl.id_picking_container_line IS NOT NULL
JOIN wh_scope ws ON src_w.partner_warehouse_code = ws.wh_code
WHERE pj.created_at >= biz_date_start AND pj.created_at < biz_date_end
GROUP BY 1,2,3,4,5
),

```

```

pick AS (
SELECT
    wh_code,
    id_user,
    biz_date,
    'pick'                                AS fn,
    SUM(units)                            AS units,
    -- effective time: if first scan >10min after job open, strip idle prefix
    SUM(
        CASE
            WHEN DATETIME_DIFF(first_scan, job_start, SECOND) > 600
            THEN DATETIME_DIFF(job_end, first_scan, SECOND)
            ELSE DATETIME_DIFF(job_end, job_start, SECOND)
        END
    )
)
effective_sec,
MIN(CASE
    WHEN DATETIME_DIFF(first_scan, job_start, SECOND) > 600
    THEN first_scan ELSE job_start

```

```

        END) AS
day_first_event,
        MAX(job_end) AS
day_last_event
FROM pick_jobs
GROUP BY 1,2,3,4
),

/*
=====
FUNCTION 2 – PUTAWAY
Source : noondwh.mxdcss_dcscs
Units : distinct completed job lines (id_status = 3)
Time : job open → last completed line updated_at
Subtypes: 4=putaway_item, 5=putaway_case, 6=putaway_carton
REF : mp_productivity.sql (dcscs CTE)
=====
*/

put AS (
SELECT
    w.partner_warehouse_code AS wh_code,
    j.id_user,
    DATE(DATETIME(j.created_at, 'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
    'put' AS fn,
    COUNT(DISTINCT jl.id_job_line) AS units,
    DATETIME_DIFF(
        MAX(DATETIME(jl.updated_at, 'Asia/Dubai')),
        MIN(DATETIME(j.created_at, 'Asia/Dubai')),
        SECOND
    ) AS
effective_sec,
    MIN(DATETIME(j.created_at, 'Asia/Dubai')) AS
day_first_event,
    MAX(DATETIME(jl.updated_at, 'Asia/Dubai')) AS
day_last_event
FROM noondwh.mxdcss_dcscs.job j

```

```

JOIN noondwh.mxdcss_dcscs.job_line jl ON jl.id_job = j.id_job
JOIN noondwh.mxdcss_dcscs.warehouse w ON j.id_warehouse = w.id_warehouse
JOIN wh_scope ws ON w.partner_warehouse_code = ws.wh_code
WHERE j.id_job_subtype IN (4,5,6)      -- putaway subtypes only
      AND jl.id_status = 3             -- completed lines only
      AND j.created_at >= biz_date_start AND j.created_at < biz_date_end
      AND j.id_user IS NOT NULL
GROUP BY 1,2,3,4
),

```

```

/*

```

FUNCTION 3 – RECEIVE (INBOUND)

Source : noondwh.scinbound_inbound

Units : distinct inbound boxes (id_status IN 13, 14)

Time : first box created → last box updated for user/wh/day

NOTE : scinbound.warehouse.partner_code maps to wh_scope.wh_code
(not partner_warehouse_code – different column name)

REF : mp_productivity.sql (recv CTE)

```

*/

```

```

recv AS (
SELECT
    w.partner_code                                AS wh_code,
    ib.id_user,
    DATE(DATETIME(ib.created_at, 'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
    'receive'                                     AS fn,
    COUNT(DISTINCT ib.id_inbound_box)             AS units,
    DATETIME_DIFF(
        MAX(DATETIME(ib.updated_at, 'Asia/Dubai')),
        MIN(DATETIME(ib.created_at, 'Asia/Dubai')),
        SECOND
    )                                              AS
effective_sec,
    MIN(DATETIME(ib.created_at, 'Asia/Dubai'))    AS
day_first_event,

```

```

        MAX(DATETIME(ib.updated_at, 'Asia/Dubai')) AS
day_last_event
FROM noondwh.scinbound_inbound.inbound_box ib
JOIN noondwh.scinbound_inbound.warehouse w ON ib.id_warehouse = w.id_warehouse
JOIN wh_scope ws ON w.partner_code = ws.wh_code -- note: partner_code, not
partner_warehouse_code
WHERE ib.id_status IN (13,14)
      AND ib.created_at >= biz_date_start AND ib.created_at < biz_date_end
      AND ib.id_user IS NOT NULL
GROUP BY 1,2,3,4
),

/*
=====
FUNCTION 4 – SORT
Source : noondwh.scgoms_goms.audit_log
Units : sort scan events (id_audit_action=21, to='pending_putaway')
Time : sum of per-scan gaps capped at 300s each
      (LEAST(gap, 300) prevents idle/break inflation)
      last scan of day gets COALESCE(gap, 300) = 300s default
NOTE : only covers GOMS Sort WHs; DXBID01 uses cwpicking Non-Sort
REF : mp_productivity.sql (sort / sort_base CTEs)
=====
*/

sort_raw AS (
SELECT
    w.partner_warehouse_code AS wh_code,
    al.id_user,
    DATE(DATETIME(al.created_at, 'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
    DATETIME(al.created_at, 'Asia/Dubai') AS scan_at,
    LEAD(DATETIME(al.created_at, 'Asia/Dubai')) OVER (
        PARTITION BY al.id_user, w.partner_warehouse_code,
                     DATE(DATETIME(al.created_at, 'Asia/Dubai') - INTERVAL 6 HOUR)
        ORDER BY al.created_at
    ) AS next_scan
FROM noondwh.scgoms_goms.audit_log al

```

```

JOIN noondwh.scgoms_goms.input_container_line icl
    ON icl.id_input_container_line = al.action_key
JOIN noondwh.scgoms_goms.input_container ic
    ON ic.id_input_container = icl.id_input_container
JOIN noondwh.scgoms_goms.warehouse w ON w.id_warehouse = ic.id_warehouse
JOIN wh_scope ws ON w.partner_warehouse_code = ws.wh_code
WHERE al.id_audit_action = 21
    AND JSON_EXTRACT_SCALAR(al.misc,'$.to') = 'pending_putaway'
    AND al.created_at >= biz_date_start AND al.created_at < biz_date_end
    AND al.id_user IS NOT NULL
),

sort AS (
SELECT
    wh_code,
    id_user,
    biz_date,
    'sort' AS fn,
    COUNT(*) AS units,
    -- capped gap sum = effective sort time without idle inflation
    SUM(LEAST(COALESCE(DATETIME_DIFF(next_scan, scan_at, SECOND), 300), 300))
    AS
effective_sec,
    MIN(scan_at) AS
day_first_event,
    MAX(scan_at) AS
day_last_event
FROM sort_raw
GROUP BY 1,2,3,4
),

```

```

/*

```

FUNCTION 5 – STOCKTAKE

Source : noondwh.mxdcss_dcsc.job + audit_log

Units : distinct job lines (excluding excess-source lines)

Time : job opened_at → pending_confirmation_at

(falls back to closed_at if no pending_confirmation exists)

Exclusion: force_closed jobs entirely

REF : mp_productivity.sql (stocktake_job_times, inv_stocktake CTEs)

*/

```
st_process_type AS (  
  -- Pre-join process_type per job to avoid subquery in JOIN (BigQuery restriction)  
  SELECT j.id_job, pt.code AS pt_code  
  FROM `noondwh.mxdcss_dcsc.job` j  
  JOIN `noondwh.mxdcss_dcsc.process` p ON p.id_process = j.id_process  
  JOIN `noondwh.mxdcss_dcsc.process_type` pt ON pt.id_process_type = p.id_process_type  
  WHERE j.id_job_subtype = 10 -- stock_take jobs only  
  AND j.created_at >= biz_date_start AND j.created_at < biz_date_end  
)
```

```
st_job_times AS (  
  SELECT  
    j.id_job, j.id_user, j.id_warehouse,  
    MIN(CASE WHEN JSON_EXTRACT_SCALAR(al.misc, '$.to') = 'opened'  
      THEN al.created_at END) AS  
opened_at,  
    MIN(CASE WHEN JSON_EXTRACT_SCALAR(al.misc, '$.to') = 'pending_confirmation'  
      THEN al.created_at END) AS  
pending_at,  
    MIN(CASE WHEN JSON_EXTRACT_SCALAR(al.misc, '$.to') = 'closed'  
      THEN al.created_at END) AS  
closed_at  
  FROM `noondwh.mxdcss_dcsc.job` j  
  JOIN `noondwh.mxdcss_dcsc.status` s ON s.id_status = j.id_status  
  JOIN st_process_type jpt ON jpt.id_job = j.id_job  
  JOIN `noondwh.mxdcss_dcsc.audit_log` al  
    ON al.id_audit_action = 1  
    AND al.action_key = j.id_job  
    AND al.created_at >= biz_date_start AND al.created_at < biz_date_end  
  WHERE s.code = 'closed' -- force_closed excluded entirely
```

```

    AND j.created_at >= biz_date_start AND j.created_at < biz_date_end
    AND j.id_user IS NOT NULL
GROUP BY 1,2,3
),

stocktake AS (
SELECT
    w.partner_warehouse_code AS wh_code,
    jt.id_user,
    DATE(DATETIME(jt.opened_at, 'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,
    'stocktake' AS fn,
    COUNT(CASE
        WHEN jl.id_job_line IS NULL
            OR JSON_EXTRACT_SCALAR(jl.misc, '$.source') = 'excess' THEN NULL
        ELSE jl.id_job_line
    END) AS units,
    SUM(TIMESTAMP_DIFF(
        COALESCE(jt.pending_at, jt.closed_at),
        jt.opened_at, SECOND
    )) AS
effective_sec,
    MIN(DATETIME(jt.opened_at, 'Asia/Dubai')) AS
day_first_event,
    MAX(DATETIME(COALESCE(jt.pending_at, jt.closed_at), 'Asia/Dubai')) AS
day_last_event
FROM st_job_times jt
LEFT JOIN `noondwh.mxdcss_dcsc.job_line` jl ON jl.id_job = jt.id_job
LEFT JOIN `noondwh.mxdcss_dcsc.warehouse` w ON w.id_warehouse = jt.id_warehouse
JOIN wh_scope ws ON w.partner_warehouse_code = ws.wh_code
WHERE jt.opened_at IS NOT NULL
    AND COALESCE(jt.pending_at, jt.closed_at) IS NOT NULL
GROUP BY 1,2,3,4
),

```

```

/*

```

FUNCTION 6 – INTERNAL MOVEMENT

Source : noondwh.mxdcss_dcsc.transaction_line + job

Units : SUM(tl.qty) – volume moved (can be > 1 per row)

Time : MIN-MAX of tl.created_at per user/wh/day (day-level,
not per-job – understates time if multiple jobs per day)

Filter : deep locations only (misc.is_deep = '1') to exclude
forward-pick-face micro-moves

NOTE : id_user sourced from job table (not transaction_line)

REF : mp_productivity.sql (inv_movement CTE)

*/

movement AS (

SELECT

w.partner_warehouse_code AS wh_code,

j.id_user,

DATE(DATETIME(tl.created_at, 'Asia/Dubai') - INTERVAL 6 HOUR) AS biz_date,

'movement' AS fn,

SUM(tl.qty) AS units,

DATETIME_DIFF(

MAX(DATETIME(tl.created_at, 'Asia/Dubai')),

MIN(DATETIME(tl.created_at, 'Asia/Dubai')),

SECOND

) AS

effective_sec,

MIN(DATETIME(tl.created_at, 'Asia/Dubai')) AS

day_first_event,

MAX(DATETIME(tl.created_at, 'Asia/Dubai')) AS

day_last_event

FROM noondwh.mxdcss_dcsc.transaction_line tl

-- source location: only deep storage (is_deep = '1')

JOIN noondwh.mxdcss_dcsc.location l

ON ('location', l.id_location) = (tl.src_key, CAST(tl.src_value AS INT64))

LEFT JOIN noondwh.mxdcss_dcsc.warehouse w ON w.id_warehouse = l.id_warehouse

JOIN noondwh.mxdcss_dcsc.job j ON j.id_job = tl.id_job

JOIN noondwh.mxdcss_dcsc.process p ON p.id_process = j.id_process


```

JOIN noondwh.mxdcss_dcsc.process_type pt      ON pt.id_process_type =
p.id_process_type
JOIN wh_scope ws ON w.partner_warehouse_code = ws.wh_code
WHERE pt.code IN ('internal_movement', 'migration')
      AND tl.created_at >= biz_date_start AND tl.created_at < biz_date_end
      AND JSON_EXTRACT_SCALAR(l.misc, '$.is_deep') = '1'
      AND j.id_user IS NOT NULL
GROUP BY 1,2,3,4
),

```

```
/*
```

```

UNION ALL 6 FUNCTIONS

```

```

Common schema: wh_code, id_user, biz_date, fn,
                units, effective_sec,
                day_first_event, day_last_event

```

```
*/
```

```

all_fns AS (
  SELECT * FROM pick
  UNION ALL SELECT * FROM put
  UNION ALL SELECT * FROM recv
  UNION ALL SELECT * FROM sort
  UNION ALL SELECT * FROM stocktake
  UNION ALL SELECT * FROM movement
),

```

```
/*
```

```

UTILIZATION WINDOW – per user × wh × day ACROSS ALL FUNCTIONS
first_event_all = earliest event across all 6 functions
last_event_all  = latest  event across all 6 functions
raw_shift_sec   = last_event_all - first_event_all (seconds)

```

```

Cap at 43200s (12h):

```

```

• If raw_shift_sec > 43200: set utilization_sec = 43200,

```

flag over_12h = TRUE for ops review

- Otherwise: utilization_sec = raw_shift_sec

Why cap? A picker could open a job at 06:01 and close the last job at 21:59 – 16h raw, but the actual shift is 12h. Values > 12h almost always indicate a forgotten open job or a double shift that should be reviewed.

```
*/
utilization_window AS (
  SELECT
    wh_code,
    id_user,
    biz_date,
    MIN(day_first_event) AS
first_event_all,
    MAX(day_last_event) AS
last_event_all,
    DATETIME_DIFF(
      MAX(day_last_event), MIN(day_first_event), SECOND
    ) AS
raw_shift_sec,
    -- cap at 12h (43200s)
    LEAST(
      DATETIME_DIFF(MAX(day_last_event), MIN(day_first_event), SECOND),
      43200
    ) AS
utilization_sec,
    -- flag: this user's shift window exceeded 12h – review recommended
    CASE
      WHEN DATETIME_DIFF(MAX(day_last_event), MIN(day_first_event), SECOND) > 43200
      THEN TRUE ELSE FALSE
    END AS
over_12h_flag
  FROM all_fns
  GROUP BY 1,2,3
```

```
),
```

```
/*
```

```
USER × FUNCTION LEVEL – detail layer
```

```
Joins utilization_window for the utilization denominator.
```

```
IPH_effective  = units / (effective_sec / 3600)
```

```
IPH_utilization = units / (utilization_sec / 3600)
```

```
                (same denominator for all functions of a user)
```

```
*/
```

```
usernames AS (
```

```
-- Prefer ops.idp.noon.team accounts (local WH logins) over noon.com SSO
```

```
-- Falls back to raw id_user if not found in either table
```

```
SELECT id_user,
```

```
  COALESCE(
```

```
    MIN(CASE WHEN username LIKE '%ops.idp.noon.team%'
```

```
          THEN SPLIT(username, '@')[OFFSET(0)] END),
```

```
    MIN(SPLIT(username, '@')[OFFSET(0)])
```

```
  ) AS username
```

```
FROM (
```

```
  SELECT id_user, username FROM noondwh.scgoms_cwpicking.user
```

```
  UNION ALL
```

```
  SELECT id_user, username FROM noondwh.scgoms_goms.user
```

```
)
```

```
GROUP BY id_user
```

```
),
```

```
wh_group AS (
```

```
-- Warehouse group label from wh_code prefix – mirrors mp_productivity logic
```

```
SELECT wh_code,
```

```
  CASE
```

```
    WHEN wh_code LIKE 'AUHFKZ%' THEN 'FZ Kezad'
```

```
    WHEN wh_code LIKE 'AUHID%' THEN 'Kezad'
```

```
    WHEN wh_code LIKE 'DXBID%' THEN 'TP'
```

```
    WHEN wh_code LIKE 'DXSID%' OR wh_code LIKE 'DXSLQ%' THEN 'Dubai South'
```

```

        WHEN wh_code = 'DXBFNV01' THEN 'F&V'
        ELSE 'Other'
    END AS warehouse_group
FROM wh_scope
),

detail AS (
    SELECT
        f.biz_date,
        f.wh_code,
        wg.warehouse_group,
        COALESCE(u.username, CAST(f.id_user AS STRING)) AS username,
        f.id_user,
        f.fn AS
function_name,
        f.units,
        f.effective_sec,

        -- — IPH WITHOUT UTILIZATION (job/scan time only) —————
        -- Denominator = active working time, idle stripped per function logic
        ROUND(SAFE_DIVIDE(f.units * 3600.0, f.effective_sec), 1) AS
iph_effective,

        -- — UTILIZATION WINDOW —————
        uw.utilization_sec,
        uw.raw_shift_sec,
        uw.over_12h_flag,      -- TRUE = shift window > 12h; ops should verify
        ROUND(uw.utilization_sec / 3600.0, 2) AS
utilization_hrs,

        -- — IPH WITH UTILIZATION (full shift window) —————
        -- Denominator = total shift time (first event → last event, max 12h)
        -- This is always ≤ iph_effective because the denominator is larger
        ROUND(SAFE_DIVIDE(f.units * 3600.0, uw.utilization_sec), 1) AS
iph_utilization,

```

```

-- -- UTILIZATION RATE -----
-- What fraction of shift window was active working time?
ROUND(SAFE_DIVIDE(f.effective_sec, uw.utilization_sec) * 100, 1) AS
utilization_pct,

    uw.first_event_all,
    uw.last_event_all
FROM all_fns f
JOIN utilization_window uw USING (wh_code, id_user, biz_date)
LEFT JOIN usernames u ON u.id_user = f.id_user
LEFT JOIN wh_group wg ON wg.wh_code = f.wh_code
),

/*
=====

SUMMARY - warehouse × function level
Aggregates detail rows. IPH at summary level computed from
SUM(units) / SUM(time) - not AVG(iph) per user - to avoid
giving equal weight to low-volume users.

iph_effective_summary    = total_units / (total_effective_sec / 3600)
iph_utilization_summary = total_units / (total_utilization_sec / 3600)

total_utilization_sec: sum of EACH USER's capped utilization_sec
(not a single window across all users)

over_12h_users: count of users flagged for shift > 12h - ops alert
=====
*/
summary AS (
SELECT
    biz_date,
    wh_code,
    warehouse_group,
    function_name,
    COUNT(DISTINCT id_user) AS headcount,

```

```

SUM(units) AS total_units,
SUM(effective_sec) AS
total_effective_sec,
-- Sum each user's capped utilization (not one big window across all users)
SUM(utilization_sec) AS
total_utilization_sec,
ROUND(SAFE_DIVIDE(SUM(units) * 3600.0, SUM(effective_sec)), 1) AS
iph_effective,
ROUND(SAFE_DIVIDE(SUM(units) * 3600.0, SUM(utilization_sec)), 1) AS
iph_utilization,
ROUND(SAFE_DIVIDE(SUM(effective_sec), SUM(utilization_sec)) * 100, 1)
AS
avg_utilization_pct,
-- Users whose raw shift window exceeded 12h - review these
COUNTIF(over_12h_flag = TRUE) AS
over_12h_users,
-- Average shift hours (capped) per user
ROUND(SUM(utilization_sec) / COUNT(DISTINCT id_user) / 3600.0, 2) AS avg_shift_hrs
FROM detail
GROUP BY 1,2,3,4
)

/*
=====
FINAL OUTPUTS - pick the block you need
=====
*/

-- — OUTPUT A: SUMMARY - warehouse × function (default output) —
-- One row per warehouse × function for the biz_date window.
-- iph_effective = active job time only (no idle)
-- iph_utilization = total shift window (first-last, max 12h)
-- over_12h_users = flag count for ops review
SELECT
biz_date,
warehouse_group,

```

```

wh_code,
function_name,
headcount,
total_units,
ROUND(total_effective_sec / 3600.0, 2) AS
total_effective_hrs,
ROUND(total_utilization_sec / 3600.0, 2) AS
total_utilization_hrs,
iph_effective,
iph_utilization,
avg_utilization_pct,
avg_shift_hrs,
over_12h_users
FROM summary
ORDER BY biz_date, warehouse_group, wh_code, function_name;

```

```

-- — OUTPUT B: USER DETAIL - user × function (uncomment to use) —
-- One row per user × function.
-- Shows both IPH flavours + over_12h flag per user.
/*
SELECT
  biz_date,
  warehouse_group,
  wh_code,
  function_name,
  username,
  id_user,
  units,
  ROUND(effective_sec / 3600.0, 2) AS effective_hrs,
  iph_effective,
  ROUND(utilization_hrs, 2) AS shift_hrs_capped,
  iph_utilization,
  utilization_pct,
  over_12h_flag, -- TRUE = needs ops review
  raw_shift_sec, -- actual window before 12h cap

```

```

    first_event_all,
    last_event_all
FROM detail
ORDER BY biz_date, warehouse_group, wh_code, function_name, username;
*/

```

```

-- — OUTPUT C: OVER-12H CASES (uncomment to use) —————

```

```

-- Shows only users where shift window > 12h.

```

```

-- Use this as the ops review alert list.

```

```

/*

```

```

SELECT DISTINCT

```

```

    biz_date,

```

```

    warehouse_group,

```

```

    wh_code,

```

```

    username,

```

```

    id_user,

```

```

    ROUND(raw_shift_sec / 3600.0, 2)          AS raw_shift_hrs,

```

```

    ROUND(utilization_hrs, 2)                AS capped_shift_hrs,

```

```

    first_event_all,

```

```

    last_event_all,

```

```

    '⚠ shift window > 12h – verify if double shift or open job'

```

```

                                AS review_note

```

```

FROM detail

```

```

WHERE over_12h_flag = TRUE

```

```

ORDER BY raw_shift_sec DESC;

```

```

*/

```


/*

WH PICKER METRICS PACE TABLE — User-level CW Picking daily metrics

-- Destination: noonbinimops.fulfillment.wh_picker_metrics_pace
-- Partitioned by date for efficient downstream queries.
-- Covers UAE (id_country=1), KSA (2,5), Egypt (3).
-- Lookback = 15 days.

*/

#

FILE : CWH_PACE/wh_picker_metrics_pace.toml
PURPOSE: Builds and exposes the wh_picker_metrics_pace BigQuery
table — the canonical user-level CW Picking metrics table.
Contains two tasks:
1. wh_picker_metrics_pace — CREATE OR REPLACE TABLE
partitioned by date with per-user picking metrics
(effective time, IPH, MS/BD/BS breakdown, tenure)
2. cwh_pace_picker_metrics — reads from that table and
pushes a warehouse-day summary to Google Sheets
DAG : Warehouse_Metrics_uae (runs daily at 05:40 UTC)
SOURCE : noondwh.scgoms_cwpicking (picking_job, picklist_line,
picking_container, warehouse, user)
noonbinimops.fulfillment.wh_code_wh_category_mapping
DEST : noonbinimops.fulfillment.wh_picker_metrics_pace (BQ table)
Google Sheets picker_metrics tab
AUTHOR : noon Minutes Ops
#

```

#
# VARIABLE DEFINITIONS
# _____
# dag_id      : Parent DAG linking both tasks.
# task_id     : Unique name within the DAG.
# project_id  : GCP billing project for BigQuery.
# sql         : BigQuery DDL — CREATE OR REPLACE TABLE statement.
# query       : BigQuery SQL for the Sheets push task.
# depends     : List of task_ids that must complete before this task.
# url         : Destination Google Spreadsheet URL.
# worksheet   : Target tab name.
#
# KEY SQL TERMS — wh_picker_metrics_pace
# _____
# pjst.code = 'closed'      : only closed CW picking jobs
# INTERVAL 6 HOUR          : business day cutoff offset
# effective_time_sec        : adjusted picking time (excludes idle)
# Warehouse_group          : city-level grouping from wh_code prefix
# AUH → Kezad, DXB → TP, RUH → Riyadh, JED → Jeddah, etc.
# wh_codes                 : STRING_AGG of all sub-warehouse categories
# user_ageing              : days since user account created
# MS_* / BD_* / BS_*       : per-job-type (Multi Store, Batch Dynamic,
#                           Batch Static) metrics
#
# KEY SQL TERMS — cwh_pace_picker_metrics
# _____
# working_hrs              : effective picking hours per user
# active_hrs               : first-to-last scan duration per user
# utilization              : working_hrs / active_hrs ratio
# iph                     : items per effective working hour
# avg_working_hrs          : working_hrs / pickers
# avg_active_hrs           : active_hrs / pickers
#

```

WITH base AS (

-- — CW Picking job-level grain _____

SELECT

```

src_w.partner_warehouse_code AS warehouse_code,
pj.job_type,
pj.id_user,
pj.job_nr,

```

```

-- Business date (country aware, 6 AM cutoff)
DATE(
  DATETIME(
    pj.created_at,
    CASE
      WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
      WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
      WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
    END
  ) - INTERVAL 6 HOUR
) AS date,

-- Job start in local timezone
DATETIME(
  pj.created_at,
  CASE
    WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
    WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
    WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
  END
) AS job_accepted_at,

-- Job end in local timezone
DATETIME(
  pj.finsihed_picking_at,
  CASE
    WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
    WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
    WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
  END
) AS job_closed_at,

-- Raw UTC delta in seconds
DATETIME_DIFF(
  pj.finsihed_picking_at,
  pj.created_at,
  SECOND
) AS delta,

-- Distinct picked lines
COUNT(DISTINCT pll.id_picklist_line) AS picked_qty,

-- Unique containers per job

```

```

COUNT(
  DISTINCT COALESCE(pcr.container_barcode, pcr.tote_code)
) AS containers_used,

-- First scan timestamp
MIN(
  DATETIME(
    pcl.created_at,
    CASE
      WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
      WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
      WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
    END
  )
) AS min_picked_at

FROM noondwh.scgoms_cwpicking.picking_job pj
JOIN noondwh.scgoms_cwpicking.status pjst
  ON pj.id_status = pjst.id_status
JOIN noondwh.scgoms_cwpicking.picklist_line pll
  ON pll.id_picking_job = pj.id_picking_job
JOIN noondwh.scgoms_cwpicking.warehouse src_w
  ON pll.id_warehouse = src_w.id_warehouse
JOIN noondwh.scgoms_cwpicking.picking_container_line pcl
  ON pcl.id_picklist_line = pll.id_picklist_line
JOIN noondwh.scgoms_cwpicking.picking_container pcr
  ON pcr.id_picking_container = pcl.id_picking_container

WHERE
  pjst.code = 'closed'
  AND pcl.id_picking_container_line IS NOT NULL
  AND DATE(
    DATETIME(
      pj.created_at,
      CASE
        WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
        WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
        WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
      END
    ) - INTERVAL 6 HOUR
  ) >= CURRENT_DATE - 15 -- 15-day lookback for the table

GROUP BY ALL
),

```

```

base_jobs AS (
-- — Effective time per job —————
SELECT *,
(CASE
  WHEN DATETIME_DIFF(min_picked_at, job_accepted_at, SECOND) > 600
  THEN DATETIME_DIFF(job_closed_at, min_picked_at, SECOND)
  ELSE delta
END) AS effective_time_sec
FROM base
),

user_level AS (
-- — User-day aggregation with per-job-type breakdown —————
SELECT
  date,
  -- City-level warehouse grouping from warehouse_code prefix
  CASE
    WHEN b.warehouse_code LIKE '%AUHFKZ%' THEN 'FZ Kezad'
    WHEN b.warehouse_code LIKE '%AUH%'   THEN 'Kezad'
    WHEN b.warehouse_code LIKE '%DXB%'   THEN 'TP'
    WHEN b.warehouse_code LIKE '%RUH%'   THEN 'Riyadh'
    WHEN b.warehouse_code LIKE '%JED%'   THEN 'Jeddah'
    WHEN b.warehouse_code LIKE '%BAH%'   THEN 'Bahrain'
    WHEN b.warehouse_code LIKE '%CAI%'   THEN 'Cairo'
    ELSE 'Other'
  END AS Warehouse_group,
  -- Aggregated list of detailed sub-warehouse categories for this user
  STRING_AGG(DISTINCT COALESCE(c.warehouse_catrgory, 'Other'), ', ' ORDER BY
COALESCE(c.warehouse_catrgory, 'Other')) AS wh_codes,
  id_user,
  SUM(picked_qty)      AS picked_qty,
  SUM(effective_time_sec) AS effective_time_sec,
  SUM(containers_used) AS containers_used,
  -- First scan: use min_picked_at if idle >10min before it, else job_accepted_at
  MIN(CASE WHEN DATETIME_DIFF(min_picked_at, job_accepted_at, SECOND) > 600
    THEN min_picked_at ELSE job_accepted_at END) AS first_scan_time,
  MAX(job_closed_at) AS last_scan_time,

  -- Multi Store Picking
  SUM(CASE WHEN job_type = 'multi_store_picking' THEN picked_qty END)      AS
MS_PickedQty,
  SUM(CASE WHEN job_type = 'multi_store_picking' THEN containers_used END) AS
MS_UniqueContainers,

```

```

SUM(CASE WHEN job_type = 'multi_store_picking' THEN effective_time_sec END) AS
MS_TimeSec,
SAFE_DIVIDE(
3600 * SUM(CASE WHEN job_type = 'multi_store_picking' THEN picked_qty END),
SUM(CASE WHEN job_type = 'multi_store_picking' THEN effective_time_sec END)
) AS MS_IPH,

-- Batch Dynamic
SUM(CASE WHEN job_type = 'batch_dynamic' THEN picked_qty END) AS
BD_PickedQty,
SUM(CASE WHEN job_type = 'batch_dynamic' THEN containers_used END) AS
BD_UniqueContainers,
SUM(CASE WHEN job_type = 'batch_dynamic' THEN effective_time_sec END) AS
BD_TimeSec,
SAFE_DIVIDE(
3600 * SUM(CASE WHEN job_type = 'batch_dynamic' THEN picked_qty END),
SUM(CASE WHEN job_type = 'batch_dynamic' THEN effective_time_sec END)
) AS BD_IPH,

-- Batch Static
SUM(CASE WHEN job_type = 'batch_static' THEN picked_qty END) AS BS_PickedQty,
SUM(CASE WHEN job_type = 'batch_static' THEN containers_used END) AS
BS_UniqueContainers,
SUM(CASE WHEN job_type = 'batch_static' THEN effective_time_sec END) AS BS_TimeSec,
SAFE_DIVIDE(
3600 * SUM(CASE WHEN job_type = 'batch_static' THEN picked_qty END),
SUM(CASE WHEN job_type = 'batch_static' THEN effective_time_sec END)
) AS BS_IPH

FROM base_jobs b
LEFT JOIN noonbinimops.fulfillment.wh_code_wh_category_mapping c
ON b.warehouse_code = c.warehouse_code
GROUP BY ALL
)

```

```

-- — Final: add user tenure info —————
SELECT
a.*,
u.username,
DATE(u.created_at) AS user_created_at,
DATETIME_DIFF(date, DATE(u.created_at), DAY) AS user_ageing -- days since account
creation
FROM user_level a
LEFT JOIN noondwh.scgoms_cwpicking.user u ON a.id_user = u.id_user

```

/*

=====

TRANSFER TRACKER (SUPPLY CHAIN) — ITEM-LEVEL STATUS DASHBOARD

=====

PURPOSE : Tracks all inter-warehouse (hub transfer) items per day, broken down by source warehouse, destination darkstore, and country. Reports each item's stage in the transfer pipeline, from pending pick through putaway / cancellation.

SOURCE : noondwh.scgoms_goms (core OMS), noondwh.scgoms_cwpicking (WMS)

GRAIN : One row per (creation_date, source_warehouse, destination_darkstore)

FILTERS : Last 7 days | Marketplace id_mp = 9 (Noon Instant)

AUTHOR : —

LAST : (auto-refresh via refresh_time column)

=====

*/

/*

VARIABLE DEFINITIONS

BigQuery doesn't support SQL DECLARE in SELECT context, so logical constants are documented here as comments and embedded inline below.

DATE_WINDOW : CURRENT_DATE() - 7 (rolling 7-day lookback)

MARKETPLACE_ID : 9	(Noon Instant marketplace)
MP_STATUS_ACTIVE : 30	(item is live / not cancelled)
MP_STATUS_CANCELLED : 31	(item has been cancelled)
STATUS_PACKED : 17	(item is packed)
STATUS_SORTING : 36	(item is in sorting/staging area)
STATUS_RECEIVED : 69	(item received at destination DS)
STATUS_PENDING_PICK : 44 OR 1	(item awaiting picker assignment)
STATUS_DELETED : 32	(item record soft-deleted)
HT_STATUS_STAGED : 1	(hub transfer line = staged)
HT_STATUS_RELEASED : 57	(hub transfer line = released from staging)
HT_STATUS_DISPATCHED : 36	(hub transfer line = dispatched)
HT_HEADER_EN_ROUTE : 26	(hub transfer header = in transit)
HT_HEADER_RECEIVING : 13	(hub transfer header = at receiving dock)

CANCEL REASON CODES (scgoms_cwpicking.reason, type = 'item_cancellation'):

ops_marked_missing : JLM

oos_stock : Stock_Unavailable_OOS | cutoff_passed_with_no_stock

sla_breach : Picking_Not_Completed_Within_SLA | cutoff_passed

*/

SELECT

/* — DIMENSION: Creation date, localised per country timezone — */

CASE

WHEN src_w.id_country = 1 THEN DATE(i.created_at, 'Asia/Dubai') -- UAE

WHEN src_w.id_country IN (2, 6) THEN DATE(i.created_at, 'Asia/Riyadh') -- KSA / BH

WHEN src_w.id_country = 3 THEN DATE(i.created_at, 'Africa/Cairo') -- EGY

END AS creation_date,

/* — DIMENSION: Source warehouse (hub / DC)

*/

src_w.partner_warehouse_code AS source_warehouse,

/* — DIMENSION: Warehouse category (DC, Hub, etc.) — */

COALESCE(c.warehouse_catrgory, 'Other') AS warehouse_category,

/* — DIMENSION: ISO country code

*/

CASE

WHEN src_w.id_country = 1 THEN 'AE'

WHEN src_w.id_country = 2 THEN 'SA'

WHEN src_w.id_country = 6 THEN 'BH'

WHEN src_w.id_country = 3 THEN 'EG'

END AS country,

/* — DIMENSION: Destination darkstore

----- */
dst_w.partner_warehouse_code AS destination_darkstore,
dst_w.name AS destination_name,

/*

METRIC BLOCK 1 — VOLUME OVERVIEW

----- */

/* All distinct items created in the date window for this transfer leg */
COUNT(DISTINCT i.id_item) AS total_qty,

/* Items whose marketplace status = 31 (cancelled) */
COUNT(DISTINCT CASE
WHEN i.id_mp_status = 31 THEN i.id_item
END) AS cancelled_qty,

/*

METRIC BLOCK 2 — PIPELINE STAGE COUNTERS

Each bucket is mutually exclusive in intent, though an item may technically satisfy more than one condition if data is inconsistent.

----- */

/* — Stage: Pending Picking

Item is active (mp_status=30) and awaiting picker assignment.
id_status 44 = picker not yet assigned; 1 = newly injected item. */
COUNT(DISTINCT CASE
WHEN (i.id_status = 44 OR i.id_status = 1)
AND i.id_mp_status = 30
THEN i.id_item
END) AS pending_picking,

/* — Stage: Pending Staging

Item has been picked (status 36=sorting or 17=packed),
is active (mp_status=30), and its hub-transfer line is staged (htl=1). */
COUNT(DISTINCT CASE
WHEN i.id_status IN (36, 17)
AND i.id_mp_status = 30
AND htl.id_status = 1
THEN i.id_item
END) AS pending_staging,

/* — Stage: Pending Release from Staging —————
Item is packed (17), active (30), and HTL shows released (57).
Awaiting consolidation / load-out authorisation. */
COUNT(DISTINCT CASE
WHEN i.id_status = 17
AND i.id_mp_status = 30
AND htl.id_status = 57
THEN i.id_item
END) AS pending_release_staging,

/* — Stage: Pending Dispatch

HTL is dispatched (36) but hub-transfer header is still en-route (26).
Items are loaded on vehicle but trip not yet confirmed at destination. */
COUNT(DISTINCT CASE
WHEN i.id_mp_status = 30
AND htl.id_status = 36
AND ht.id_status = 26
THEN i.id_item
END) AS pending_dispatch,

/* — Stage: Pending Receiving

HTL dispatched (36), hub-transfer header at receiving dock (13).
Vehicle has arrived; GRN not yet confirmed. */
COUNT(DISTINCT CASE
WHEN i.id_mp_status = 30
AND htl.id_status = 36
AND ht.id_status = 13
THEN i.id_item
END) AS pending_receiving,

/* — Stage: In Transit (catch-all) —————
Not cancelled, not deleted, and putaway has NOT been recorded.
Covers items somewhere in the transfer pipeline with no end-state yet. */

```
COUNT(DISTINCT CASE
  WHEN i.id_mp_status != 31
    AND i.id_status != 32 -- exclude soft-deleted records
    AND JSON_EXTRACT_SCALAR(i.misc, '$.putaway_status') IS NULL
  THEN i.id_item
END) AS transit,
```

```
/* — Stage: Received at Darkstore
```

```
  Item is packed (17), active (30), HTL shows received at DS (69).
  GRN completed; item is physically at destination but not yet put away. */
COUNT(DISTINCT CASE
  WHEN i.id_status = 17
    AND i.id_mp_status = 30
    AND htl.id_status = 69
  THEN i.id_item
END) AS received_at_darkstore,
```

```
/*
```

METRIC BLOCK 3 — PUTAWAY OUTCOMES

```
*/
```

```
/* Putaway confirmed complete (misc.putaway_status = 'completed') */
COUNT(DISTINCT CASE
  WHEN i.id_mp_status = 30
    AND JSON_EXTRACT_SCALAR(i.misc, '$.putaway_status') = 'completed'
  THEN i.id_item
END) AS putaway_completed,
```

```
/* Item could not be located during putaway (misc.putaway_status = 'missing') */
COUNT(DISTINCT CASE
  WHEN i.id_mp_status = 30
    AND JSON_EXTRACT_SCALAR(i.misc, '$.putaway_status') = 'missing'
  THEN i.id_item
END) AS missing,
```

```
/*
```

METRIC BLOCK 4 — CANCELLATION REASON BREAKDOWN

All three counters require:

- mp_status = 31 (cancelled)
- picklist cancelled (pllst.code = 'cancelled')

Then split by cancellation reason code (rr.code).

```
*/

/* Ops-marked missing — picker confirmed item not found on shelf (reason: JLM) */
COUNT(DISTINCT CASE
  WHEN i.id_mp_status = 31
    AND pllst.code = 'cancelled'
    AND rr.code = 'JLM'
  THEN i.id_item
END) AS ops_marked_missing,

/* Out-of-stock cancellation — no sellable stock available at pick time */
COUNT(DISTINCT CASE
  WHEN i.id_mp_status = 31
    AND pllst.code = 'cancelled'
    AND rr.code IN ('Stock_Unavailable_OOS', 'cutoff_passed_with_no_stock')
  THEN i.id_item
END) AS oos_stock,

/* SLA breach cancellation — pick not completed within the promised window */
COUNT(DISTINCT CASE
  WHEN i.id_mp_status = 31
    AND pllst.code = 'cancelled'
    AND rr.code IN ('Picking_Not_Completed_Within_SLA', 'cutoff_passed')
  THEN i.id_item
END) AS sla_breach,

/* — Metadata: last refresh timestamp (per-country timezone) — */
CASE
  WHEN src_w.id_country = 1 THEN DATETIME(CURRENT_TIMESTAMP(),
'Asia/Dubai')
  WHEN src_w.id_country IN (2, 6) THEN DATETIME(CURRENT_TIMESTAMP(),
'Asia/Riyadh')
  WHEN src_w.id_country = 3 THEN DATETIME(CURRENT_TIMESTAMP(),
'Africa/Cairo')
END AS refresh_time
```

/*

FROM / JOIN BLOCK

Join order:

1. item (fact table)
 2. src_w / dst_w — warehouse master for source and destination
 3. c — warehouse-category mapping (noonbinimops, external)
 4. sl / sh — shipment line → shipment header (links item to HT)
 5. htl / htg / ht — hub-transfer line → group → header (physical trip)
 6. tr / t / pll — WMS: transfer request → line → picklist line
 7. rr / pllst — WMS: cancellation reason & picklist status
-

*/

FROM

noondwh.scgoms_goms.item i

/* Source warehouse: resolves country, timezone, and warehouse code */

LEFT JOIN noondwh.scgoms_goms.warehouse src_w

ON i.id_warehouse = src_w.id_warehouse

/* Destination darkstore: stored as JSON in item.misc → cast to INT64 */

LEFT JOIN noondwh.scgoms_goms.warehouse dst_w

ON CAST(JSON_EXTRACT_SCALAR(i.misc, '\$.id_dst_warehouse') AS INT64) =
dst_w.id_warehouse

/* Shipment line (item → shipment mapping) */

LEFT JOIN noondwh.scgoms_goms.shipment_line sl

USING (id_item)

/* Shipment header (needed to link to hub_transfer_line) */

LEFT JOIN noondwh.scgoms_goms.shipment sh

USING (id_shipment)

/* Hub transfer LINE: links the shipment to a physical transfer trip

Only 'forward_nim' type transfers (DC → darkstore direction) */

LEFT JOIN noondwh.scgoms_goms.hub_transfer_line htl

ON (htl.type, htl.line_type, htl.line_id)
= ('forward_nim', 'shipment', sh.id_shipment)

/* Hub transfer GROUP (intermediate grouping layer, joined for completeness) */

LEFT JOIN noondwh.scgoms_goms.hub_transfer_group htg

ON htg.id_hub_transfer_group = htl.id_hub_transfer_group

```

/* Hub transfer HEADER (vehicle-level trip record; carries overall trip status) */
LEFT JOIN noonwh.scgoms_goms.hub_transfer ht
  USING (id_hub_transfer)

/* Warehouse category mapping (external reference, noonbinimops project) */
LEFT JOIN noonbinimops.fulfillment.wh_code_wh_category_mapping c
  ON src_w.partner_warehouse_code = c.warehouse_code

/* — WMS (cwpicking) joins — filtered to last 7 days to limit scan cost — */

/* Transfer REQUEST header in the WMS (matches by transfer_nr on the item) */
LEFT JOIN noonwh.scgoms_cwpicking.transfer_request tr
  ON tr.transfer_nr = i.transfer_nr
  AND DATE(tr.created_at) >= CURRENT_DATE() - 7

/* Transfer REQUEST LINE (SKU-level; joined by request + SKU) */
LEFT JOIN noonwh.scgoms_cwpicking.transfer_request_line t
  ON tr.id_transfer_request = t.id_transfer_request
  AND i.catalog_sku = t.sku
  AND DATE(t.created_at) >= CURRENT_DATE() - 7

/* Picklist LINE (actual pick task; 1-to-1 with transfer request line) */
LEFT JOIN noonwh.scgoms_cwpicking.picklist_line pll
  ON t.id_transfer_request_line = pll.id_transfer_request_line
  AND DATE(pll.created_at) >= CURRENT_DATE() - 7

/* Cancellation REASON master (only item-level cancellation reasons) */
LEFT JOIN noonwh.scgoms_cwpicking.reason rr
  ON rr.id_reason = pll.id_cancellation_reason
  AND rr.type = 'item_cancellation'

/* Picklist STATUS master (e.g. 'cancelled', 'completed') */
LEFT JOIN noonwh.scgoms_cwpicking.status pllst
  ON pll.id_status = pllst.id_status

```

```

/*

```

```

  FILTER:

```

- Rolling 7-day window on item creation date (aligns with WMS join filter)
- Only Noon Instant marketplace (id_mp = 9)

```

  */

```

```

WHERE

```

```
DATE(i.created_at) >= CURRENT_DATE() - 7
AND i.id_mp = 9
```

```
/* GROUP BY ALL: collapses all non-aggregated columns automatically (BigQuery) */
GROUP BY
ALL
```

```
/*
=====
=====
```

DRIVER TASK SUMMARY (MIDDLE MILE) — TRIP-LEVEL OPERATIONAL DASHBOARD

```
=====
=====
```

PURPOSE : Tracks all outbound driver tasks over the last 15 days for UAE central warehouses. Reports trip count, pallet count, and item volume per warehouse group per day.

SOURCE : noondwh.scgoms_goms

GRAIN : One row per (warehouse_group, driver_task_created_date)

FILTERS : Last 15 days | UAE only (id_country = 1)

AUTHOR : —

LAST : (auto-refresh via CURRENT_DATE)

```
=====
=====
```

```
*/
```

```
/*
```

VARIABLE DEFINITIONS

BigQuery doesn't support SQL DECLARE in SELECT context, so logical constants are documented here and embedded inline below.

DATE_WINDOW : CURRENT_DATE() - 15 (rolling 15-day lookback)

UAE_COUNTRY_ID : 1 (id_country for UAE)
ENTITY_TYPE_PALLET : 'pallet' (driver_task_line.entity_type filter)
HUB_TRANSFER_LINE_TYPE : 'shipment' (hub_transfer_line.line_type filter)

AUDIT ACTION IDs (scgoms_goms — do NOT reuse across datasets):

AUDIT_TASK_COMPLETED : 18 (driver_task → completed)
AUDIT_ROUTE_ARRIVED : 16 (driver_task_route → arrived)
AUDIT_TRANSFER_CLOSED : 11 (hub_transfer → closed)

WAREHOUSE GROUP MAPPING (source_warehouse → business label):

DXBID01, DXBID02 → TP_ambient
DXBID03, DXBID04 → TP_chiller
DXB → TP_fnv
FKZ → Frozen
AUHID01, AUHID02 → kezaad_ambient
AUHID03, AUHID04 → kezaad_chiller
AUH → Kezaad

STATUS CODE MAPPING (driver_task_line.id_status → business label):

pending → In Transit
handed_over → Completed
skipped → Returned to WH

NOTE: Source warehouse is resolved from shipment.id_warehouse,
NOT from hub_transfer polymorphic IDs (source_id / destination_id).

_____ */

-- =====

-- CTE 1: base

-- Grain: one row per task × route × pallet × shipment × item

-- Resolves warehouse, audit timestamps, shipment & item counts

-- =====

WITH base AS (

SELECT
dt.id_driver_task,
dt.task_nr,
u.email,
u.username,
ht.canter_code AS pallet_code,
CASE
WHEN s.code = 'pending' THEN 'In Transit'
WHEN s.code = 'handed_over' THEN 'Completed'


```

    WHEN s.code = 'skipped' THEN 'Returned to WH'
    ELSE s.code
END
    AS line_status,
r.code
    AS hold_reason,
dt.created_at
    AS task_created_at,
MAX(al.created_at)
    AS task_completed_at,
dst_w.partner_warehouse_code
    AS destination_dark_store,
src_w.partner_warehouse_code
    AS source_warehouse,
IF(dtl.id_hold_reason IS NOT NULL, alu.username, NULL) AS
person_who_terminated_the_task,
MIN(al1.created_at)
    AS mark_arrived_timestamp, -- first 'arrived' on route
MAX(al2.created_at)
    AS received_at_darkstore, -- last hub_transfer 'closed'
COUNT(DISTINCT sh.id_shipment)
    AS qty_shipments,
COUNT(DISTINCT it.id_item)
    AS qty_items

FROM `noondwh.scgoms_goms.driver_task_line` dtl
LEFT JOIN `noondwh.scgoms_goms.driver_task_route` dtr
    USING (id_driver_task_route)
LEFT JOIN `noondwh.scgoms_goms.driver_task` dt
    USING (id_driver_task)
LEFT JOIN `noondwh.scgoms_goms.warehouse` dst_w
    ON dst_w.id_warehouse = dtr.id_destination
LEFT JOIN `noondwh.scgoms_goms.status` s
    ON s.id_status = dtl.id_status
LEFT JOIN `noondwh.scgoms_goms.reason` r
    ON r.id_reason = dtl.id_hold_reason
-- entity_type/entity_id is polymorphic; scoped to 'pallet' rows only
LEFT JOIN `noondwh.scgoms_goms.hub_transfer` ht
    ON (dtl.entity_type, dtl.entity_id) = ('pallet', ht.id_hub_transfer)
LEFT JOIN `noondwh.scgoms_goms.hub_transfer_line` htl
    USING (id_hub_transfer)
LEFT JOIN `noondwh.scgoms_goms.hub_transfer_group` htg
    USING (id_hub_transfer_group)
LEFT JOIN `noondwh.scgoms_goms.shipment` sh
    ON (htl.line_id, htl.line_type) = (sh.id_shipment, 'shipment')
LEFT JOIN `noondwh.scgoms_goms.shipment_line` shl
    ON shl.id_shipment = sh.id_shipment
-- Source warehouse from shipment — hub_transfer IDs are polymorphic
LEFT JOIN `noondwh.scgoms_goms.warehouse` src_w
    ON src_w.id_warehouse = CAST(sh.id_warehouse AS INT64)
LEFT JOIN `noondwh.scgoms_goms.item` it
    USING (id_item)
LEFT JOIN `noondwh.scgoms_goms.user` u
    USING (id_user)

```

```

-- Audit: task completed timestamp (action 18)
LEFT JOIN `noondwh.scgoms_goms.audit_log` al
  ON al.action_key    = dt.id_driver_task
  AND al.id_audit_action = 18
  AND JSON_EXTRACT_SCALAR(al.misc, '$.to') = 'completed'
LEFT JOIN `noondwh.scgoms_goms.user` alu
  ON alu.id_user = al.id_user
-- Audit: route arrived timestamp (action 16)
LEFT JOIN `noondwh.scgoms_goms.audit_log` al1
  ON al1.action_key    = dtr.id_driver_task_route
  AND al1.id_audit_action = 16
  AND JSON_EXTRACT_SCALAR(al1.misc, '$.to') = 'arrived'
-- Audit: hub_transfer closed = received at darkstore (action 11)
LEFT JOIN `noondwh.scgoms_goms.audit_log` al2
  ON al2.action_key    = ht.id_hub_transfer
  AND al2.id_audit_action = 11
  AND JSON_EXTRACT_SCALAR(al2.misc, '$.to') = 'closed'

WHERE DATE(dtl.created_at, 'Asia/Dubai') >= DATE_SUB(CURRENT_DATE('Asia/Dubai'),
INTERVAL 15 DAY)
  AND src_w.id_country = 1 -- UAE only

GROUP BY ALL
),

-- =====
-- CTE 2: final_df
-- Grain: one row per task × destination darkstore
-- Maps source warehouses to business group labels
-- =====
final_df AS (
  SELECT
    CASE
      WHEN source_warehouse IN ('DXBID01', 'DXBID02') THEN 'TP_ambient'
      WHEN source_warehouse IN ('DXBID03', 'DXBID04') THEN 'TP_chiller'
      WHEN source_warehouse LIKE '%DXB%'           THEN 'TP_fnv'
      WHEN source_warehouse LIKE '%FKZ%'           THEN 'Frozen'
      WHEN source_warehouse IN ('AUHID01', 'AUHID02') THEN 'kezaad_ambient'
      WHEN source_warehouse IN ('AUHID03', 'AUHID04') THEN 'kezaad_chiller'
      WHEN source_warehouse LIKE '%AUH%'           THEN 'Kezaad'
    END
    AS warehouse,
    id_driver_task,
    task_nr,
    DATE(task_created_at, 'Asia/Dubai') AS driver_task_created_time,

```

```

    DATETIME(mark_arrived_timestamp, 'Asia/Dubai') AS mark_arrived_timestamp,
    destination_dark_store AS dst_warehouse,
    COUNT(DISTINCT pallet_code) AS qty_pallets,
    SUM(qty_shipments) AS qty_shipments,
    SUM(qty_items) AS qty_items

FROM base
GROUP BY ALL
)

```

```

-- =====
-- FINAL OUTPUT
-- Grain: one row per (warehouse_group, driver_task_created_date)
-- =====
SELECT
    warehouse,
    driver_task_created_time,
    COUNT(DISTINCT task_nr) AS trips,
    SUM(qty_pallets) AS pallets,
    SUM(qty_items) AS items
FROM final_df
WHERE warehouse IS NOT NULL
    AND warehouse <> "
GROUP BY ALL;

```

```

/*

```

```

=====
=====

```

ULTRAFRESH PIPELINE STATUS

```

=====
=====

```

PURPOSE : Tracks all outbound transfer items for the last ~24 hours across
 Ultrafresh warehouses (DXBID07, AUHID07), broken down by slot,
 type, source warehouse, and destination darkstore.
 Reports item counts at each pipeline stage from pending pick
 through putaway / cancellation.

SOURCE : noondwh.scgoms_goms
GRAIN : One row per (created_at, slot, type, source_warehouse,
destination_darkstore)
FILTERS : Rolling ~24h window | Marketplace id_mp = 9 (Noon Instant)
| UAE only (id_country = 1) | Ultrafresh warehouses only

AUTHOR : —
LAST : (auto-refresh via CURRENT_TIMESTAMP)

=====

=====

*/

/*

VARIABLE DEFINITIONS

BigQuery doesn't support SQL DECLARE in SELECT context, so logical constants are documented here and embedded inline below.

DATE_WINDOW_NON_EGG : CURRENT_TIMESTAMP() - 1 DAY + 13 HOUR (non-egg
lookback start)

DATE_WINDOW_EGG : CURRENT_TIMESTAMP() - 1 DAY + 5 HOUR (egg lookback
start — earlier window)

MARKETPLACE_ID : 9 (Noon Instant)

MP_STATUS_ACTIVE : 30 (item is live / not cancelled)

MP_STATUS_CANCELLED : 31 (item has been cancelled)

STATUS_PACKED : 17 (item is packed)

STATUS_SORTING : 36 (item is in sorting/staging area)

STATUS_PENDING_PICK : 44 (item awaiting picker assignment)

STATUS_DELETED : 32 (item record soft-deleted)

HT_STATUS_STAGED : 1 (hub_transfer_line = staged)

HT_STATUS_RELEASED : 57 (hub_transfer_line = released from

staging)

HT_STATUS_DISPATCHED : 36 (hub_transfer_line = dispatched)

HT_STATUS_RECEIVED : 69 (hub_transfer_line = received at

darkstore)

HT_HEADER_EN_ROUTE : 26 (hub_transfer header = in transit)

HT_HEADER_RECEIVING : 13 (hub_transfer header = at receiving

dock)

UAE_COUNTRY_ID : 1 (id_country for UAE)

SLOT LOGIC (non-egg, DXB/AUH/SHJ warehouses):

Slot_1 : item.created_at in [T-1 13:00, T-1 21:00)

Slot_2 : item.created_at in [T-1 21:00, T 13:00)
Non DXB/AUH/SHJ destinations always → Slot_2

WAREHOUSE CATEGORIES:

AUHFZ01/02/03 → Frozen_TP
DXBID01/02 → Ambient_TP
DXBID03/04 → Chiller_TP
DXBID07 → Ultrafresh_TP
DXBFNV01 → FnV
AUHID01/02 → Ambient_AUH
AUHID03/04 → Chiller_AUH
AUHID07 → Ultrafresh_AUH

TYPE LOGIC:

crossdoc job + no transfer_nr → CROSS_DOC
all others → PICKING

NOTE: egg SKUs use a wider time window (5h vs 13h offset) to capture earlier morning slots. The egg_base CTE currently mirrors non_egg_base output — extend egg-specific filters here if needed.

____ */

```
-- =====  
-- CTE 1: item_base_non_egg  
-- Raw item rows for non-egg SKUs — last ~24h (13h offset)  
-- Excludes a hardcoded list of egg/excluded catalog_skus  
-- =====
```

```
WITH item_base_non_egg AS (  
  SELECT  
    created_at,  
    id_mp_status,  
    id_status,  
    misc,  
    id_warehouse,  
    id_item,  
    id_mp,  
    transfer_nr  
  FROM `noondwh.scgoms_goms.item`  
  WHERE created_at >= (CURRENT_TIMESTAMP() - INTERVAL 1 DAY + INTERVAL 13  
    HOUR)  
    AND catalog_sku NOT IN (
```

'ZDBD96B2A31B90DA3535BZ-1','ZF70A41840211D44119A1Z-1','Z68E570CD7B584F2CEFFDZ-1',

'Z57C9E4E200DA0C9957ACZ-1','ZFE1F92583C9094B4E78DZ-1','ZFE40744FFD25831970A5Z-1',

'Z1902BB576BE48EE08D00Z-1','Z2BB15284D64530C1C05BZ-1','ZC9A51E5FA6608F752F70Z-1',

'Z4523F9F7005238D843D5Z-1','Z3789EBEB6595CE35DCCEZ-1','Z237C5055361C50A7A71FZ-1',

'Z6698D2D2BDBA41D58CB3Z-1','Z6FDBF76C2C322B067228Z-1','ZF26DC7CC594C8894A817Z-1',

'ZB35F71329465C90AEFFAZ-1','Z4BDD0FC2EEF96B2C5187Z-1','Z47FC5A39CB05156F8136Z-1',

'ZE65EEC6C1D8053F9E0D2Z-1','ZEEE3AB1C9858EB118FD2Z-1','Z397668FD6BFB5340E488Z-1',

'Z570B4BC1E994BBCAF2ABZ-1','Z6399810E6C33F913DA30Z-1','ZC695BAB888341A6915CEZ-1',

'ZF8BA5447E2C0C0B9574DZ-1','Z12C1214A9D7BB114F1C8Z-1','ZC14E5682390F8D7CC77AZ-1',

'Z59E57306C5391D31E795Z-1','ZD7C9862E3EBF56D7FD61Z-1','Z8679040FE648972CF10DZ-1',

'Z8979B9749AECF808C551Z-1','ZD30C97A6B25E02B50C1CZ-1','ZC04BA8EB8027D85D8CC4Z-1',

'ZD871DE485C3BF2CA33FDZ-1','Z756CF10F309513A05413Z-1','ZD53F0499F4CE32BB4E2CZ-1',

'Z3CCB27EA61F1AC63D820Z-1','Z814D23C0A6E15FD13E80Z-1','Z4D24B5134F244B218C2BZ-1',

'Z8523783B8F4D6BDF8DD2Z-1','Z55E99103039232BC1CACZ-1','ZEBBAF3BC29FDDB4E6501Z-1',

'Z3AF090856C0AA17C2D37Z-1','ZD66D8E379074BBC7099DZ-1','ZAD994290A1EB7059436CZ-1',

'Z24B7645810B458876CC9Z-1','Z5FEFC8C8E20639E6085CZ-1','ZFE681C5586B02F65C655Z-1',

'ZC6FAD4EDE55A6FF5BF82Z-1','Z438DC9F97D4FFEBF1BFFZ-1','Z2908E1774A24BFC597F8Z-1',

'ZBF92153759AE09FFE5FDZ-1','Z6F7937F3835EFBCAF5BEZ-1','ZD2198350DCBB144AEE1CZ-1',

'Z761ED2FC15A72F9D2285Z-1','ZD40F5EBAAD1869822DE8Z-1','Z7A6056149E2650D5A499Z-1',

'ZA9E905C93F2CD7735DD5Z-1','Z2752C87A1FB276102FA2Z-1','Z0243C9A53B31BA60665EZ-1',

'Z8D1C3044DA02EE1B1504Z-1','ZE3FC97EAF83207323499Z-1','Z80B4A702166E486F7244Z-1',

'Z75C6462D07D5B46E0DD2Z-1','Z7339D47BC259B160C535Z-1','Z51D8B79D3639A4F2A954Z-1',

'ZB434379F40CCA7D6C128Z-1','ZC110A4A8D5344F0982FAZ-1','Z20E34F716D800FD1FDE5Z-1',

'Z583AAE1E3EC9099B4B8EZ-1','Z5D0883A3BD2E4C658F12Z-1','Z782823BDF3B01C2DBEACZ-1',

'ZCC844D9C96729D6F905CZ-1','Z7B50EEDD4EFF9DB9E14CZ-1','Z69EC54C9B6830134C1DCZ-1',

'Z394AABB73EF98DA53702Z-1','ZEB1D66B02FD45C7F5235Z-1','ZBC0FA136F91D649B9FD6Z-1',

'Z9C159F269EA9F8384DC6Z-1','ZD9F490E66DB1F80AFCC1Z-1','Z6322C97105AD82323AA3Z-1',

'ZFBD594063B7289164CF5Z-1','Z961C8C56BE447410EB72Z-1','ZADD27C4506C3814DBC3EZ-1',

'Z26C1063E13C954FB1CDEZ-1','Z3CAC17258A9C7D4887C2Z-1','ZD8391FC74A2FC5EF033EZ-1',

'Z83FE44DC84B7E21BE121Z-1','ZFF3DCF1E224ECE397268Z-1','ZA86B12E6CBCDF73511D5Z-1',

'ZBB7797DD410A05E7F449Z-1','ZBBB3E724F92070B22A5AZ-1','ZFD0FA25041E79BCAE3E1Z-1',

'ZEF728A0BE09E273B418DZ-1','ZE531AF6D2A1ADBCBF2A7Z-1','ZE16B3BBA9BE4AAA16223Z-1',

'ZE94FDDDD8E6BF88E7E25EZ-1','Z02932FC383C4D64C0531Z-1','Z3CBD70E2D219737FC600Z-1',

'Z0D54A1885494FD2FA756Z-1','ZAAED4556AB260734B90DZ-1','Z29D9CCA7D6121F24DF2DZ-1',

'ZAB807779B07579349A58Z-1','Z1F68261FC47CAC3C4988Z-1','Z88D4A83B064B9D777B8CZ-1',

'ZBA02034FAE84A5BCF4EDZ-1','ZF2A257FD0D4939D096DFZ-1','Z2ADE110495895FDC8952Z-1',

'Z6CC0F17FC7A014E14873Z-1','Z827EE80F7E6737FF1FB6Z-1','ZD3D939B457D73445F431Z-1',

'Z2F5326AE08471E123834Z-1','Z2998C777C770F2FFADD9Z-1','Z0C8912E4EDBDE2DEF2E4Z-1',

'Z906F2D48887E03E1EC6CZ-1','Z122EDA23764DF26A9882Z-1','ZA1C9E5092E061A0F714DZ-1',

'Z31B9D833F20568BBB8BBZ-1','Z2E15E0C2E491F6186756Z-1','Z50750AAEBCCBFCDF6A0CZ-1',

'ZCC79DB1122F11ACB520BZ-1','Z90B82F94E100FDF43BF1Z-1','ZA96C26651697804B8DA6Z-1',

'ZBEA211F51E5FC5D76FB5Z-1','Z9795A566BC1BF4DAE756Z-1','Z42C838C5664BD41B2599Z-1',

'ZD14379EF865941E963BFZ-1','ZFF16BC15B87424A622D0Z-1','ZFBD3786077B8CFE018C3Z-1',

'Z28DBC8B68211854F9F5DZ-1','Z848C3A0663A1FE22FC77Z-1','Z1F79D9EA2C90B4DCB984Z-1',

'Z49D1CC688D229525B3B8Z-1','Z836DCE0BC1AA21E7EC5CZ-1','Z548C9370373ED56555B2Z-1',

'ZBDFDD6B799D7487AE79FZ-1','ZB3A85AEBB693F271A525Z-1','Z599FF318F96E61F2A2A4Z-1',

'ZC2FF5C33D7B3725ACF6EZ-1','ZEC02FC92EB09067209E0Z-1','ZABCA9BC2DBDDAB3E89DCZ-1',

'ZC4D68917D25557E52ECCZ-1','Z17514F8D437CA0B9D870Z-1','ZB9EA17F3A70F41208CFEZ-1',

'ZD0D6360602ED8C4BF318Z-1','Z6F41B3BB4C3E33113BECZ-1','ZAEB476E1841334F2DD27Z-1',

'ZCFE4A08A5ED7D286F2CCZ-1','ZD6E64CF9425F4EE8403DZ-1','ZA2128D6F7E375C80BEDEZ-1',

'ZE542921BEDC3CEA7470EZ-1','Z5119FD7E7DECE6670AC8Z-1','Z491CAB3905C67E8EF4E9Z-1',

'Z020B7055C55464C03933Z-1','ZA482BA3EC0E9406ABCCDZ-1','Z8CC831ACDCB82BDE8749Z-1',

'ZEB1C9A42148B8D6284B4Z-1','Z9619AAFC5B57740355D1Z-1','Z9BAA09751DA3FC054F36Z-1',

'Z05AE3CA274AAA735585EZ-1','ZA4E0ACB7A951F7809C3CZ-1','Z92F971F0103C15513D9AZ-1',

'ZCB8179FF0B5858AE5349Z-1','ZBE3CE67F2194E5C15CDCZ-1','ZEFB5204E09F0FA3826A0Z-1',

'ZCE3239884FBA49EABE55Z-1','Z79515F74AEE2E05F4BA0Z-1','ZDA2B54DE409ECCD90501Z-1',

'Z82AB8FB6F81C2C0B947CZ-1','ZAF18384B777E5B559449Z-1','Z9BC60FBCBC579C9196B2Z-1',

'Z63B9364C8C0432162830Z-1','Z38AB4AFF569D5D3F92E8Z-1','ZDD0752A9E6A0CFBA7CB0Z-1',

'ZEDD6855A2A31EDE9A905Z-1','Z526A4BD7449FF65AAD69Z-1','ZB8B5DDAFE020A482EB18Z-1',

'Z81430141E96169903B57Z-1','Z0FFC2AC57BD5D3229B59Z-1','Z9FE2DF895B2D9EC58616Z-1',

'ZDB7442E2720FEDA7467FZ-1','Z85D15006D7CFEFB06646Z-1','Z8C9257D07648347AA3C9Z-1',

'ZD7A567D470D2C3018214Z-1','Z9674EAB005A7DB637F32Z-1','ZBC31CB49AEEEC1815F9Z-1',

'Z5C0714DAF6F5E0B294E9Z-1','ZB29D84A7371394199C0EZ-1','ZDACEC85D401F5CB0CD9AZ-1',

'Z17EED84357FB5111CBDBZ-1','Z6DE62DADCAD08F3F56C1Z-1','ZEC6629475BB9AB9EB4C7Z-1',

'Z247B07CE27213A4AA0A5Z-1','Z6EFE41E6BF3532783A85Z-1','Z6297EC0B698A25BB17F3Z-1',

'Z75730E0F0E5BFCF2FFECZ-1','ZBC979EABF4C8C011861AZ-1','Z58E9D67E1BDE14D02E01Z-1',

'Z5E6C719B9712863F87D4Z-1','ZAE5F7BD2E62D05C5CA1EZ-1','Z9D615F88ED8443E54E1AZ-1',

'ZE37E628CC17FDCF29122Z-1','Z9B2AEF365383186BC0B3Z-1','ZB3B4E1EB032CDFC1047CZ-1',

'ZC257374C53F1D7613031Z-1','Z8137477DAD2F18B99695Z-1','Z29FDC9FE318558CE94F4Z-1',

'ZB6A4F7CBDB43FF595714Z-1','Z524F68E8C1381739DFD8Z-1','Z5CDC39896AB38025C31EZ-1',

'ZABF30800324F4D29FDDCZ-1','Z990ED19303C5F828F6CFZ-1','Z9F2FABF0F2B3F9465182Z-1',

'Z3C85173387FA67ABBD FBZ-1','Z86457C794211AF4355A6Z-1','ZAE7F612710ED97DB4754Z-1',

'Z6D56CD0FE45876A4D5CEZ-1','Z0156CB1269D808EF4C23Z-1','ZAD4339C2C6C5B393658AZ-1',

'Z223D420C2D5CC499620BZ-1','ZAC26B8BCD3BE5968328FZ-1','Z482D1AB5EBA17EC9D7D9Z-1',

'ZF76498DF4E628EB6A3AAZ-1','Z2159D049A442310D80C8Z-1','ZE34620A7C51ABE6A0690Z-1',

'Z4A9AAA7951914712FC76Z-1','ZB93F34C79901B9F40F68Z-1','ZFD88C8CACBA0EBE8A6BBZ-1',

'ZAC3A656FF13B449BA002Z-1','Z78F8109C8C959DE4AB16Z-1','Z1C941FB22F5106B99295Z-1',

'Z350CF77EE5BCA729FD39Z-1','Z81E1FECC3001406D6488Z-1','Z43DA18A0D8C641DE8882Z-1',

'Z420BD54F147E553E14E1Z-1','Z3C0BB415FB587A848B91Z-1','Z9CE653B504BB78C8C04EZ-1',

'ZAC636EBB9D49E8669592Z-1','ZB4525A5A9C09D42519F4Z-1','Z4876422384CFF6BE146BZ-1',

'ZA2DD46C0103195FA4374Z-1','Z3B6C0C6541F54B93F966Z-1','Z8B997BB8943EB388B583Z-1',

'ZC6093570EB319E70C953Z-1','Z7061EBD116B90208F677Z-1','Z54CFE43B802BCD2AA4CBZ-1',

'ZA6B01376E813A457C18CZ-1','ZC15FB81271EC0B657483Z-1','ZC3AED3DD2490BF0EF60FZ-1',

'Z78626810767DCA2C51E0Z-1','ZADB8B6CE5CC2EE87BE23Z-1','Z09820ABE31797AC9F249Z-1',

'ZF437B46858154307F124Z-1','Z7E2B400D194CF82085A4Z-1','ZEA8F234CFC1EFFEC7B84Z-1',

'Z59EC067F6AD3BB3F1D86Z-1','Z50D025E7F5BC39C447E6Z-1','ZCA224ACD45CA6A2CCA94Z-1',

'Z313A55BAB6F259ECB775Z-1','Z341C5E2F2AF526800481Z-1','ZAFDFB3D6E9AC593D4D8DZ-1',

'ZEBBC44A1147CFC58E408Z-1','ZEDFD918C90BD58DAF525Z-1','ZE53FCEB22E3ACA2919AAZ-1',

'ZE39E40CE2D3451046FFCZ-1','Z48C58176B4BF5DB958B3Z-1','ZC5BA05FD1E3466857127Z-1',

'Z89F7EA3CA30E4C9AD23FZ-1','Z6470E36CE799E0D6BA98Z-1','Z3C4787E69EC873F73180Z-1',

'ZAA5CBF002E3EF80B1519Z-1','Z6879D1FE2EC4FFE56B4BZ-1','Z1F0BE45931A398399A3EZ-1',

'Z502D5863B452BBC4D17BZ-1','Z0353C8CED0C1D3B833EFZ-1','Z6345F8048DC7F3379F58Z-1',

'Z6174222AA908DBBEB684Z-1','Z7A2997D200EDFCE198B6Z-1','Z3DEE26B58B6D0FC1739FZ-1',

'ZB187C98CAC2E32C4E4EFZ-1','Z54D95600E1C086FBA6CAZ-1','Z9778E2D5EE755E16E2FBZ-1',

'Z324602388C44300407DFZ-1','Z932209052FDD71474FB9Z-1','Z77096D0127B38B666908Z-1',

'Z727E8A00D8CCA115FD20Z-1','Z0D759C09A44311ACF9A2Z-1','ZD1E2D3684EE7A1CF0B88Z-1',

'Z5D83F8790297E1917A62Z-1','Z7C5FDF73BBCB418875AFZ-1','Z2D965F328D6B463E7438Z-1',

'Z4D4FD79761874F5F8024Z-1','Z142D1F142A1AA7749B42Z-1','Z3B9943380C1B1EB89292Z-1',

'Z617516EF91E73F58A2BAZ-1','ZF7450164D4DB1B99C709Z-1','Z99687DEB4AFB175D3CB1Z-1',

'Z735EF8E1E31BD70D487FZ-1','ZEC4A4F284FA18BF4A894Z-1','Z7202F3E516C5FB132F8AZ-1',

'Z605FA2228DA84F304E63Z-1','ZFB42C683F513E64C2869Z-1','Z912E9F43664FC5B45983Z-1',

'ZAC35AB997C0B6E6B72DEZ-1','ZFEB80F096CA7DB11FA4FZ-1','Z5E4DEC7D4461BCAB77DDZ-1',

'ZD1AB44C857E688F984EFZ-1','ZBE15844CC4EE1858EE3FZ-1','Z5172747D966943FD5398Z-1',

'Z1ECDE11BDFC6E4356141Z-1','ZB2517DD5D9649EB4ABC3Z-1','Z75F8BA619A2D3AF5B583Z-1',

'Z8580289CB54678E80A82Z-1','Z07472F0161DE1F93DFE3Z-1','Z209EADFE14BCAF8A2587Z-1',

'ZC8026006F52A0F5097FEZ-1','ZCBB5566B7DB08CFDEF80Z-1','Z74918247D20179C00FB3Z-1',

'Z02E03D06366B51C46438Z-1','Z414D7D18049502BCB398Z-1','Z20520042A30D1ED94909Z-1',

'ZF42613F9D0171331B961Z-1','Z4B23C2FC367794ACF722Z-1','Z00355FACD4C48EC3825AZ-1',

'ZF27720D2D8F3AE188B08Z-1','ZBFF064C122B75F100FE6Z-1','Z4B0F9B95B22CEFED0443Z-1',

'ZA92126E3B45840DECFA1Z-1','Z3AC6AFABCF959AC91E7EZ-1','Z9E183B3425CEAB7B806Z-1',

'Z6BC14C9AA59C79E4C09EZ-1','Z3EDEF2412AD15CC9D0C5Z-1','ZE647CFBE0A34EEEBF93CZ-1',

'ZD1A3D5542C31118956B8Z-1','Z3124F3F26C1D31C22757Z-1','Z23539CD57210B9E53941Z-1'

)

GROUP BY 1,2,3,4,5,6,7,8

),

```
-- =====
-- CTE 2: non_egg_base
-- Aggregates non-egg items into pipeline stage counts
-- Scoped to Ultrafresh warehouses: DXBID07, AUHID07
-- =====
non_egg_base AS (
  SELECT
    -- Bucket into today or tomorrow based on cutoff (13h offset = midnight Dubai)
    CASE
      WHEN i.created_at < (CURRENT_TIMESTAMP() + INTERVAL 13 HOUR)
      THEN DATE(CURRENT_TIMESTAMP() + INTERVAL 13 HOUR)
      ELSE DATE(CURRENT_TIMESTAMP() + INTERVAL 13 HOUR + INTERVAL 1 DAY)
    END AS created_at,

    -- Slot logic: non DXB/AUH/SHJ destinations always Slot_2
    CASE
      WHEN (dst_w.partner_warehouse_code NOT LIKE 'DXB%'
        AND dst_w.partner_warehouse_code NOT LIKE 'AUH%'
        AND dst_w.partner_warehouse_code NOT LIKE 'SHJ%') THEN 'Slot_2'
      WHEN i.created_at >= (CURRENT_TIMESTAMP() - INTERVAL 1 DAY + INTERVAL 13
        HOUR)
        AND i.created_at < (CURRENT_TIMESTAMP() - INTERVAL 1 DAY + INTERVAL 21
        HOUR)
      THEN 'Slot_1'
      WHEN i.created_at >= (CURRENT_TIMESTAMP() - INTERVAL 1 DAY + INTERVAL 21
        HOUR)
        AND i.created_at < (CURRENT_TIMESTAMP() + INTERVAL 13 HOUR)
      THEN 'Slot_2'
      ELSE 'Slot_1'
    END AS slot,

    -- crossdoc job with no transfer_nr = CROSS_DOC; everything else = PICKING
    CASE
      WHEN sj.job_type = 'crossdoc' AND i.transfer_nr IS NULL THEN 'CROSS_DOC'
      ELSE 'PICKING'
    END AS type,

    src_w.partner_warehouse_code AS source_warehouse,
    dst_w.partner_warehouse_code AS destination_darkstore,
    dst_w.name AS destination_name,
```

```

CASE
  WHEN src_w.partner_warehouse_code IN ('AUHFKZ01','AUHFKZ02','AUHFKZ03') THEN
'Frozen_TP'
  WHEN src_w.partner_warehouse_code IN ('DXBID01','DXBID02')          THEN
'Ambient_TP'
  WHEN src_w.partner_warehouse_code IN ('DXBID03','DXBID04')          THEN
'Chiller_TP'
  WHEN src_w.partner_warehouse_code IN ('DXBID07')                    THEN 'Ultrafresh_TP'
  WHEN src_w.partner_warehouse_code IN ('DXBFNV01')                    THEN 'FnV'
  WHEN src_w.partner_warehouse_code IN ('AUHID01','AUHID02')          THEN
'Ambient_AUH'
  WHEN src_w.partner_warehouse_code IN ('AUHID03','AUHID04')          THEN
'Chiller_AUH'
  WHEN src_w.partner_warehouse_code IN ('AUHID07')                    THEN
'Ultrafresh_AUH'
END AS wh_category,

```

```

COUNT(DISTINCT i.id_item)                AS total_qty,
COUNT(DISTINCT IF(i.id_mp_status = 31, i.id_item, NULL))    AS cancelled_qty,

```

```

-- Awaiting picker assignment (status 44, active mp_status 30)

```

```

COUNT(DISTINCT CASE
  WHEN i.id_status = 44 AND i.id_mp_status = 30 THEN i.id_item
END)
AS pending_picking,

```

```

-- Packed/sorting but not yet in a hub_transfer_line, or htl staged (1)

```

```

COUNT(DISTINCT IF(
  i.id_status IN (36, 17)
  AND i.id_mp_status = 30
  AND (htl.id_status = 1 OR htl.id_hub_transfer_line IS NULL),
  i.id_item, NULL))
AS pending_staging,

```

```

-- Packed, htl released from staging (57)

```

```

COUNT(DISTINCT IF(
  i.id_status = 17 AND i.id_mp_status = 30 AND htl.id_status = 57,
  i.id_item, NULL))
AS pending_release_staging,

```

```

-- htl dispatched (36) + hub_transfer en route (26)

```

```

COUNT(DISTINCT IF(
  i.id_mp_status = 30 AND htl.id_status = 36 AND ht.id_status = 26,
  i.id_item, NULL))
AS pending_dispatch,

```

```

-- htl dispatched (36) + hub_transfer at receiving dock (13)

```

```

COUNT(DISTINCT IF(

```

```

i.id_mp_status = 30 AND htl.id_status = 36 AND ht.id_status = 13,
i.id_item, NULL))                AS pending_receiving,

-- Active, not deleted, putaway not started
COUNT(DISTINCT IF(
  i.id_mp_status != 31
  AND i.id_status != 32
  AND JSON_EXTRACT_SCALAR(i.misc, '$.putaway_status') IS NULL,
  i.id_item, NULL))                AS transit,

-- Packed, htl received at darkstore (69)
COUNT(DISTINCT IF(
  i.id_status = 17 AND i.id_mp_status = 30 AND htl.id_status = 69,
  i.id_item, NULL))                AS received_at_darkstore,

COUNT(DISTINCT IF(
  i.id_mp_status = 30
  AND JSON_EXTRACT_SCALAR(i.misc, '$.putaway_status') = 'completed',
  i.id_item, NULL))                AS putaway_completed,

COUNT(DISTINCT IF(
  i.id_mp_status = 30
  AND JSON_EXTRACT_SCALAR(i.misc, '$.putaway_status') = 'missing',
  i.id_item, NULL))                AS missing,

-- Cancelled items that were recreated via NSV3
COUNT(DISTINCT IF(
  i.id_mp_status = 31
  AND JSON_EXTRACT_SCALAR(i.misc, '$.unfulfillable_reason') LIKE '%recreate_nsv3%',
  i.id_item, NULL))                AS recreate_nsv3

FROM item_base_non_egg i
LEFT JOIN `noondwh.scgoms_goms.warehouse` src_w
  ON i.id_warehouse = src_w.id_warehouse
LEFT JOIN `noondwh.scgoms_goms.warehouse` dst_w
  ON JSON_EXTRACT_SCALAR(i.misc, '$.id_dst_warehouse') = CAST(dst_w.id_warehouse
AS STRING)
LEFT JOIN `noondwh.scgoms_goms.shipment_line` sl USING (id_item)
LEFT JOIN `noondwh.scgoms_goms.shipment` sh USING (id_shipment)
-- hub_transfer_line: scoped to forward_nim shipment lines only
LEFT JOIN `noondwh.scgoms_goms.hub_transfer_line` htl
  ON (htl.type, htl.line_type, htl.line_id) = ('forward_nim', 'shipment', sh.id_shipment)
LEFT JOIN `noondwh.scgoms_goms.hub_transfer_group` htg
  ON htg.id_hub_transfer_group = htl.id_hub_transfer_group

```



```

LEFT JOIN `noondwh.scgoms_goms.hub_transfer` ht USING (id_hub_transfer)
LEFT JOIN `noondwh.scgoms_goms.input_container_line` icl ON icl.id_item = i.id_item
LEFT JOIN `noondwh.scgoms_goms.input_container` ic ON icl.id_input_container =
ic.id_input_container
LEFT JOIN `noondwh.scgoms_goms.sortation_job` sj ON ic.id_sortation_job =
sj.id_sortation_job

```

```

WHERE i.id_mp = 9 -- Noon Instant only
AND src_w.id_country = 1 -- UAE only
AND src_w.partner_warehouse_code IN ('AUHID07', 'DXBID07') -- Ultrafresh only

```

```

GROUP BY 1,2,3,4,5,6,7
),

```

```

-- =====
-- CTE 3: item_base_egg
-- Raw item rows for egg SKUs — wider window (5h offset)
-- No catalog_sku exclusion applied
-- =====

```

```

item_base_egg AS (
  SELECT
    created_at,
    id_mp_status,
    id_status,
    misc,
    id_warehouse,
    id_item,
    id_mp,
    transfer_nr
  FROM `noondwh.scgoms_goms.item`
  WHERE created_at >= (CURRENT_TIMESTAMP() - INTERVAL 1 DAY + INTERVAL 5 HOUR)
  GROUP BY 1,2,3,4,5,6,7,8
),

```

```

-- =====
-- CTE 4: egg_base
-- Placeholder — currently mirrors non_egg_base output
-- Extend here when egg-specific pipeline logic is needed
-- =====

```

```

egg_base AS (
  SELECT * FROM non_egg_base
)

```

```
-- =====
-- FINAL OUTPUT
-- Union egg and non-egg pipelines into a single result set
-- =====
SELECT * FROM egg_base
UNION ALL
SELECT * FROM non_egg_base;
```

```
/*
```

```
=====
=====
```

KEZAD (AUHID01/02) INVENTORY & PICKING HEALTH DASHBOARD

```
=====
=====
```

PURPOSE : Comprehensive inventory and picking health analysis for Kezad warehouses. Covers:

1. SKU-level inventory with DOC, DRR, location type
2. Average SKUs per bin broken down by zone
3. Location catalogue (deep vs forward)
4. Picking performance — picked vs skipped, deep vs forward, top vs BAU SKUs
5. Daily cancellations by reason
6. Forward-to-deep migration suggestions (excess DOC > 14 days)
7. Batch issue detection (deep expiry older than forward)

SOURCE : noondwh.mxdcss_dcscs, noondwh.scgoms_cwpicking,
noondwh.scgoms_goms, noonbinimops.fulfillment,
noonbiinstnt.minutes

GRAIN : Varies per output block (see individual SELECT comments)

FILTERS : UAE (country_code = 'ae') | AUHID01 / AUHID02

AUTHOR : —

LAST : (auto-refresh via CURRENT_DATE / CURRENT_TIMESTAMP)

Structure — outputs are commented out blocks so you run only what you need without re-computing the shared base CTEs

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*/

/*

VARIABLE DEFINITIONS

WAREHOUSE_FILTER : AUHID01, AUHID02 (Kezad warehouses)
COUNTRY_CODE : 'ae' (UAE)
UAE_COUNTRY_ID : 1 (id_country for UAE)
SA_COUNTRY_ID : 2 / 5 (Saudi — used in date shift)
TOP_SKU_FLAG : is_top = 1 (top 2500 SKUs)
DRR_WINDOW : last 7 days (CURRENT_DATE-7 to CURRENT_DATE-1)
PICKING_WINDOW : last 8 business days (pj.created_at - 6h30m)
TRANSFER_JOIN_WINDOW : last 10 days (transfer_request.created_at)
BUSINESS_DAY_CUTOFF : 06:00 local (6h offset for AE, 6h for SA)
AE_CUTOFF_OFFSET : INTERVAL 6 HOUR + 30 MIN (Dubai business day)
SA_CUTOFF_OFFSET : INTERVAL 6 HOUR (Riyadh business day)

DOC_BUCKETS (Days of Cover):

Zero : DOC = 0
00-03 : DOC ≤ 3
04-07 : DOC ≤ 7
07-14 : DOC ≤ 14
14++ : DOC > 14 ← candidates for forward→deep migration

SKU TYPE:

Slow_Moving : DRR < 1
BAU : DRR ≥ 1

LOCATION TYPE:

Deep_location : misc.is_deep = '1'
Forward_location : misc.is_deep != '1'

SKIP REASONS COUNTED AS REAL MISS: 'missing', 'damaged', 'expired'

MP_STATUS_CANCELLED : 31 (item cancelled in goms)

PICKING_JOB_CLOSED : 'closed'

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-- =====
-- SHARED CTE 1: top_2500
-- Top SKUs for UAE (is_top = 1 flag from instock snapshot)
-- =====
WITH top_2500 AS (
  SELECT sku
  FROM noonbiinstnt.minutes.instock_views_current_snapshot
  WHERE country_code = 'ae'
  AND is_top = 1
  GROUP BY ALL
),

-- =====
-- SHARED CTE 2: deep_loc
-- All deep locations (misc.is_deep = '1') in AUHID01/02
-- Used across picking, migration, and batch issue analyses
-- =====
deep_loc AS (
  SELECT
    w.partner_warehouse_code AS wh_code,
    z.zone_code,
    l.location_code,
    JSON_EXTRACT_SCALAR(l.misc, '$.is_deep') AS is_deep
  FROM noondwh.mxdcss_dcsc.country c
  LEFT JOIN noondwh.mxdcss_dcsc.warehouse w USING (id_country)
  LEFT JOIN noondwh.mxdcss_dcsc.tenant t USING (id_warehouse)
  LEFT JOIN noondwh.mxdcss_dcsc.zone z USING (id_tenant)
  LEFT JOIN noondwh.mxdcss_dcsc.location l USING (id_zone)
  LEFT JOIN noondwh.mxdcss_dcsc.location_item li USING (id_location)
  JOIN noondwh.mxdcss_dcsc.item i USING (id_item)
  WHERE w.partner_warehouse_code IN ('AUHID01', 'AUHID02')
  AND z.is_saleable = 1
  AND JSON_EXTRACT_SCALAR(l.misc, '$.is_deep') = '1'
  GROUP BY ALL
),

-- =====
-- SHARED CTE 3: deep_zones
-- Distinct deep zone codes per warehouse (derived from deep_loc)
-- =====
deep_zones AS (
  SELECT wh_code, zone_code

```

```
-- =====
-- SHARED CTE 4: inv_base
-- Saleable inventory snapshot for AUHID01, with top SKU flag
-- Source: fulfillment materialized stock table
-- =====
```

```

=====
-- SHARED CTE 5: drr
-- Daily run rate per SKU — last 7 days delivered orders
-- dr = total_sales / 7 (used for DOC calculation)
-- Source: goms.item (more accurate than sales_order for this use case)
=====

```

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drr AS (
SELECT
    catalog_sku                                AS sku,
    COUNT(DISTINCT i.id_item)                  AS total_sales,
    COUNT(DISTINCT i.id_item)
        / COUNT(DISTINCT DATE(i.created_at, 'Asia/Dubai')) AS dr
FROM noondwh.scgoms_goms.item i
LEFT JOIN noondwh.scgoms_goms.warehouse src_w
    ON i.id_warehouse = src_w.id_warehouse
WHERE DATE(i.created_at) BETWEEN CURRENT_DATE() - 7 AND CURRENT_DATE() - 1
    AND i.id_mp = 9                                -- Noon Instant only

```

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        AND src_w.id_country = 1
        AND src_w.partner_warehouse_code = 'AUHID01'
    GROUP BY ALL
),

-- =====
-- SHARED CTE 6: inv_with_doc
-- Inventory joined with DRR and location type — central enriched base
-- for all inventory-facing analyses (DOC, migration, zone breakdown)
-- =====
inv_with_doc AS (
    SELECT
        a.partner_warehouse_code,
        a.zone_code,
        a.location_code,
        a.zsku,
        a.sku_flag,
        a.category,
        a.product_title,
        CASE WHEN d.dr < 1.0 THEN 'Slow_Moving' ELSE 'BAU' END          AS sku_type,
        d.dr                                                            AS drr,
        CASE WHEN b.is_deep = '1' THEN 'Deep_location'
              ELSE 'Forward_location' END                             AS location_type,
        SUM(a.gross_qty)                                                AS gross_qty,
        SUM(a.gross_qty) / GREATEST(d.dr, 0.1)                        AS doc_kept,
        COUNT(DISTINCT a.location_code)                                AS locations_kept,
        COUNT(DISTINCT a.zone_code)                                    AS zones_kept,
        CASE
            WHEN SUM(a.gross_qty) / GREATEST(d.dr, 0.1) = 0 THEN 'Zero'
            WHEN SUM(a.gross_qty) / GREATEST(d.dr, 0.1) <= 3 THEN '00-03'
            WHEN SUM(a.gross_qty) / GREATEST(d.dr, 0.1) <= 7 THEN '04-07'
            WHEN SUM(a.gross_qty) / GREATEST(d.dr, 0.1) <= 14 THEN '07-14'
            ELSE '14++'
        END                                                            AS doc_bucket
    FROM inv_base a
    LEFT JOIN deep_loc b
        ON a.partner_warehouse_code = b.wh_code
        AND a.zone_code = b.zone_code
        AND a.location_code = b.location_code
    LEFT JOIN drr d ON d.sku = a.zsku
    GROUP BY ALL
),

-- =====

```

```

-- SHARED CTE 7: picking_base
-- All closed picking jobs in last 8 business days for AUHID01/02
-- Enriched with skip reason, location type, top SKU flag
-- Used by: picking performance, JLM, daily cancellations
-- =====
picking_base AS (
  SELECT
    src_w.partner_warehouse_code          AS wh_code,
    wz.code                               AS zone_code,
    pll.sku,
    pll.location_code,
    pll.id_picklist_line,
    CASE WHEN tt.sku IS NOT NULL THEN 'Top' ELSE 'BAU' END AS sku_type,
    CASE
      WHEN src_w.id_country = 1
      THEN DATE(DATETIME(pj.created_at, 'Asia/Dubai')
        - INTERVAL 6 HOUR - INTERVAL 30 MINUTE)
      WHEN src_w.id_country = 2
      THEN DATE(DATETIME(pj.created_at, 'Asia/Riyadh')
        - INTERVAL 6 HOUR)
    END AS date,
    plc.id_picking_container_line,
    r.code                               AS skip_reason,
    dl.location_code                     AS deep_location_code,
    i.id_item,
    i.id_mp_status
  FROM noondwh.scgoms_cwpicking.picking_job pj
  LEFT JOIN noondwh.scgoms_cwpicking.status pjst
    ON pj.id_status = pjst.id_status
  LEFT JOIN noondwh.scgoms_cwpicking.picklist_line pll
    USING (id_picking_job)
  LEFT JOIN noondwh.scgoms_cwpicking.status pllst
    ON pll.id_status = pllst.id_status
  LEFT JOIN noondwh.scgoms_cwpicking.warehouse src_w
    ON pll.id_warehouse = src_w.id_warehouse
  LEFT JOIN noondwh.scgoms_cwpicking.reason r
    ON r.id_reason = pll.id_skip_reason
  LEFT JOIN noondwh.scgoms_cwpicking.warehouse_zone wz
    ON pll.id_warehouse_zone = wz.id_warehouse_zone
  LEFT JOIN noondwh.scgoms_cwpicking.picking_container_line plc
    USING (id_picklist_line)
  LEFT JOIN deep_loc dl
    ON dl.location_code = pll.location_code
  AND dl.wh_code = src_w.partner_warehouse_code

```

```

LEFT JOIN top_2500 tt ON tt.sku = pll.sku
-- Transfer join scoped to last 10 days to limit scan cost
LEFT JOIN noondwh.scgoms_cwpicking.transfer_request_line t
  ON t.id_transfer_request_line = pll.id_transfer_request_line
  AND DATE(t.created_at, 'Asia/Dubai') >= CURRENT_DATE() - 10
LEFT JOIN noondwh.scgoms_cwpicking.transfer_request tr
  ON tr.id_transfer_request = t.id_transfer_request
  AND DATE(tr.created_at, 'Asia/Dubai') >= CURRENT_DATE() - 10
LEFT JOIN noondwh.scgoms_goms.item i
  ON tr.transfer_nr = i.transfer_nr
  AND i.catalog_sku = t.sku
  AND DATE(i.created_at, 'Asia/Dubai') >= CURRENT_DATE() - 10
WHERE pjst.code = 'closed'
  AND src_w.partner_warehouse_code IN ('AUHID01', 'AUHID02')
  AND DATE(
    DATETIME(pj.created_at, 'Asia/Dubai')
    - INTERVAL 6 HOUR - INTERVAL 30 MINUTE
  ) >= CURRENT_DATE() - 8
),

-- =====
-- SHARED CTE 8: picking_agg
-- Aggregates picking_base to (wh, zone, date) grain
-- Computes picked/skipped counts split by top/deep dimensions
-- =====
picking_agg AS (
  SELECT
    wh_code,
    zone_code,
    date,

    -- Overall
    COUNT(DISTINCT CASE
      WHEN skip_reason IN ('missing', 'damaged', 'expired')
      AND id_picking_container_line IS NULL THEN id_picklist_line END) AS skipped_items,
    COUNT(DISTINCT CASE
      WHEN id_picking_container_line IS NOT NULL THEN id_picklist_line END) AS
    picked_items,

    -- Unique SKU×location combos
    COUNT(DISTINCT CASE
      WHEN id_picking_container_line IS NOT NULL
      THEN CONCAT(sku, location_code) END) AS picked_sku_loc,
    COUNT(DISTINCT CASE

```



```
WHEN skip_reason IN ('missing','damaged','expired')
AND id_picking_container_line IS NULL
THEN CONCAT(sku, location_code) END) AS skipped_sku_loc,
```

```
-- Top SKU split
COUNT(DISTINCT CASE
  WHEN sku_type = 'Top'
  AND skip_reason IN ('missing','damaged','expired')
  AND id_picking_container_line IS NULL THEN id_picklist_line END) AS
skipped_items_top,
COUNT(DISTINCT CASE
  WHEN sku_type = 'Top'
  AND id_picking_container_line IS NOT NULL THEN id_picklist_line END) AS
picked_items_top,
```

```
-- Deep location split
COUNT(DISTINCT CASE
  WHEN deep_location_code IS NOT NULL
  AND skip_reason IN ('missing','damaged','expired')
  AND id_picking_container_line IS NULL THEN id_picklist_line END) AS
skipped_items_deep,
COUNT(DISTINCT CASE
  WHEN deep_location_code IS NOT NULL
  AND id_picking_container_line IS NOT NULL THEN id_picklist_line END) AS
picked_items_deep,
```

```
-- Top × Deep
COUNT(DISTINCT CASE
  WHEN sku_type = 'Top' AND deep_location_code IS NOT NULL
  AND skip_reason IN ('missing','damaged','expired')
  AND id_picking_container_line IS NULL THEN id_picklist_line END) AS
skipped_items_deep_top,
COUNT(DISTINCT CASE
  WHEN sku_type = 'Top' AND deep_location_code IS NOT NULL
  AND id_picking_container_line IS NOT NULL THEN id_picklist_line END) AS
picked_items_deep_top,
```

```
-- Top SKU×location combos
COUNT(DISTINCT CASE
  WHEN sku_type = 'Top'
  AND id_picking_container_line IS NOT NULL
  THEN CONCAT(sku, location_code) END) AS picked_sku_loc_top,
COUNT(DISTINCT CASE
  WHEN sku_type = 'Top'
```

```

AND skip_reason IN ('missing','damaged','expired')
AND id_picking_container_line IS NULL
THEN CONCAT(sku, location_code) END)                AS skipped_sku_loc_top,

-- Cancellations (id_mp_status = 31 in goms)
COUNT(DISTINCT CASE
  WHEN skip_reason IN ('missing','damaged','expired')
    AND id_mp_status = 31 THEN id_item END)          AS
cancelled_due_to_missing,
COUNT(DISTINCT CASE
  WHEN sku_type = 'Top'
    AND skip_reason IN ('missing','damaged','expired')
    AND id_mp_status = 31 THEN id_item END)          AS
cancelled_due_to_missing_top,
COUNT(DISTINCT CASE
  WHEN deep_location_code IS NOT NULL
    AND skip_reason IN ('missing','damaged','expired')
    AND id_mp_status = 31 THEN id_item END)          AS
cancelled_due_to_missing_deep

```

```

FROM picking_base
GROUP BY ALL
),

```

```

-- =====
-- SHARED CTE 9: sku_qty_raw
-- Location-level expiry and quantity detail for AUHID01
-- Used by batch issue detection (deep expiry < forward expiry)
-- =====

```

```

sku_qty_raw AS (
  SELECT
    w.partner_warehouse_code                AS wh_code,
    li.barcode,
    i.id_partner,
    l.id_location,
    l.location_code,
    JSON_EXTRACT_SCALAR(l.misc, '$.is_deep') AS is_deep,
    li.qty,
    i.expiry_date,
    CASE WHEN JSON_EXTRACT_SCALAR(l.misc, '$.is_deep') = '1'
      THEN li.qty ELSE 0 END                AS deep_qty,
    CASE WHEN JSON_EXTRACT_SCALAR(l.misc, '$.is_deep') != '1'
      THEN i.expiry_date END                AS forward_expiry_date,
    CASE WHEN JSON_EXTRACT_SCALAR(l.misc, '$.is_deep') = '1'

```

```

        THEN i.expiry_date END                                AS deep_expiry_date
FROM noondwh.mxdcss_dcsc.country c
LEFT JOIN noondwh.mxdcss_dcsc.warehouse w USING (id_country)
LEFT JOIN noondwh.mxdcss_dcsc.tenant t USING (id_warehouse)
LEFT JOIN noondwh.mxdcss_dcsc.zone z USING (id_tenant)
LEFT JOIN noondwh.mxdcss_dcsc.location l USING (id_zone)
LEFT JOIN noondwh.mxdcss_dcsc.location_item li USING (id_location)
JOIN noondwh.mxdcss_dcsc.item i USING (id_item)
WHERE w.partner_warehouse_code = 'AUHID01'
      AND z.is_saleable = 1
      AND i.expiry_date IS NOT NULL
),

-- Earliest forward expiry per barcode+partner — batch issue anchor
forward_min AS (
  SELECT wh_code, barcode, id_partner,
         MIN(forward_expiry_date) AS min_forward_expiry
  FROM sku_qty_raw
  WHERE forward_expiry_date IS NOT NULL
  GROUP BY ALL
),

-- Total deep qty with older expiry than forward anchor (true batch issue qty)
batch_issue_calc AS (
  SELECT
    s.wh_code, s.barcode, s.id_partner,
    SUM(
      CASE WHEN s.deep_expiry_date < f.min_forward_expiry
            THEN s.deep_qty ELSE 0 END
    ) AS true_batch_issue_qty
  FROM sku_qty_raw s
  LEFT JOIN forward_min f
    ON s.wh_code = f.wh_code AND s.barcode = f.barcode AND s.id_partner = f.id_partner
  GROUP BY ALL
),

-- SKU → barcode mapping from fulfillment materialized table
sku_map AS (
  SELECT warehouse AS partner_warehouse_code, pbarcode_canonical, id_partner, zsku
  FROM noonbinimuae.minutes.minutes_dcsc_stock_AE
  WHERE warehouse IN ('AUHID01', 'AUHID02')
  GROUP BY ALL
),

```

```

-- =====
-- MIGRATION: excess forward inventory candidates (DOC > 14 days)
-- SKUs with forward qty exceeding 14 days of cover
-- =====
migration_candidates AS (
  SELECT
    *,
    gross_qty - GREATEST(COALESCE(drr, 0.1) * 14, 2) AS qty_to_be_moved
  FROM inv_with_doc
  WHERE doc_bucket = '14++'
    AND location_type = 'Forward_location'
    AND gross_qty - GREATEST(COALESCE(drr, 0.1) * 14, 2) > 0
)

/* =====
   OUTPUT 1: SKU inventory detail — DOC, DRR, location type
   Grain: one row per (warehouse, sku, zone, location)
   ===== */
SELECT
  partner_warehouse_code,
  zsku,
  zone_code,
  location_code,           -- use for forward→deep planning
  sku_flag,
  sku_type,
  drr,
  category,
  product_title,
  location_type,
  gross_qty,
  doc_kept,
  locations_kept,
  zones_kept
FROM inv_with_doc

--
--
-- Uncomment the block you need. Only one SELECT runs at a time.
--
--

```

```

/* =====
OUTPUT 2: Average SKUs per bin broken down by zone
Grain: one row per (warehouse, zone)
=====
SELECT
  partner_warehouse_code,
  zone_code,
  COUNT(DISTINCT location_code)          AS locations,
  COUNT(DISTINCT zsku)                   AS skus,
  COUNT(DISTINCT zsku) / COUNT(DISTINCT location_code) AS sku_per_location
FROM inv_with_doc
GROUP BY ALL;
*/

```

```

/* =====
OUTPUT 3: Location catalogue — deep vs forward flag
Grain: one row per (warehouse, zone, location)
=====
SELECT DISTINCT
  wh_code,
  zone_code,
  location_code,
  CASE WHEN is_deep = '1' THEN 'Deep_location' ELSE 'Forward_location' END AS
location_type
FROM deep_loc
ORDER BY 1, 2, 3;
*/

```

```

/* =====
OUTPUT 4: Picking performance — picked/skipped by zone/day
Grain: one row per (warehouse, zone, date)
=====
SELECT * FROM picking_agg;
*/

```

```

/* =====
OUTPUT 5: Daily cancellations (only SKUs with cancellations)
Adds SKU and sku_type dimension to picking_agg
Grain: one row per (warehouse, zone, sku, date)
=====
SELECT
  pb.wh_code,
  pb.zone_code,
  pb.sku,

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pb.sku_type,
pb.date,
COUNT(DISTINCT CASE
  WHEN pb.skip_reason IN ('missing','damaged','expired')
  AND pb.id_picking_container_line IS NULL
  THEN pb.id_picklist_line END)          AS skipped_items,
COUNT(DISTINCT CASE
  WHEN pb.id_picking_container_line IS NOT NULL
  THEN pb.id_picklist_line END)          AS picked_items,
COUNT(DISTINCT CASE
  WHEN pb.skip_reason IN ('missing','damaged','expired')
  AND pb.id_mp_status = 31
  THEN pb.id_item END)                    AS cancelled_due_to_missing
FROM picking_base pb
GROUP BY ALL
HAVING cancelled_due_to_missing > 0;
*/

```

```

/* =====
OUTPUT 6a: Forward→Deep migration — deep zones only
Grain: one row per (warehouse, sku, location) to move
=====

```

```

SELECT
a.*,
SUM(stock_gross - stock_reserved)
  OVER (PARTITION BY a.warehouse, a.zsku
        ORDER BY stock_gross - stock_reserved DESC, a.location_code) AS cum_qty,
b.qty_to_be_moved
FROM noonbinimops.fulfillment.minutes_dcsl_stock_AE a
JOIN migration_candidates b
  ON a.warehouse = b.partner_warehouse_code AND a.zsku = b.zsku
LEFT JOIN deep_loc d
  ON a.warehouse = d.wh_code AND a.zone_code = d.zone_code AND a.location_code =
d.location_code
JOIN deep_zones dz ON dz.zone_code = a.zone_code AND dz.wh_code = a.warehouse
WHERE d.location_code IS NULL
  AND is_saleable = '1'
  AND a.warehouse = 'AUHID01'
QUALIFY cum_qty <= qty_to_be_moved + 20
ORDER BY cum_qty DESC;
*/

```

```

/* =====
OUTPUT 6b: Forward→Deep migration — all zones

```

Same as 6a but without the deep_zones JOIN constraint

```
=====
SELECT
  a.*,
  SUM(stock_gross - stock_reserved)
    OVER (PARTITION BY a.warehouse, a.zsku
          ORDER BY stock_gross - stock_reserved DESC, a.location_code) AS cum_qty,
  b.qty_to_be_moved
FROM noonbinimops.fulfillment.minutes_dcscs_stock_AE a
JOIN migration_candidates b
  ON a.warehouse = b.partner_warehouse_code AND a.zsku = b.zsku
LEFT JOIN deep_loc d
  ON a.warehouse = d.wh_code AND a.zone_code = d.zone_code AND a.location_code =
d.location_code
WHERE d.location_code IS NULL
  AND is_saleable = '1'
  AND a.warehouse = 'AUHID01'
QUALIFY cum_qty <= qty_to_be_moved + 20
ORDER BY cum_qty DESC;
*/
```

```
/* =====
  OUTPUT 7: Batch issue — deep locations with older expiry
            than the earliest forward stock for same barcode
  Grain: one row per (sku, barcode, location)
  =====
```

```
SELECT
  sm.partner_warehouse_code,
  sm.zsku,
  CASE WHEN t2k.sku IS NOT NULL THEN 'Top_2500' ELSE 'bau' END AS sku_flag,
  r.location_code,
  r.id_location,
  r.barcode,
  r.id_partner,
  r.expiry_date      AS location_expiry_date,
  f.min_forward_expiry,
  r.qty              AS location_qty,
  r.deep_qty,
  bi.true_batch_issue_qty
FROM sku_map sm
JOIN sku_qty_raw r
  ON sm.partner_warehouse_code = r.wh_code
  AND sm.pbarcode_canonical = r.barcode
  AND sm.id_partner = r.id_partner
```

```

LEFT JOIN forward_min f
  ON f.wh_code = r.wh_code AND f.barcode = r.barcode AND f.id_partner = r.id_partner
LEFT JOIN batch_issue_calc bi
  ON bi.wh_code = r.wh_code AND bi.barcode = r.barcode AND bi.id_partner = r.id_partner
LEFT JOIN top_2500 t2k ON t2k.sku = sm.zsku
WHERE r.is_deep = '1'
  AND r.expiry_date < f.min_forward_expiry
  AND bi.true_batch_issue_qty > 0
ORDER BY sm.zsku, r.barcode, r.expiry_date;
*/

```

```

/*

```

Zone × Aisle health dashboard — AUHID01

```

GRAIN      : One row per warehouse × zone × aisle
            Deep / Forward metrics are COLUMNS, not rows
DATE SCOPE: Stock take >= 2026-04-14 (operational go-live date)
            Skip metrics = rolling last 7 days
KEY FIXES :
  1. Date filter = >= '2026-04-14' (not MTD – Apr 1 had bulk cancellations)
  2. location_dim for job matching: is_active removed – inactive locations
    still receive stock-take tasks and must be counted
  3. location_dim for location COUNTS: is_active = 1 kept (display only)

```

```

*/

```

```

--
-- CTE 1a : Location dim for JOB MATCHING
--          No is_active filter – inactive locs still get ST tasks
--

```

```

WITH location_dim_jobs AS (
  SELECT DISTINCT
    w.partner_warehouse_code      AS warehouse_code,
    z.zone_code,
    l.location_code,

```



```

        SPLIT(l.location_code, '-') [SAFE_OFFSET(1)]      AS aisle,
        JSON_EXTRACT_SCALAR(l.misc, '$.is_deep')          AS is_deep
FROM `noondwh.mxdcss_dcsc.warehouse` w
JOIN `noondwh.mxdcss_dcsc.zone`      z ON z.id_warehouse = w.id_warehouse
JOIN `noondwh.mxdcss_dcsc.location`  l ON l.id_zone      = z.id_zone
WHERE w.partner_warehouse_code = 'AUHID01'
      AND z.is_saleable = 1
      AND l.is_virtual  = 0
      and is_active = 1
      -- is_active intentionally excluded: inactive locs still receive tasks
),

-- -----
-- CTE 1b : Location dim for COUNTS (display: active locations only)
-- -----

location_dim_counts AS (
  SELECT DISTINCT
    w.partner_warehouse_code      AS warehouse_code,
    z.zone_code,
    l.id_location,
    l.location_code,
    SPLIT(l.location_code, '-') [SAFE_OFFSET(1)]      AS aisle,
    JSON_EXTRACT_SCALAR(l.misc, '$.is_deep')          AS is_deep
FROM `noondwh.mxdcss_dcsc.warehouse` w
JOIN `noondwh.mxdcss_dcsc.zone`      z ON z.id_warehouse = w.id_warehouse
JOIN `noondwh.mxdcss_dcsc.location`  l ON l.id_zone      = z.id_zone
WHERE w.partner_warehouse_code = 'AUHID01'
      AND z.is_saleable = 1
      AND l.is_active   = 1
      AND l.is_virtual  = 0
),

-- -----
-- CTE 2 : Location counts - overall / deep / forward (active only)
-- -----

location_counts AS (

```

```

SELECT
    warehouse_code,
    zone_code,
    aisle,
    COUNT(DISTINCT id_location) AS
total_locations,
    COUNT(DISTINCT CASE WHEN is_deep = '1' THEN id_location END) AS deep_locations,
    COUNT(DISTINCT CASE WHEN is_deep = '0' THEN id_location END) AS
forward_locations
FROM location_dim_counts
GROUP BY 1, 2, 3
),

-- -----
-- CTE 3 : Stock-take jobs from 2026-04-14 onwards
--         Excludes force_closed & cancelled
--         Uses location_dim_jobs (no is_active filter)
-- -----

stock_take_base AS (
SELECT
    w.partner_warehouse_code AS warehouse_code,
    ld.zone_code,
    ld.aisle,
    ld.is_deep,
    ld.location_code AS job_location_code,
    j.job_nr,
    s.code AS job_status
FROM `noondwh.mxdcss_dcss.job` j
JOIN `noondwh.mxdcss_dcss.status` s USING (id_status)
JOIN `noondwh.mxdcss_dcss.warehouse` w USING (id_warehouse)
JOIN `noondwh.mxdcss_dcss.process` p USING (id_process)
JOIN `noondwh.mxdcss_dcss.process_type` pt USING (id_process_type)
JOIN location_dim_jobs ld
    ON ld.warehouse_code = w.partner_warehouse_code
    AND ld.location_code = JSON_EXTRACT_SCALAR(j.misc, '$.location_code')
WHERE j.id_job_subtype = 10

```

```

AND pt.code = 'routine_stock_take'
AND w.partner_warehouse_code = 'AUHID01'
AND s.code NOT IN ('force_closed', 'cancelled')
AND DATE(DATETIME(j.created_at, 'Asia/Dubai')) >= DATE '2026-04-14'
),

stock_take_agg AS (
SELECT
    warehouse_code,
    zone_code,
    aisle,
    COUNT(DISTINCT job_nr) AS st_created,
    COUNT(DISTINCT CASE WHEN job_status = 'closed' THEN job_nr END) AS st_closed,
    COUNT(DISTINCT CASE WHEN is_deep = '1' THEN job_nr END) AS
st_created_deep,
    COUNT(DISTINCT CASE WHEN is_deep = '1' AND job_status = 'closed'
                        THEN job_nr END) AS
st_closed_deep,
    COUNT(DISTINCT CASE WHEN is_deep = '0' THEN job_nr END) AS
st_created_fwd,
    COUNT(DISTINCT CASE WHEN is_deep = '0' AND job_status = 'closed'
                        THEN job_nr END) AS
st_closed_fwd,
    ROUND(SAFE_DIVIDE(
        COUNT(DISTINCT job_nr),
        COUNT(DISTINCT job_location_code)
    ), 2) AS w2w_overall,
    ROUND(SAFE_DIVIDE(
        COUNT(DISTINCT CASE WHEN is_deep = '1' THEN job_nr END),
        COUNT(DISTINCT CASE WHEN is_deep = '1' THEN job_location_code END)
    ), 2) AS w2w_deep,
    ROUND(SAFE_DIVIDE(
        COUNT(DISTINCT CASE WHEN is_deep = '0' THEN job_nr END),
        COUNT(DISTINCT CASE WHEN is_deep = '0' THEN job_location_code END)
    ), 2) AS w2w_fwd
FROM stock_take_base

```

```

GROUP BY 1, 2, 3
),

-- -----
-- CTE 4 : Skip metrics - rolling 7 days (cwpicking)
-- -----

skip_base AS (
SELECT
    src_w.partner_warehouse_code                AS warehouse_code,
    DATE(
        DATETIME(pj.created_at, 'Asia/Dubai')
        - INTERVAL 6 HOUR - INTERVAL 30 MINUTE
    )                                            AS biz_day,
    ld.zone_code,
    ld.aisle,
    ld.is_deep,
    pll.id_picklist_line
FROM `noondwh.scgoms_cwpicking.picking_job`    pj
JOIN `noondwh.scgoms_cwpicking.status`        pjst
    ON pjst.id_status = pj.id_status AND pjst.code = 'closed'
JOIN `noondwh.scgoms_cwpicking.picklist_line`  pll
    ON pll.id_picking_job = pj.id_picking_job
    AND pll.id_old_picklist_line IS NULL
JOIN `noondwh.scgoms_cwpicking.status`        pllst
    ON pllst.id_status = pll.id_status AND pllst.code = 'skipped'
JOIN `noondwh.scgoms_cwpicking.reason`        r
    ON r.id_reason = pll.id_skip_reason AND r.code IN ('missing', 'damaged', 'expired')
JOIN `noondwh.scgoms_cwpicking.warehouse`    src_w
    ON src_w.id_warehouse = pll.id_warehouse
    AND src_w.partner_warehouse_code = 'AUHID01'
LEFT JOIN `noondwh.scgoms_cwpicking.picking_container_line` plc
    ON plc.id_picklist_line = pll.id_picklist_line
JOIN location_dim_jobs                        ld
    ON ld.warehouse_code = src_w.partner_warehouse_code
    AND ld.location_code = pll.location_code
WHERE plc.id_picking_container_line IS NULL

```

```

AND DATE(
    DATETIME(pj.created_at, 'Asia/Dubai')
    - INTERVAL 6 HOUR - INTERVAL 30 MINUTE
    ) between CURRENT_DATE('Asia/Dubai') - 7 and CURRENT_DATE('Asia/Dubai') - 1
),

skip_wh_totals AS (
SELECT
    warehouse_code,
    COUNT(DISTINCT id_picklist_line) AS wh_total_7d,
    COUNT(DISTINCT CASE WHEN biz_day = CURRENT_DATE('Asia/Dubai') - 1
        THEN id_picklist_line END) AS wh_total_d1
FROM skip_base
GROUP BY 1
),

skip_agg AS (
SELECT
    sb.warehouse_code, sb.zone_code, sb.aisle,
    COUNT(DISTINCT sb.id_picklist_line) AS
skip_items_7d,
    ROUND(SAFE_DIVIDE(COUNT(DISTINCT sb.id_picklist_line)*100.0,
        MAX(wt.wh_total_7d)),2) AS
skip_contrib_7d_pct,
    ROUND(SAFE_DIVIDE(COUNT(DISTINCT id_picklist_line), 7), 1) AS avg_skips_per_day_7d,
    COUNT(DISTINCT CASE WHEN sb.is_deep='1' THEN sb.id_picklist_line END) AS
skip_items_7d_deep,
    ROUND(SAFE_DIVIDE(COUNT(DISTINCT CASE WHEN sb.is_deep='1'
        THEN sb.id_picklist_line END)*100.0,
        MAX(wt.wh_total_7d)),2) AS
skip_contrib_7d_pct_deep,
    COUNT(DISTINCT CASE WHEN sb.is_deep='0' THEN sb.id_picklist_line END) AS
skip_items_7d_fwd,
    ROUND(SAFE_DIVIDE(COUNT(DISTINCT CASE WHEN sb.is_deep='0'
        THEN sb.id_picklist_line END)*100.0,

```

```

                MAX(wt.wh_total_7d)),2)                                AS
skip_contrib_7d_pct_fwd,
    COUNT(DISTINCT CASE WHEN sb.biz_day = CURRENT_DATE('Asia/Dubai')-1
                THEN sb.id_picklist_line END)                        AS
skip_items_d1,
    ROUND(SAFE_DIVIDE(COUNT(DISTINCT CASE WHEN sb.biz_day =
CURRENT_DATE('Asia/Dubai')-1
                THEN sb.id_picklist_line END)*100.0,
                MAX(wt.wh_total_d1)),2)                                AS
skip_contrib_d1_pct,
    COUNT(DISTINCT CASE WHEN sb.biz_day = CURRENT_DATE('Asia/Dubai')-1
                AND sb.is_deep='1' THEN sb.id_picklist_line END) AS
skip_items_d1_deep,
    COUNT(DISTINCT CASE WHEN sb.biz_day = CURRENT_DATE('Asia/Dubai')-1
                AND sb.is_deep='0' THEN sb.id_picklist_line END) AS
skip_items_d1_fwd
FROM skip_base sb
JOIN skip_wh_totals wt USING (warehouse_code)
GROUP BY 1, 2, 3
)

-- -----
-- FINAL
-- -----

SELECT
    lc.warehouse_code                                AS warehouse,
    lc.zone_code,
    lc.aisle,
    lc.total_locations,
    lc.deep_locations,
    lc.forward_locations,
    COALESCE(st.st_created, 0)                        AS st_created,
    COALESCE(st.st_closed, 0)                        AS st_closed,
    ROUND(SAFE_DIVIDE(COALESCE(st.st_closed,0),
        NULLIF(COALESCE(st.st_created,0),0)),4)      AS st_adherence_pct,
    COALESCE(st.st_created_deep, 0)                  AS st_created_deep,

```

```

COALESCE(st.st_closed_deep, 0) AS st_closed_deep,
ROUND(SAFE_DIVIDE(COALESCE(st.st_closed_deep, 0),
    NULLIF(COALESCE(st.st_created_deep, 0), 0)), 4) AS st_adherence_deep_pct,
COALESCE(st.st_created_fwd, 0) AS st_created_fwd,
COALESCE(st.st_closed_fwd, 0) AS st_closed_fwd,
ROUND(SAFE_DIVIDE(COALESCE(st.st_closed_fwd, 0),
    NULLIF(COALESCE(st.st_created_fwd, 0), 0)), 4) AS st_adherence_fwd_pct,
COALESCE(st.w2w_overall, 0) AS w2w_overall,
COALESCE(st.w2w_deep, 0) AS w2w_deep,
COALESCE(st.w2w_fwd, 0) AS w2w_fwd,
COALESCE(sk.skip_items_7d, 0) AS skip_items_7d,
COALESCE(sk.skip_contrib_7d_pct, 0) AS skip_contrib_7d_pct,
COALESCE(sk.avg_skips_per_day_7d, 0) AS avg_skips_per_day_7d,
COALESCE(sk.skip_items_7d_deep, 0) AS skip_items_7d_deep,
COALESCE(sk.skip_contrib_7d_pct_deep, 0) AS skip_contrib_7d_pct_deep,
COALESCE(sk.skip_items_7d_fwd, 0) AS skip_items_7d_fwd,
COALESCE(sk.skip_contrib_7d_pct_fwd, 0) AS skip_contrib_7d_pct_fwd,
COALESCE(sk.skip_items_d1, 0) AS skip_items_d1,
COALESCE(sk.skip_contrib_d1_pct, 0) AS skip_contrib_d1_pct,
COALESCE(sk.skip_items_d1_deep, 0) AS skip_items_d1_deep,
COALESCE(sk.skip_items_d1_fwd, 0) AS skip_items_d1_fwd
FROM location_counts lc
LEFT JOIN stock_take_agg st USING (warehouse_code, zone_code, aisle)
LEFT JOIN skip_agg sk USING (warehouse_code, zone_code, aisle)
ORDER BY lc.zone_code, lc.aisle

```

/*

ACTUAL MANPOWER — DAY-ON-DAY BY FUNCTION & OVERALL

Purpose : Count of employees actually present per date,
split by warehouse × Function, with an OVERALL row.

Source : noonbinimksa.Stores.wh_Biometric_base

Grain : date × wh_code × Function (function-level rows)
date × wh_code × 'OVERALL' (total row per warehouse)

DEFINITIONS

total_roster : all employees scheduled (working_day = 1)
present : valid_in_time IS NOT NULL
absent : on roster but no valid in-punch
half_punch : valid_in_time exists, valid_out_time is NULL (overlap
documented clearly in comments)
avg_worked_hrs : avg(worked_hours) for present employees only
dod_present_change : present count today vs yesterday (LAG)
dod_pct_change : % change in present count day-on-day

NOTE: working_day = 1 filters to scheduled working days only.
Remove this filter if you want to see all calendar days
including week-offs.

*/

```
WITH bio AS (  
  SELECT  
    created_date, biz_date AS biz_date,  
    wh_code,  
    wh_name,
```



```

Function                                     AS function_name,
employee_id,
CASE WHEN valid_in_time IS NOT NULL THEN 1 ELSE 0 END AS is_present,
-- NOTE: half_punch is a SUBSET of present (in but no out).
--      present + absent = total_roster, but half_punch overlaps present.
CASE WHEN valid_in_time IS NOT NULL
      AND valid_out_time IS NULL THEN 1 ELSE 0 END AS is_half_punch,
worked_hours
FROM `noonbinimksa.Stores.wh_Biometric_base`
WHERE working_day = 1
      AND created_date >= CURRENT_DATE('Asia/Riyadh') - 30
),

-- Function-level aggregation
fn_agg AS (
SELECT
  biz_date, wh_code, wh_name,
  function_name,
  COUNT(DISTINCT employee_id) AS total_roster,
  COUNT(DISTINCT CASE WHEN is_present = 1 THEN employee_id END) AS present,
  COUNT(DISTINCT CASE WHEN is_present = 0 THEN employee_id END) AS absent,
  COUNT(DISTINCT CASE WHEN is_half_punch = 1 THEN employee_id END) AS half_punch,
  ROUND(AVG(CASE WHEN is_present = 1 THEN worked_hours END), 2) AS avg_worked_hrs
FROM bio
GROUP BY biz_date, wh_code, wh_name, function_name
),

-- Overall-level aggregation (all functions combined)
overall_agg AS (
SELECT
  biz_date, wh_code, wh_name,
  'OVERALL' AS function_name,
  COUNT(DISTINCT employee_id) AS total_roster,
  COUNT(DISTINCT CASE WHEN is_present = 1 THEN employee_id END) AS present,
  COUNT(DISTINCT CASE WHEN is_present = 0 THEN employee_id END) AS absent,
  COUNT(DISTINCT CASE WHEN is_half_punch = 1 THEN employee_id END) AS half_punch,

```

```

    ROUND(AVG(CASE WHEN is_present = 1 THEN worked_hours END), 2) AS avg_worked_hrs
FROM bio
GROUP BY biz_date, wh_code, wh_name    -- literal 'OVERALL' excluded from GROUP BY
),

-- Union both layers
combined AS (
    SELECT * FROM fn_agg
    UNION ALL
    SELECT * FROM overall_agg
),

lagged AS (
    SELECT
        *,
        LAG(present) OVER (
            PARTITION BY wh_code, function_name
            ORDER BY biz_date
        )
        AS prev_day_present
    FROM combined
)

-- Final output
SELECT
    biz_date,
    wh_code,
    wh_name,
    function_name,
    total_roster,
    present,
    absent,
    half_punch,
    ROUND(SAFE_DIVIDE(present, total_roster) * 100, 1)    AS attendance_pct,
    avg_worked_hrs,
    prev_day_present,
    present - prev_day_present    AS dod_change,

```

```
ROUND(  
    SAFE_DIVIDE(present - prev_day_present,  
                NULLIF(prev_day_present, 0)) * 100, 1  
) AS dod_pct_change
```

```
FROM lagged
```

```
ORDER BY wh_code, function_name, biz_date
```

/*

=====

Vehicle GPS Distance & Driving Time Analysis

Purpose:

Measures distance traveled and effective driving time per vehicle per day, linked back to which warehouse(s) the vehicle served, using GPS pings from scgoms_trakker.

Grain: One row per vehicle per task-day

Sources:

- scgoms_goms.driver_task_line → task events, entity linkage
- scgoms_goms.driver_task_route → route grouping
- scgoms_goms.driver_task → vehicle + user assignment
- scgoms_goms.user → driver name
- scgoms_goms.hub_transfer → pallet → hub_transfer link
- scgoms_goms.hub_transfer_line → hub_transfer → shipment link
- scgoms_goms.shipment → source warehouse
- scgoms_goms.hub_transfer_group → grouping (unused here, kept for context)
- scgoms_trakker.trip → GPS trip sessions
- scgoms_trakker.trip_detail → GPS segment-level distance & time

Key filters:

- Last 8 days (rolling window)
- hub_transfer.type = 'forward_nim' (exclude RT0 / reverse logistics)

- distance_kilometers > 2 for driving time (noise threshold)
- distance_kilometers > 2 for distance too (consistent with time filter)

Known behavior:

- Warehouse column uses STRING_AGG – a vehicle serving multiple warehouses in a day will show comma-separated values (e.g. 'DXB, FNV')
- next_day_task is NULL for a vehicle's last working day in the window; those days use end-of-day as the GPS window ceiling instead of NULL

```
=====
*/

-- =====
-- CTE 1: vehicle_base
-- One row per vehicle per task-day.
-- Determines which warehouse(s) a vehicle served and
-- the time window of its tasks.
-- =====

WITH vehicle_base AS (
  SELECT
    DATE(dtl.created_at, 'Asia/Dubai') AS task_created_at,
    src_w.id_country,

    -- Warehouse label: derived from partner_warehouse_code
    -- STRING_AGG intentionally de-dupes; multi-warehouse vehicles get
    -- comma-separated values like 'DXB, FNV'
    STRING_AGG(
      DISTINCT CASE
        WHEN src_w.partner_warehouse_code LIKE '%FNV%' THEN 'FNV'
        WHEN src_w.partner_warehouse_code LIKE '%FKZ%' THEN 'FKZ'
        WHEN src_w.partner_warehouse_code LIKE '%DXB%' THEN 'DXB'
        WHEN src_w.partner_warehouse_code LIKE '%AUH%' THEN 'AUH'
        WHEN src_w.partner_warehouse_code LIKE '%RUH%' THEN 'RUH'
        WHEN src_w.partner_warehouse_code LIKE '%JED%' THEN 'JED'
        ELSE 'Others'
      END,
      ', '
    ) AS warehouse_label
  FROM dtl
  JOIN src_w ON dtl.vehicle_id = src_w.vehicle_id
  WHERE dtl.created_at >= '2023-01-01'
  GROUP BY dtl.vehicle_id, src_w.id_country
)
```

```

)
AS warehouse,

dt.id_vehicle,
MIN(u.username) AS driver_name,

-- Window start: earliest task event on this day for GPS join
MIN(DATETIME(dtl.created_at, 'Asia/Dubai')) AS first_task_created_at,

-- Window end: latest task event on this day (used as fallback ceiling)
MAX(DATETIME(dtl.created_at, 'Asia/Dubai')) AS last_task_created_at,

COUNT(DISTINCT dtl.id_driver_task) AS task_num

FROM `noondwh.scgoms_goms.driver_task_line` dtl
LEFT JOIN `noondwh.scgoms_goms.driver_task_route` dtr USING (id_driver_task_route)
LEFT JOIN `noondwh.scgoms_goms.driver_task` dt USING (id_driver_task)
LEFT JOIN `noondwh.scgoms_goms.user` u USING (id_user)

-- Link task line entity to a pallet-level hub_transfer
LEFT JOIN `noondwh.scgoms_goms.hub_transfer` ht
  ON dtl.entity_type = 'pallet'
  AND dtl.entity_id = ht.id_hub_transfer
  AND ht.type = 'forward_nim' -- exclude RTO / reverse logistics

LEFT JOIN `noondwh.scgoms_goms.hub_transfer_line` htl USING (id_hub_transfer)
LEFT JOIN `noondwh.scgoms_goms.hub_transfer_group` htg USING (id_hub_transfer_group)

-- Resolve shipment from hub_transfer_line (only shipment-type lines)
LEFT JOIN `noondwh.scgoms_goms.shipment` sh
  ON htl.line_id = sh.id_shipment
  AND htl.line_type = 'shipment'

-- Source warehouse from shipment (not from polymorphic hub_transfer fields)
LEFT JOIN `noondwh.scgoms_goms.warehouse` src_w
  ON src_w.id_warehouse = CAST(sh.id_warehouse AS INT64)

```

```

WHERE DATE(dtl.created_at, 'Asia/Dubai') >= CURRENT_DATE('Asia/Dubai') - 8

GROUP BY
    DATE(dtl.created_at, 'Asia/Dubai'),
    src_w.id_country,
    dt.id_vehicle
),

-- =====
-- CTE 2: final_vehicle
-- Adds the next working day's start time per vehicle using LEAD.
-- Used to bound the GPS trip join: trips that START between
-- this day's first task and the next day's first task belong
-- to this day's run.
-- For the last day in the window, next_day_task is NULL -
-- we fall back to last_task_created_at + 1 hour as the ceiling.
-- =====
final_vehicle AS (
    SELECT
        *,
        LEAD(first_task_created_at)
            OVER (PARTITION BY id_vehicle ORDER BY first_task_created_at) AS next_day_task
    FROM vehicle_base
),

-- =====
-- CTE 3: driver_gps_base
-- Joins GPS trip data to the vehicle's task window.
--
-- GPS join logic:
--   trip_start (DATE) must fall within [first_task_created_at, ceiling)
--   ceiling = next_day_task if available, else last_task_created_at + 1hr
--
-- Distance filter:
--   Segments < 2 km excluded from BOTH distance and driving time
--   to avoid counting idle GPS noise / parking lot pings.

```

```

-- =====

driver_gps_base AS (
SELECT
    v.task_created_at,
    v.id_country,
    v.warehouse,
    v.id_vehicle,
    v.driver_name,
    v.task_num,

    -- Total distance, excluding short idle segments (< 2 km)
    SUM(
        CASE WHEN d.distance_kilometers > 2 THEN d.distance_kilometers ELSE 0 END
    )
        AS distance_km,

    -- Effective driving time in hours, same 2 km noise threshold
    SUM(
        CASE WHEN d.distance_kilometers > 2 THEN d.driving_time ELSE 0 END
    ) / 3600.0
        AS total_driving_time_hrs

FROM final_vehicle v

-- Join GPS trips that started within this vehicle's task window for the day
LEFT JOIN `noondwh.scgoms_trakker.trip` t
    ON v.id_vehicle = t.id_vehicle
    AND DATE(t.trip_start, 'Asia/Dubai') >= DATE(v.first_task_created_at)
    AND DATE(t.trip_start, 'Asia/Dubai') < DATE(
        COALESCE(v.next_day_task, DATETIME_ADD(v.last_task_created_at, INTERVAL 1
HOUR))
    )

JOIN `noondwh.scgoms_trakker.trip_detail` d
    ON t.id_trip = d.id_trip

GROUP BY
    v.task_created_at,

```



```

        v.id_country,
        v.warehouse,
        v.id_vehicle,
        v.driver_name,
        v.task_num
    )

-- =====
-- Final output
-- Adds a driving time bucket for quick segmentation.
-- Sorted by: country → date → warehouse → vehicle
-- =====

SELECT
    *,
    CASE
        WHEN total_driving_time_hrs < 5 THEN 'LT_5hrs'
        ELSE 'GT_5hrs'
    END AS time_bucket
FROM driver_gps_base
ORDER BY
    id_country,
    task_created_at,
    warehouse,
    Id_vehicle

```

```
/*
--
=====
=====
```

Putaway Pendency : Open for more than 2 days, with remaining items

```
--      not yet completed. Used to track aging inbound backlogs across warehouses.
--
=====
=====
*/
--
=====
=====
-- CTE 1: pending_asn
-- PURPOSE: Get distinct ASN (process) numbers for inbound jobs that are still
--      in 'pending' or 'opened' status, created within the last 60 days.
--      This acts as a filter — we only want ASNs that are still active/unfinished.
--
=====
=====
WITH pending_asn AS (
  SELECT
    p.process_nr AS asn_nr    -- ASN number (e.g., ASN-000123); unique ID for each
    inbound shipment
```

```

FROM noondwh.mxdcss_dcsc.job j                                -- Core jobs table (each
scan/task at the DC)
LEFT JOIN noondwh.mxdcss_dcsc.warehouse w ON j.id_warehouse = w.id_warehouse --
Warehouse where job is happening
LEFT JOIN noondwh.mxdcss_dcsc.status s ON j.id_status = s.id_status -- Status
lookup (pending, opened, closed, etc.)
LEFT JOIN noondwh.mxdcss_dcsc.process p ON j.id_process = p.id_process -- The
parent process (ASN) this job belongs to
LEFT JOIN noondwh.mxdcss_dcsc.process_type pt ON p.id_process_type =
pt.id_process_type -- Process type (inbound, outbound, etc.)

WHERE
j.id_job_subtype IN (4, 5, 6)                                -- Filter: only inbound-related job subtypes
(receive/putaway/label)
AND DATE(j.created_at, 'Asia/Dubai') >= CURRENT_DATE() - INTERVAL 60 DAY -- Only
jobs created in last 60 days (UAE timezone)
AND s.code IN ('pending', 'opened')                          -- Only ASNs that are still active (not
closed/cancelled)
AND pt.code IN ('inbound')                                    -- Only inbound process types (not returns,
transfers, etc.)
AND w.id_country = 1                                          -- UAE only (country ID = 1)
GROUP BY 1                                                    -- Deduplicate: one row per ASN number
),

--
=====
=====
-- CTE 2: base
-- PURPOSE: For each job within the pending ASNs, fetch job-level details including:
--   - Container barcode (the tote or carton being processed)
--   - User who worked on it
--   - Item barcode
--   - Count of remaining vs completed job lines
--   - Earliest job date for the ASN (used for aging calculation)
--
=====
=====
base AS (
SELECT
j.job_nr,                                -- Unique job number (e.g., JOB-000456); one job =
one scan task
w.partner_warehouse_code,                -- External/partner code for the warehouse
(e.g., DXBID01)
p.process_nr AS asn_nr,                  -- The ASN number this job belongs to

```

```

        COALESCE(jc.tote_code, jc.barcode) AS container_barcode, -- Container identifier; prefer
tote_code, fall back to barcode
        u.username, -- The DC operator/user who worked on this job
        i.barcode AS pbarcode, -- Product/item barcode for catalog enrichment
later (used for product title join)

        -- Earliest job creation date across all jobs in this ASN (Dubai timezone)
        -- Used as the "ASN received date" for aging calculation
        MIN(DATE(j.created_at, 'Asia/Dubai')) OVER (PARTITION BY p.process_nr) AS
min_asn_job_date,

        -- Count of job lines still not completed (i.e., job is active and line isn't done)
        -- id_status = 3 is "completed"; all others are pending/in-progress
        COUNT(CASE WHEN s.code IN ('pending', 'opened') AND jl.id_status != 3 THEN 1 ELSE
NULL END) AS items_remaining,

        -- Count of job lines that have been fully processed/completed
        COUNT(CASE WHEN jl.id_status = 3 THEN 1 ELSE NULL END) AS items_completed

FROM noondwh.mxdcss_dcss.job j
LEFT JOIN noondwh.mxdcss_dcss.warehouse w ON j.id_warehouse = w.id_warehouse
LEFT JOIN noondwh.mxdcss_dcss.status s ON j.id_status = s.id_status
LEFT JOIN noondwh.mxdcss_dcss.user u ON j.id_user = u.id_user -- Who
performed the job
LEFT JOIN noondwh.mxdcss_dcss.job_subtype js ON j.id_job_subtype = js.id_job_subtype
-- Subtype details (not used in output but available)
LEFT JOIN noondwh.mxdcss_dcss.process p ON j.id_process = p.id_process
LEFT JOIN noondwh.mxdcss_dcss.process_type pt ON p.id_process_type =
pt.id_process_type
LEFT JOIN noondwh.mxdcss_dcss.job_container jc ON jc.id_job = j.id_job --
Container (tote/carton) assigned to this job
LEFT JOIN noondwh.mxdcss_dcss.job_line jl ON jl.id_job = j.id_job -- Individual
line items within the job (1 line = 1 unit/SKU task)
LEFT JOIN noondwh.mxdcss_dcss.item i USING (id_item) --
Item/product details (barcode)

WHERE
    j.id_job_subtype IN (4, 5, 6) -- Inbound job subtypes only
    AND DATE(j.created_at, 'Asia/Dubai') >= CURRENT_DATE() - INTERVAL 60 DAY -- Last
60 days (UAE timezone)
    AND w.id_country = 1 -- UAE only
    AND p.process_nr IN (SELECT * FROM pending_asn) -- Only jobs belonging
to still-pending ASNs

```

```

GROUP BY
    1, 2, 3, 4, 5, 6, j.created_at -- Group at job+container+user+item level
)

--
=====
=====
-- FINAL SELECT
-- PURPOSE: Enrich base data with:
--     1. WH_Category: human-readable warehouse grouping for reporting
--     2. product_title: product name from catalog join
--     3. ageing: how many days since the ASN first arrived at the DC
-- FILTERS: Only ASNs older than 2 days (min_asn_job_date <= today - 2)
--     AND that still have unprocessed items (items_remaining > 0)
--
=====
=====
SELECT
    b.*, -- All columns from base CTE

    -- Warehouse category mapping: groups partner warehouse codes into business-friendly
    labels
    -- Helps report by fulfilment zone (FZ Kezad, TP Chiller, FNV, etc.)
    CASE
        WHEN partner_warehouse_code IN ('AUHFKZ01', 'AUHFKZ02', 'AUHFKZ03') THEN 'FZ
        Kezad' -- Free Zone Kezad warehouses
        WHEN partner_warehouse_code IN ('AUHID01', 'AUHID02') THEN
        'Kezad_Ambient' -- Kezad ambient-temp storage
        WHEN partner_warehouse_code IN ('AUHID03', 'AUHID04') THEN
        'Kezad_Chiller' -- Kezad chiller/cold-chain storage
        WHEN partner_warehouse_code IN ('DXBID01', 'DXBID02') THEN 'TP_Ambient'
        -- Third Party Dubai ambient
        WHEN partner_warehouse_code IN ('DXBID03', 'DXBID04') THEN 'TP_Chiller'
        -- Third Party Dubai chiller
        WHEN partner_warehouse_code IN ('DXBID07') THEN 'TP_Ultrafresh'
        -- Ultra-fresh (very short shelf-life) Dubai
        WHEN partner_warehouse_code IN ('DXBFNV01') THEN 'FNV' --
        Fruits & Vegetables dedicated warehouse
        ELSE 'Other'
    END AS WH_Category,

    -- Product title from the Noon catalog; joined via pbarcode (canonical product barcode)
    -- Source: noonbinimops.fulfillment.mmmcaa_darkstore — the merchandise master / catalog
    table

```

MAX(product_title) AS product_title, -- MAX used to collapse duplicates from GROUP BY ALL

-- Aging in days: how long this ASN has been sitting unprocessed

-- Calculated from the earliest job date associated with this ASN to today

DATE_DIFF(CURRENT_DATE, min_asn_job_date, DAY) AS ageing -- ageing = number of days since first inbound job for this ASN

FROM base b

LEFT JOIN noonbinimops.fulfillment.mmmcaa_darkstore m

ON b.pbarcode = m.pbarcode_canonical -- Join on canonical product barcode to get product title

WHERE

min_asn_job_date <= CURRENT_DATE - 2 -- Only ASNs that have been pending for more than 2 days (exclude very recent)

AND items_remaining > 0 -- Only ASNs that still have work to do (exclude fully completed ones)

GROUP BY ALL -- BigQuery syntax: group by all non-aggregated columns automatically

ORDER BY

partner_warehouse_code, -- Primary sort: group by warehouse

min_asn_job_date -- Secondary sort: oldest ASNs first within each warehouse

```
/* =====
```

WH - Picking, Sorting, and QPL metrics

It includes:

- Picking productivity (IPH, time, qty)
- Sorting productivity (time, qty, IPH)
- QPL (Quantity per line)
- Breakdown by job types (MS, BD, BS, Batch)

BUSINESS LOGIC:

- Business day = local time - 6 hours (6 AM cutoff)
- Timezones vary by country
- Only closed jobs are considered
- Last ~8-10 days of data included

```
===== */
```

WITH base AS (

```
/* =====
```

BASE LAYER (Job-level aggregation)

- Each row represents a picking job

```

- Calculates time, picked qty, containers used
===== */
SELECT
src_w.partner_warehouse_code AS warehouse_code,    -- Warehouse identifier
pj.job_type,                                         -- Type of picking job
pj.id_user,                                         -- Picker ID
pj.job_nr,                                         -- Job number

/* -----
BUSINESS DATE LOGIC:
Convert UTC → local timezone → subtract 6 hours
This shifts business day to start at 6 AM
----- */

DATE(
  DATETIME(
    pj.created_at,
    CASE
      WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
      WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
      WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
    END
  ) - INTERVAL 6 HOUR
) AS date,

/* Job accepted timestamp (localized) */
DATETIME(
  pj.created_at,
  CASE
    WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
    WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
    WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
  END
) AS job_accepted_at,

/* Job closed timestamp (localized) */
DATETIME(

```



```

    pj.finsihed_picking_at,
CASE
    WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
    WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
    WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
END
) AS job_closed_at,

/* -----
TOTAL JOB DURATION (in seconds)
Raw difference between job start and end
----- */

DATETIME_DIFF(
    pj.finsihed_picking_at,
    pj.created_at,
    SECOND
) AS delta,

/* Total picked items (distinct lines) */
COUNT(DISTINCT pll.id_picklist_line) AS picked_items,

/* Unique containers used per job */
COUNT(
    DISTINCT COALESCE(pcr.container_barcode, pcr.tote_code)
) AS containers_used,

/* First picking scan timestamp */
MIN(
    DATETIME(
        pcl.created_at,
        CASE
            WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
            WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
            WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
        END
    )
)

```

```

) AS min_picked_at

FROM noondwh.scgoms_cwpicking.picking_job pj

/* Join job status → filter only closed jobs */
JOIN noondwh.scgoms_cwpicking.status pjst
  ON pj.id_status = pjst.id_status

/* Picklist lines (items picked) */
JOIN noondwh.scgoms_cwpicking.picklist_line pll
  ON pll.id_picking_job = pj.id_picking_job

/* Warehouse details */
JOIN noondwh.scgoms_cwpicking.warehouse src_w
  ON pll.id_warehouse = src_w.id_warehouse

/* Container line (scan events) */
JOIN noondwh.scgoms_cwpicking.picking_container_line pcl
  ON pcl.id_picklist_line = pll.id_picklist_line

/* Container details */
JOIN noondwh.scgoms_cwpicking.picking_container pcr
  ON pcr.id_picking_container = pcl.id_picking_container

WHERE
  pjst.code = 'closed' -- Only completed jobs
  AND pcl.id_picking_container_line IS NOT NULL

/* Filter last 8 business days */
AND DATE(
  DATETIME(
    pj.created_at,
    CASE
      WHEN src_w.id_country = 1 THEN 'Asia/Dubai'
      WHEN src_w.id_country IN (2,5) THEN 'Asia/Riyadh'
      WHEN src_w.id_country = 3 THEN 'Africa/Cairo'
    )
  )

```

```

        END
    ) - INTERVAL 6 HOUR
) >= CURRENT_DATE - 8

GROUP BY ALL
),

/* =====
USER LEVEL AGGREGATION
- Aggregates multiple jobs per user per day
===== */
user_level AS (
SELECT
    date,
    warehouse_code,
    job_type,
    id_user,

    /* Total picked quantity */
    SUM(picked_items) AS picked_qty,

    /* Total containers used */
    SUM(containers_used) AS containers_used,

    /* -----
    EFFECTIVE TIME CALCULATION:
    If delay > 10 min before first scan → remove idle time
    Else use full job duration
    ----- */
    SUM(
        CASE
            WHEN DATETIME_DIFF(min_picked_at, job_accepted_at, SECOND) > 600
            THEN DATETIME_DIFF(job_closed_at, min_picked_at, SECOND)
            ELSE delta
        END
    ) AS effective_time_sec

```

```

FROM base
GROUP BY ALL
),

/* =====
PICKING METRICS (Warehouse Level)
===== */
pick_df AS (
SELECT
date,
warehouse_code,

/* Overall totals */
SUM(picked_qty) AS Total_Picked_Qty,
SUM(effective_time_sec) AS total_time_sec,
SUM(containers_used) AS pick_containers_used,

/* ===== MULTI STORE PICKING ===== */
SUM(CASE WHEN job_type = 'multi_store_picking' THEN picked_qty END) AS MS_PickedQty,
SUM(CASE WHEN job_type = 'multi_store_picking' THEN containers_used END) AS
MS_UniqueContainers,
SUM(CASE WHEN job_type = 'multi_store_picking' THEN effective_time_sec END) AS
MS_TimeSec,

/* IPH = Items per hour */
SAFE_DIVIDE(
3600 * SUM(CASE WHEN job_type = 'multi_store_picking' THEN picked_qty END),
SUM(CASE WHEN job_type = 'multi_store_picking' THEN effective_time_sec END)
) AS MS_IPH,

/* ===== BATCH DYNAMIC ===== */
SUM(CASE WHEN job_type = 'batch_dynamic' THEN picked_qty END) AS BD_PickedQty,
SUM(CASE WHEN job_type = 'batch_dynamic' THEN containers_used END) AS
BD_UniqueContainers,
SUM(CASE WHEN job_type = 'batch_dynamic' THEN effective_time_sec END) AS BD_TimeSec,

```

```

SAFE_DIVIDE(
    3600 * SUM(CASE WHEN job_type = 'batch_dynamic' THEN picked_qty END),
    SUM(CASE WHEN job_type = 'batch_dynamic' THEN effective_time_sec END)
) AS BD_IPH,

/* ===== BATCH STATIC ===== */
SUM(CASE WHEN job_type = 'batch_static' THEN picked_qty END) AS BS_PickedQty,
SUM(CASE WHEN job_type = 'batch_static' THEN containers_used END) AS
BS_UniqueContainers,
SUM(CASE WHEN job_type = 'batch_static' THEN effective_time_sec END) AS BS_TimeSec,

SAFE_DIVIDE(
    3600 * SUM(CASE WHEN job_type = 'batch_static' THEN picked_qty END),
    SUM(CASE WHEN job_type = 'batch_static' THEN effective_time_sec END)
) AS BS_IPH,

/* ===== COMBINED BATCH ===== */
SUM(CASE WHEN job_type IN ('batch_static', 'batch_dynamic') THEN picked_qty END) AS
Batch_PickedQty,
SUM(CASE WHEN job_type IN ('batch_static', 'batch_dynamic') THEN containers_used END)
AS Batch_UniqueContainers,
SUM(CASE WHEN job_type IN ('batch_static', 'batch_dynamic') THEN effective_time_sec
END) AS Batch_TimeSec,

SAFE_DIVIDE(
    3600 * SUM(CASE WHEN job_type IN ('batch_static', 'batch_dynamic') THEN picked_qty
END),
    SUM(CASE WHEN job_type IN ('batch_static', 'batch_dynamic') THEN effective_time_sec
END)
) AS Batch_IPH,

/* Number of unique pickers */
COUNT(DISTINCT id_user) AS pickers

FROM user_level

```

```

GROUP BY date, warehouse_code
),

/* =====
   SORTING LOGIC
===== */
sort_base AS (
SELECT
/* Business date logic (same as picking) */
DATE(
    DATETIME(
        al3.created_at,
        CASE
            WHEN w.id_country = 1 THEN 'Asia/Dubai'
            WHEN w.id_country IN (2,5) THEN 'Asia/Riyadh'
            WHEN w.id_country = 3 THEN 'Africa/Cairo'
        END
    ) - INTERVAL 6 HOUR
) AS date,

w.partner_warehouse_code AS warehouse_code,
sj.id_user,
pj.job_type AS picking_job_type,
it.id_item,

/* Putaway timestamp */
DATETIME(
    al3.created_at,
    CASE
        WHEN w.id_country = 1 THEN 'Asia/Dubai'
        WHEN w.id_country IN (2,5) THEN 'Asia/Riyadh'
        WHEN w.id_country = 3 THEN 'Africa/Cairo'
    END
) AS putaway_time

FROM noondwh.scgoms_goms.input_container ic

```

```

JOIN noondwh.scgoms_goms.sortation_job sj
  ON ic.id_sortation_job = sj.id_sortation_job
JOIN noondwh.scgoms_goms.warehouse w
  ON ic.id_warehouse = w.id_warehouse
JOIN noondwh.scgoms_goms.input_container_line icl
  ON ic.id_input_container = icl.id_input_container
JOIN noondwh.scgoms_goms.item it
  ON icl.id_item = it.id_item
JOIN noondwh.scgoms_goms.audit_log al3
  ON al3.id_audit_action = 21
  AND al3.action_key = icl.id_input_container_line
  AND JSON_EXTRACT_SCALAR(al3.misc, '$.to') = 'putaway'
  AND DATE(al3.created_at) >= CURRENT_DATE - 9
JOIN noondwh.scgoms_cwpicking.picking_container pc
  ON pc.container_barcode = ic.container_barcode
JOIN noondwh.scgoms_cwpicking.picking_job pj
  ON pj.id_picking_job = pc.id_picking_job

WHERE
  sj.job_type = 'nonsort'
  AND it.transfer_nr IS NOT NULL
  AND sj.id_user IS NOT NULL

/* Filter last 8 days */
AND DATE(
  DATETIME(al3.created_at,
    CASE
      WHEN w.id_country = 1 THEN 'Asia/Dubai'
      WHEN w.id_country IN (2,5) THEN 'Asia/Riyadh'
      WHEN w.id_country = 3 THEN 'Africa/Cairo'
    END
  ) - INTERVAL 6 HOUR
) >= CURRENT_DATE - 8
),

/* Calculate time gap between scans */

```

```

sort_scan AS (
SELECT
    *,
    DATETIME_DIFF(
        putaway_time,
        LAG(putaway_time) OVER (PARTITION BY id_user, date ORDER BY putaway_time),
        SECOND
    ) AS item_time
FROM sort_base
),

/* Aggregate sorting metrics */
sort_df AS (
SELECT
    date,
    warehouse_code,

    /* Batch Dynamic sorting */
    COUNT(DISTINCT CASE WHEN picking_job_type = 'batch_dynamic' THEN id_item END) AS
    BD_sorting_qty,
    SUM(
        CASE WHEN picking_job_type = 'batch_dynamic' THEN
            CASE
                WHEN item_time <= 300 THEN item_time
                WHEN item_time IS NULL THEN 300
                ELSE 0
            END
        END
    ) AS BD_sort_time_sec,

    /* Batch Static sorting */
    COUNT(DISTINCT CASE WHEN picking_job_type = 'batch_static' THEN id_item END) AS
    BS_sorting_qty,
    SUM(
        CASE WHEN picking_job_type = 'batch_static' THEN
            CASE

```



```

        WHEN item_time <= 300 THEN item_time
        WHEN item_time IS NULL THEN 300
        ELSE 0
    END
END
) AS BS_sort_time_sec,

/* Overall sorting */
COUNT(DISTINCT id_item) AS sorting_qty_overall,
SUM(
    CASE
        WHEN item_time <= 300 THEN item_time
        WHEN item_time IS NULL THEN 300
        ELSE 0
    END
) AS sorting_time_overall

FROM sort_scan
GROUP BY date, warehouse_code
),

/* =====
QPL (Quantity Per Line)
===== */
qpl_df AS (
SELECT
    DATE(
        DATETIME(
            tr.created_at,
            CASE
                WHEN w.id_country = 1 THEN 'Asia/Dubai'
                WHEN w.id_country IN (2,5) THEN 'Asia/Riyadh'
                WHEN w.id_country = 3 THEN 'Africa/Cairo'
            END
        )
    ) AS date,

```

```

w.partner_warehouse_code AS warehouse_code,

/* QPL = total qty / total lines */
SAFE_DIVIDE(
    SUM(tr1.quantity),
    COUNT(tr1.id_transfer_request_line)
) AS Overall_QPL

FROM noondwh.scgoms_cwpicking.transfer_request tr
JOIN noondwh.scgoms_cwpicking.transfer_request_line tr1
    USING (id_transfer_request)
JOIN noondwh.scgoms_cwpicking.warehouse w
    ON tr.id_src_warehouse = w.id_warehouse

WHERE
    DATE(
        DATETIME(
            tr.created_at,
            CASE
                WHEN w.id_country = 1 THEN 'Asia/Dubai'
                WHEN w.id_country IN (2,5) THEN 'Asia/Riyadh'
                WHEN w.id_country = 3 THEN 'Africa/Cairo'
            END
        )
    ) >= CURRENT_DATE - 10

GROUP BY date, warehouse_code
)

/* =====
   FINAL OUTPUT
===== */

SELECT
    p.date                AS Pick_Date,
    p.warehouse_code      AS Warehouse_ID,

```

```

COALESCE(c.warehouse_catrgory, 'Other')           AS WH_Category,

q.Overall_QPL,

Total_Picked_Qty,
total_time_sec / 3600                             AS Picking_hrs,
Total_Picked_Qty * 3600 / total_time_sec          AS Picking_IPH,

/* Multi Store */
p.MS_PickedQty,
p.MS_UniqueContainers,
ROUND(p.MS_TimeSec / 3600, 2)                     AS MS_PickedTime,
p.MS_IPH,

/* Batch Dynamic */
p.BD_PickedQty,
p.BD_UniqueContainers,
ROUND(p.BD_TimeSec / 3600, 2)                     AS BD_PickedTime,
p.BD_IPH,

/* Batch Static */
p.BS_PickedQty,
p.BS_UniqueContainers,
ROUND(p.BS_TimeSec / 3600, 2)                     AS BS_PickedTime,
p.BS_IPH,

/* Sorting */
s.BD_sorting_qty,
ROUND(s.BD_sort_time_sec / 3600, 2)               AS BD_Sort_Time,
SAFE_DIVIDE(3600 * s.BD_sorting_qty, s.BD_sort_time_sec) AS BD_Sort_IPH,

s.BS_sorting_qty,
ROUND(s.BS_sort_time_sec / 3600, 2)               AS BS_Sort_Time,
SAFE_DIVIDE(3600 * s.BS_sorting_qty, s.BS_sort_time_sec) AS BS_Sort_IPH,

pick_containers_used,

```

pickers,

Batch_PickedQty,

Batch_UniqueContainers,

Batch_TimeSec,

Batch_IPH,

sorting_qty_overall,

sorting_time_overall

FROM pick_df p

LEFT JOIN sort_df s

ON p.date = s.date AND p.warehouse_code = s.warehouse_code

LEFT JOIN qpl_df q

ON p.date = q.date AND p.warehouse_code = q.warehouse_code

LEFT JOIN noonbinimops.fulfillment.wh_code_wh_category_mapping c

ON p.warehouse_code = c.warehouse_code;

```
/* =====
```

Fake Stocktake Identification Analysis

PURPOSE:

Identify correlation between:

- Stocktake audit locations (deep vs forward)
- Skipped picks (missing/damaged/expired)
- SKU classification (Top vs BAU)

OUTPUT:

- Locations audited
- User who audited
- Whether location is Deep/Forward
- Whether SKU is Top or BAU
- If pick skip happened within 24 hours after audit

```
===== */
```

```
WITH top_2500 AS (
```

```
/* =====
```

TOP SKU LIST

- Extract top SKUs (Top 2500 items) for UAE
- Used to classify picks into:
 - 'top' → high priority SKUs
 - 'bau' → regular SKUs

```
===== */
SELECT
    sku
FROM noonbiinstnt.minutes.instock_views_current_snapshot
WHERE country_code = 'ae'
    AND is_top = 1
GROUP BY ALL
),

/* =====
DEEP LOCATION IDENTIFICATION
- Identifies whether a location is "Deep storage"
- Deep = bulk storage (back area)
- Forward = picking/front area
===== */
deep_loc AS (
SELECT
    w.partner_warehouse_code AS wh_code,    -- Warehouse ID
    l.location_code,                    -- Location identifier

    /* Extract "is_deep" flag from JSON metadata */
    JSON_EXTRACT_SCALAR(l.misc, '$.is_deep') AS is_deep

FROM noondwh.mxdcss_dcsc.country c

LEFT JOIN noondwh.mxdcss_dcsc.warehouse w USING (id_country)
LEFT JOIN noondwh.mxdcss_dcsc.tenant t USING (id_warehouse)
LEFT JOIN noondwh.mxdcss_dcsc.zone z USING (id_tenant)
LEFT JOIN noondwh.mxdcss_dcsc.location l USING (id_zone)
LEFT JOIN noondwh.mxdcss_dcsc.location_item li USING (id_location)
JOIN noondwh.mxdcss_dcsc.item i USING (id_item)
```

```

WHERE
    w.partner_warehouse_code IN ('AUHID01','AUHID02') -- Target warehouses
    AND z.is_saleable = 1 -- Only saleable zones
    AND JSON_EXTRACT_SCALAR(l.misc, '$.is_deep') = '1' -- Only deep locations

GROUP BY ALL
),

/* =====
AUDIT LOCATIONS (STOCKTAKE)
- Captures stocktake jobs performed on locations
- Determines whether location is Deep or Forward
===== */
audit_locs AS (
SELECT
    /* Business date (localized) */
    DATE(j.updated_at, 'Asia/Dubai') AS updated_date,

    /* Exact timestamp */
    DATETIME(j.updated_at, 'Asia/Dubai') AS updated_at,

    w.partner_warehouse_code AS wh_code,

    /* Extract location_code from job metadata */
    JSON_EXTRACT_SCALAR(j.misc, '$.location_code') AS location_code,

    /* Classify location type */
    CASE
        WHEN dl.location_code IS NOT NULL THEN 'Deep'
        ELSE 'Forward'
    END AS location_type,

    /* User details */
    u.username,
    u.email

```

```

FROM noondwh.mxdcss_dcsc.job j

JOIN noondwh.mxdcss_dcsc.status s USING (id_status)
JOIN noondwh.mxdcss_dcsc.warehouse w USING (id_warehouse)
JOIN noondwh.mxdcss_dcsc.process p USING (id_process)
JOIN noondwh.mxdcss_dcsc.process_type pt USING (id_process_type)

/* Join with deep locations to classify */
LEFT JOIN deep_loc dl
  ON dl.location_code = JSON_EXTRACT_SCALAR(j.misc, '$.location_code')
  AND dl.wh_code = w.partner_warehouse_code

/* User info */
LEFT JOIN noondwh.mxdcss_dcsc.user u
  ON u.id_user = j.id_user

WHERE
  DATE(j.updated_at, 'Asia/Dubai') >= CURRENT_DATE - 3 -- Last 3 days
  AND s.code IN ('closed') -- Completed jobs
  AND w.partner_warehouse_code IN ('AUHID01', 'AUHID02')
  AND j.id_job_subtype = 10 -- Stocktake jobs

GROUP BY ALL
),

/* =====
PICK SKIP DATA
- Captures skipped picks due to:
    missing, damaged, expired
- Identifies SKU type (Top vs BAU)
===== */
pick AS (
SELECT
  src_w.partner_warehouse_code AS wh_code,

```



```

/* Pick date and timestamp */
DATE(p11.updated_at, 'Asia/Dubai') AS created_date,
DATETIME(p11.updated_at, 'Asia/Dubai') AS created_at,

/* SKU classification */
CASE
    WHEN tt.sku IS NOT NULL THEN 'top'
    ELSE 'bau'
END AS sku_type,

p11.location_code

FROM noondwh.scgoms_cwpicking.picking_job pj

LEFT JOIN noondwh.scgoms_cwpicking.status pjst
    ON pj.id_status = pjst.id_status

LEFT JOIN noondwh.scgoms_cwpicking.picklist_line p11
    USING (id_picking_job)

LEFT JOIN noondwh.scgoms_cwpicking.status p11st
    ON p11.id_status = p11st.id_status

LEFT JOIN noondwh.scgoms_cwpicking.warehouse src_w
    ON p11.id_warehouse = src_w.id_warehouse

LEFT JOIN noondwh.scgoms_cwpicking.reason r
    ON r.id_reason = p11.id_skip_reason

LEFT JOIN noondwh.scgoms_cwpicking.warehouse_zone wz
    ON p11.id_warehouse_zone = wz.id_warehouse_zone

LEFT JOIN noondwh.scgoms_cwpicking.warehouse dst_w
    ON p11.id_destination = dst_w.id_warehouse

LEFT JOIN noondwh.scgoms_cwpicking.picking_container_line plc

```

```

USING (id_picklist_line)

LEFT JOIN noondwh.scgoms_cwpicking.picking_container pcr
  ON pcr.id_picking_container = plc.id_picking_container

LEFT JOIN noondwh.scgoms_cwpicking.cluster cl
  ON cl.id_cluster = pll.id_cluster

/* Join top SKU list */
LEFT JOIN top_2500 tt
  ON tt.sku = pll.sku

WHERE
  /* Business date logic (6.5 hour offset) */
  DATE(DATETIME(pj.created_at, 'Asia/Dubai')
    - INTERVAL '6' HOUR
    - INTERVAL '30' MINUTE) >= CURRENT_DATE - 4

AND pjst.code = 'closed' -- Completed jobs
AND src_w.partner_warehouse_code IN ('AUHID01', 'AUHID02')

/* Only skipped picks */
AND pllst.code = 'skipped'

/* Only specific skip reasons */
AND r.code IN ('missing', 'damaged', 'expired')

/* Ensure no container assigned (true skip) */
AND plc.id_picking_container_line IS NULL

GROUP BY ALL
)

/* =====
FINAL OUTPUT
- Joins audit events with skipped picks

```

```

- Matches same location within 24 hours after audit
===== */
SELECT
a.wh_code,

/* Audit date */
a.updated_date AS audited_date,

a.location_code,

/* Auditor info */
a.username,
a.email,

/* Deep vs Forward */
location_type,

/* SKU classification */
sku_type

FROM audit_locs a

/* Join pick events within 24h after audit */
JOIN pick p
ON a.wh_code = p.wh_code
AND a.location_code = p.location_code
AND p.created_at BETWEEN a.updated_at
AND a.updated_at + INTERVAL '24' HOUR

GROUP BY ALL

```

```
/* =====
```

Fake JLM Identification and Analysis

PURPOSE:

Identify "Fake JLM" behavior:

- A picker skips an item (missing/damaged/expired)
- Then picks the SAME SKU from SAME LOCATION within 4 hours

This helps detect:

- False skips
- Incorrect inventory reporting
- Potential process gaps or misuse

```
===== */
```

WITH overall AS (

```
/* =====
```

BASE DATA (LINE-LEVEL PICK ACTIVITY)

- Each row represents SKU activity at location level

- Captures both:
 - skipped items
 - successfully picked items

```
===== */
SELECT
src_w.partner_warehouse_code AS wh_code,    -- Warehouse ID

/* -----
BUSINESS DATE LOGIC:
- Convert to Dubai timezone
- Shift by 6.5 hours (6h + 30m)
- Defines business day cutoff
----- */

DATE(
    DATETIME(pj.created_at, 'Asia/Dubai')
    - INTERVAL '6' HOUR
    - INTERVAL '30' MINUTE
) AS date,

pj.id_user,                -- Picker ID
pll.location_code,         -- Location of pick
pll.sku,                   -- SKU
pll.updated_at,            -- Event timestamp

/* -----
SKIPPED ITEMS:
Conditions:
- Status = skipped
- Reason = missing / damaged / expired
- No container assigned (true skip)
----- */

COUNT(DISTINCT CASE
    WHEN pllst.code = 'skipped'
        AND r.code IN ('missing', 'damaged', 'expired')
        AND plc.id_picking_container_line IS NULL
    THEN pll.id_picklist_line
```

```

END) AS skipped_items,

/* -----
PICKED ITEMS:
Condition:
- Container line exists → item successfully picked
----- */

COUNT(DISTINCT CASE
  WHEN plc.id_picking_container_line IS NOT NULL
  THEN pll.id_picklist_line
END) AS picked_items

FROM noondwh.scgoms_cwpicking.picking_job pj

/* Job status */
LEFT JOIN noondwh.scgoms_cwpicking.status pjst
  ON pj.id_status = pjst.id_status

/* Picklist lines (SKU-level detail) */
LEFT JOIN noondwh.scgoms_cwpicking.picklist_line pll
  USING (id_picking_job)

/* Line status */
LEFT JOIN noondwh.scgoms_cwpicking.status pllst
  ON pll.id_status = pllst.id_status

/* Warehouse */
LEFT JOIN noondwh.scgoms_cwpicking.warehouse src_w
  ON pll.id_warehouse = src_w.id_warehouse

/* Skip reason */
LEFT JOIN noondwh.scgoms_cwpicking.reason r
  ON r.id_reason = pll.id_skip_reason

/* Additional joins (context, not directly used in logic) */
LEFT JOIN noondwh.scgoms_cwpicking.warehouse_zone wz

```

```

ON pll.id_warehouse_zone = wz.id_warehouse_zone

LEFT JOIN noondwh.scgoms_cwpicking.warehouse dst_w
ON pll.id_destination = dst_w.id_warehouse

LEFT JOIN noondwh.scgoms_cwpicking.picking_container_line plc
USING (id_picklist_line)

LEFT JOIN noondwh.scgoms_cwpicking.picking_container pcr
ON pcr.id_picking_container = plc.id_picking_container

LEFT JOIN noondwh.scgoms_cwpicking.cluster cl
ON cl.id_cluster = pll.id_cluster

/* Transfer joins (enrichment layer) */
LEFT JOIN noondwh.scgoms_cwpicking.transfer_request_line t
ON t.id_transfer_request_line = pll.id_transfer_request_line
AND DATE(t.created_at, 'Asia/Dubai') >= CURRENT_DATE - 10

LEFT JOIN noondwh.scgoms_cwpicking.transfer_request tr
ON tr.id_transfer_request = t.id_transfer_request
AND DATE(tr.created_at, 'Asia/Dubai') >= CURRENT_DATE - 10

LEFT JOIN noondwh.scgoms_goms.item i
ON tr.transfer_nr = i.transfer_nr
AND i.catalog_sku = t.sku
AND DATE(i.created_at, 'Asia/Dubai') >= CURRENT_DATE - 10

WHERE
/* Last 7 business days */
DATE(
    DATETIME(pj.created_at, 'Asia/Dubai')
    - INTERVAL '6' HOUR
    - INTERVAL '30' MINUTE
) >= CURRENT_DATE - 7

```

```

AND pjst.code = 'closed'    -- Completed jobs only
AND src_w.partner_warehouse_code IN ('AUHID01')

GROUP BY ALL
),

/* =====
FAKE DETECTION LOGIC
- Match skipped → later picked (same SKU + location)
- Within 4 hours window
===== */
raw_data AS (
SELECT
a.*,

/* -----
FAKE FLAG:
If same SKU & location gets picked after skip within 4 hours
----- */
MAX(COALESCE(b.picked_items, 0)) AS fake

FROM overall a

LEFT JOIN overall b
ON a.wh_code = b.wh_code
AND a.location_code = b.location_code
AND a.sku = b.sku

/* Must happen AFTER skip */
AND b.updated_at > a.updated_at

/* Within 4-hour window */
AND b.updated_at < a.updated_at + INTERVAL '4' HOUR

/* Same business date */
AND a.date = b.date

```



```

/* Ensure valid conditions */
AND b.picked_items > 0      -- Later picked
AND a.skipped_items > 0     -- Initially skipped

GROUP BY ALL
)

/* =====
   FINAL OUTPUT (USER LEVEL METRICS)
   ===== */
SELECT
    date,

    r.id_user,                -- Picker ID
    u.username AS email,      -- User email

    /* -----
       FAKE JLM COUNT:
       Number of timestamps where fake detected
       ----- */
    COUNT(DISTINCT CASE
        WHEN fake > 0 THEN r.updated_at
    END) AS fake_jlm,

    /* Unique locations involved in fake cases */
    COUNT(DISTINCT CASE
        WHEN fake > 0 THEN location_code
    END) AS fake_jlm_locations,

    /* Total skipped items that became fake */
    SUM(CASE
        WHEN fake > 0 THEN skipped_items
    END) AS fake_jlm_skipped_items,

    /* Overall skipped items */

```

```
SUM(skipped_items) AS total_skipped_items,

/* Total picked items */
SUM(picked_items) AS picked_items

FROM raw_data r

/* User info */
JOIN noondwh.scgoms_goms.user u
  ON u.id_user = r.id_user

GROUP BY ALL

/* Only show users with fake activity */
HAVING fake_jlm > 0
```

/* =====

Dispatch and Receiving Adherence

PURPOSE:

Analyze driver task performance by comparing:

- Actual dispatch & receiving timestamps
- Planned dispatch & receiving slot timings

Also calculates:

- Dispatch delay (in minutes)
- Receiving delay (in minutes)
- Utilisation factor (pallet efficiency)

Note - This not for all the warehouses, there are different tables for different warehouses and different queries for the same

===== */

WITH

```

/* =====
DRIVER TASK DATA
- Contains actual execution data of trips
===== */

driver_task AS (
SELECT
    id_driver_task,          -- Unique driver task ID
    task_nr,                -- Task number (business identifier)
    driver_task_created_time, -- Actual dispatch timestamp
    dst_warehouse,          -- Destination warehouse
    qty_items,              -- Total items dispatched
    qty_pallets,            -- Total pallets dispatched
    mark_arrived_timestamp   -- Actual receiving timestamp

FROM noonbinimops.fulfillment.tp_driver_task

WHERE
    dst_warehouse IS NOT NULL          -- Must have warehouse assigned
    AND mark_arrived_timestamp IS NOT NULL -- Only completed deliveries
),

/* =====
DISPATCH SLOT MASTER
- Planned schedule for dispatch & receiving
===== */

dispatch_slots AS (
SELECT
    trip_key,          -- Unique trip identifier
    ds_code,           -- Warehouse code
    pallet_count,      -- Maximum pallet capacity
    dispatch_time,     -- Planned dispatch time (HH:MM:SS)
    receiving_time     -- Planned receiving time (HH:MM:SS)

FROM noonbinimops.fulfillment.tp_dispatch_slots

```

```

GROUP BY 1,2,3,4,5    -- Deduplicate
),

/* =====
DISPATCH DELTA CALCULATION
- Finds closest planned dispatch slot to actual dispatch time
- Checks:
    same day
    previous day
    next day
- Picks minimum difference
===== */
min_diff AS (
SELECT
dt.task_nr,
dt.dst_warehouse,

ds.trip_key,

ds.pallet_count AS max_pallet_cnt,    -- Planned capacity
dt.qty_pallets AS pallets_dispatched,  -- Actual pallets

dt.driver_task_created_time AS actual_dispatch_time,
dt.mark_arrived_timestamp AS actual_receive_time,

dt.qty_items AS items_dispatched,

ds.dispatch_time,
ds.receiving_time,

/* -----
DISPATCH DELTA (in minutes)
- Compare actual vs planned dispatch time
- Try 3 possibilities:
    same date
    previous date

```

```

        next date
    - Take minimum difference
----- */
LEAST(

    /* Same day match */
    ABS(TIMESTAMP_DIFF(
        TIMESTAMP(SAFE.PARSE_DATETIME(
            '%Y-%m-%d %H:%M:%S',
            FORMAT_TIMESTAMP('%Y-%m-%d', dt.driver_task_created_time)
            || ' ' || ds.dispatch_time)),
        TIMESTAMP(dt.driver_task_created_time),
        MINUTE
    )),

    /* Previous day match */
    ABS(TIMESTAMP_DIFF(
        TIMESTAMP(SAFE.PARSE_DATETIME(
            '%Y-%m-%d %H:%M:%S',
            FORMAT_TIMESTAMP('%Y-%m-%d', dt.driver_task_created_time - INTERVAL 1 DAY)
            || ' ' || ds.dispatch_time)),
        TIMESTAMP(dt.driver_task_created_time),
        MINUTE
    )),

    /* Next day match */
    ABS(TIMESTAMP_DIFF(
        TIMESTAMP(SAFE.PARSE_DATETIME(
            '%Y-%m-%d %H:%M:%S',
            FORMAT_TIMESTAMP('%Y-%m-%d', dt.driver_task_created_time + INTERVAL 1 DAY)
            || ' ' || ds.dispatch_time)),
        TIMESTAMP(dt.driver_task_created_time),
        MINUTE
    ))

) AS dispatch_delta

```

```

FROM driver_task dt

/* Match slots by warehouse */
JOIN dispatch_slots ds
  ON ds.ds_code = dt.dst_warehouse
),

/* =====
   FIND MINIMUM DISPATCH DELTA
   - For each task, pick the closest slot
   ===== */
min_time_diff AS (
SELECT
  task_nr,
  dst_warehouse,
  MIN(dispatch_delta) AS min_time_diff
FROM min_diff
GROUP BY 1,2
),

/* =====
   FILTER BEST MATCHED SLOT
   ===== */
t1 AS (
SELECT md.*
FROM min_diff md

JOIN min_time_diff mtd
  ON md.task_nr = mtd.task_nr
 AND md.dst_warehouse = mtd.dst_warehouse
 AND md.dispatch_delta = mtd.min_time_diff
),

/* =====
   RECEIVING DELTA CALCULATION

```

```

===== */
t2 AS (
SELECT
    t1.*,

    /* -----
    RECEIVING DELTA (in minutes)
    ----- */

    LEAST(

        /* Same day */
        ABS(TIMESTAMP_DIFF(
            TIMESTAMP(SAFE.PARSE_DATETIME(
                '%Y-%m-%d %H:%M:%S',
                FORMAT_TIMESTAMP('%Y-%m-%d', t1.actual_receive_time)
                || ' ' || t1.receiving_time)),
            TIMESTAMP(t1.actual_receive_time),
            MINUTE
        )),

        /* Previous day */
        ABS(TIMESTAMP_DIFF(
            TIMESTAMP(SAFE.PARSE_DATETIME(
                '%Y-%m-%d %H:%M:%S',
                FORMAT_TIMESTAMP('%Y-%m-%d', t1.actual_receive_time - INTERVAL 1 DAY)
                || ' ' || t1.receiving_time)),
            TIMESTAMP(t1.actual_receive_time),
            MINUTE
        )),

        /* Next day */
        ABS(TIMESTAMP_DIFF(
            TIMESTAMP(SAFE.PARSE_DATETIME(
                '%Y-%m-%d %H:%M:%S',
                FORMAT_TIMESTAMP('%Y-%m-%d', t1.actual_receive_time + INTERVAL 1 DAY)

```



```

        || ' ' || t1.receiving_time)),
    TIMESTAMP(t1.actual_receive_time),
    MINUTE
))

) AS receiving_delta

FROM t1
),

/* =====
UTILISATION FACTOR
- Measures pallet efficiency
- Formula:
    (actual pallets / max capacity) * 100
===== */
utilisation_factor AS (
SELECT
    t2.*,

    ROUND(
        100 * SAFE_DIVIDE(
            CAST(t2.pallets_dispatched AS FLOAT64),
            CAST(t2.max_pallet_cnt AS FLOAT64)
        ),
        2
    ) AS utilisation_factor

FROM t2
)

/* =====
FINAL OUTPUT
===== */
SELECT *
FROM utilisation_factor

```

```
-- Optional filter:
-- WHERE task_nr = 'DT3045469089467A';
```

```
/*
```

STAGING USER PRODUCTIVITY — warehouse × date × staged_user

(Validation of Logic Pending)

```
SOURCE   : noondwh.scgoms_goms      (audit_log, htl, shipment)
           : noondwh.scgoms_cwpicking (picking_container, picklist_line)
```

```
GRAIN     : one row per source_warehouse × biz_date × staged_user
```

```
SCOPE     : UAE (1), KSA (2,5), Egypt (3) – central warehouses only
           : multi_store_picking jobs only
```

```
BIZ DATE  : audit_log scan time localised per country then - 6h
```

```
           : UAE → Asia/Dubai    - 6h
```

```
           : KSA → Asia/Riyadh   - 6h
```

```
           : Egypt→ Africa/Cairo - 6h
```

```
TIME LOGIC (scan-only, no job proxy):
```

- Each audit_log row (action=12, to='staged') = one tote scan by a user
- LEAD() gets next scan for same user × warehouse × biz_date
- Gap ≤ 600s → working time (productive scanning)

```

- Gap > 600s → idle (excluded from working time)
- active_time = MAX(scan) - MIN(scan) per user × wh × day
- working_time = SUM of sub-600s inter-scan gaps only
WINDOW : last 7 completed business days (today excluded)

```

```

*/

WITH

-- — TIMEZONE MAP —————
tz AS (
  SELECT id_country, tz_name FROM UNNEST([
    STRUCT(1 AS id_country, 'Asia/Dubai' AS tz_name),
    STRUCT(2 AS id_country, 'Asia/Riyadh' AS tz_name),
    STRUCT(5 AS id_country, 'Asia/Riyadh' AS tz_name),
    STRUCT(3 AS id_country, 'Africa/Cairo' AS tz_name)
  ])
),

-- — STEP 1: RAW STAGING SCAN EVENTS —————
-- One row per user × tote scan (audit_log staged event)
-- Items per tote aggregated here (cwpicking grain) to avoid fan-out downstream
-- Join chain: audit_log → htl → shipment → picking_container → picklist_line
raw_scans AS (
  SELECT
    u.username AS staged_user,
    al.id_user,
    src_w.partner_warehouse_code AS
source_warehouse,
    src_w.id_country,
    c.tz_name,
    cwpc.container_barcode AS awb_nr,
    -- Localise scan time to country tz
    DATETIME(al.created_at, c.tz_name) AS
scan_at_local,
    -- Items in this tote: only non-recreated picklist lines (skip-and-recreate guard)

```

```

COUNT(DISTINCT CASE WHEN pll.id_old_picklist_line IS NULL
                        THEN pll.id_picklist_line END) AS
items_in_tote
FROM `noondwh.scgoms_goms.audit_log` al
JOIN `noondwh.scgoms_goms.hub_transfer_line` htl
  ON htl.id_hub_transfer_line = al.action_key
  AND htl.type = 'forward_nim'
  AND htl.line_type = 'shipment'
JOIN `noondwh.scgoms_goms.shipment` sh
  ON sh.id_shipment = htl.line_id
  AND sh.created_at >= TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 8 DAY)
-- Bridge goms → cwpicking via awb_nr = container_barcode
JOIN `noondwh.scgoms_cwpicking.picking_container` cwpc
  ON cwpc.container_barcode = sh.awb_nr
JOIN `noondwh.scgoms_cwpicking.picking_job` cwpj
  ON cwpj.id_picking_job = cwpc.id_picking_job
  AND cwpj.job_type = 'multi_store_picking'
JOIN `noondwh.scgoms_cwpicking.picking_container_line` pcl
  ON pcl.id_picking_container = cwpc.id_picking_container
JOIN `noondwh.scgoms_cwpicking.picklist_line` pll
  ON pll.id_picklist_line = pcl.id_picklist_line
JOIN `noondwh.scgoms_cwpicking.warehouse` src_w
  ON src_w.id_warehouse = cwpj.id_warehouse
  AND src_w.type = 'central'
JOIN tz c
  ON c.id_country = src_w.id_country
-- Resolve staging user from goms.user (staging is a goms operation)
LEFT JOIN `noondwh.scgoms_goms.user` u
  ON u.id_user = al.id_user
WHERE al.id_audit_action = 12
  AND JSON_EXTRACT_SCALAR(al.misc, '$.to') = 'staged'
  AND al.created_at >= TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 7 DAY)
  AND DATE(DATETIME(al.created_at, c.tz_name) - INTERVAL 6 HOUR)
    < CURRENT_DATE('Asia/Dubai')
GROUP BY 1,2,3,4,5,6,7
),

```

```

-- — STEP 2: ADD BUSINESS DATE + LEAD() FOR NEXT SCAN —————
-- LEAD() partitioned by user × warehouse × biz_date
-- ensures gaps don't bleed across warehouses or days
scan_timeline AS (
  SELECT
    staged_user,
    id_user,
    source_warehouse,
    id_country,
    tz_name,
    awb_nr,
    items_in_tote,
    scan_at_local,
    -- Business date: -6h from local time aligns shift to 06:00 start
    DATE(scan_at_local - INTERVAL 6 HOUR) AS biz_date,
    LEAD(scan_at_local) OVER (
      PARTITION BY
        id_user,
        source_warehouse,
        DATE(scan_at_local - INTERVAL 6 HOUR)
      ORDER BY scan_at_local
    ) AS
  next_scan_local
  FROM raw_scans
),

```

```

-- — STEP 3: CLASSIFY INTER-SCAN GAPS —————
-- ≤ 600s (10 min) → working gap (counted toward working time)
-- > 600s          → idle (excluded; break, waiting, system delay)
-- NULL           → last scan of day, no gap to measure
scan_gaps AS (
  SELECT
    staged_user,
    id_user,
    source_warehouse,

```

```

biz_date,
awb_nr,
items_in_tote,
scan_at_local,
CASE
    WHEN next_scan_local IS NULL THEN NULL
    WHEN TIMESTAMP_DIFF(next_scan_local, scan_at_local, SECOND) <= 600
        THEN TIMESTAMP_DIFF(next_scan_local, scan_at_local, SECOND)
    ELSE NULL
END AS
working_gap_seconds
FROM scan_timeline
)

-- — STEP 4: AGGREGATE per warehouse × date × user —————
SELECT
    source_warehouse,
    biz_date,
    COALESCE(staged_user, CAST(id_user AS STRING)) AS staged_user,

    -- Volume
    COUNT(DISTINCT awb_nr) AS
totes_staged,
    SUM(items_in_tote) AS qty_staged,
    ROUND(SUM(items_in_tote) / NULLIF(COUNT(DISTINCT awb_nr), 0), 1) AS
avg_items_per_tote,

    -- Scan window timestamps
    MIN(scan_at_local) AS first_scan,
    MAX(scan_at_local) AS last_scan,

    -- Active time: first scan → last scan (includes idle gaps), in minutes
    ROUND(
        TIMESTAMP_DIFF(MAX(scan_at_local), MIN(scan_at_local), SECOND)
        / 60.0, 1

```

```

)
AS
active_time_mins,

-- Working time: sum of sub-10-min inter-scan gaps only, in minutes
ROUND(SUM(COALESCE(working_gap_seconds, 0)) / 60.0, 1)
AS
working_time_mins,

-- Utilisation: working / active (how much of session was productive)
ROUND(
    100.0 * SUM(COALESCE(working_gap_seconds, 0))
    / NULLIF(
        TIMESTAMP_DIFF(MAX(scan_at_local), MIN(scan_at_local), SECOND)
        , 0)
    , 1)
AS
utilisation_pct,

-- Throughput metrics
ROUND(
    SUM(items_in_tote)
    / NULLIF(SUM(COALESCE(working_gap_seconds, 0)) / 3600.0, 0)
    , 1)
AS
items_per_working_hour,
ROUND(
    COUNT(DISTINCT awb_nr)
    / NULLIF(SUM(COALESCE(working_gap_seconds, 0)) / 3600.0, 0)
    , 1)
AS
totes_per_working_hour

FROM scan_gaps
GROUP BY 1, 2, 3
ORDER BY biz_date DESC, source_warehouse, qty_staged DESC

```

Barcode History Dcss

Easy Guide

Purpose: Builds a barcode movement history so you can see where an item moved from, where it moved to, and who processed it.

Key aliases / tables: tl = transaction_line, j = job, p = process, pt = process_type, w = warehouse, i = item, src_l = location, src_z = zone

Main fields / variables: process_type, pbarcode_canonical, source, source_code, src_zone, destination, destination_code, dst_zone, created_at_ts, created_at_date

Barcode History

SQL

```
SELECT DISTINCT
  p.process_nr,
  pt.code AS process_type,
  i.barcode AS pbarcode_canonical,
  i.id_partner,
  w.partner_warehouse_code,
  tl.src_key AS source,
  CASE
    WHEN tl.src_key = 'location' THEN src_l.location_code
    ELSE src_zc.barcode
  END AS source_code,
  src_z.zone_code AS src_zone,
  tl.dst_key AS destination,
  CASE
    WHEN tl.dst_key = 'location' THEN dst_l.location_code
    ELSE dst_zc.barcode
  END AS destination_code,
  dst_z.zone_code AS dst_zone,
  tl.qty,
  u.username,
  u.email,
  datetime(tl.created_at, 'Asia/Dubai') as created_at_ts,
  date(tl.created_at, 'Asia/Dubai') as created_at_date,
FROM
  noondwh.mxdcss_dcsc.transaction_line tl
LEFT JOIN
  noondwh.mxdcss_dcsc.job j
ON
  tl.id_job = j.id_job
LEFT JOIN
  noondwh.mxdcss_dcsc.process p
ON
  j.id_process = p.id_process
```

```

LEFT JOIN
    noondwh.mxdcss_dcss.process_type pt
ON
    p.id_process_type = pt.id_process_type
LEFT JOIN
    noondwh.mxdcss_dcss.warehouse w
ON
    w.id_warehouse = j.id_warehouse
LEFT JOIN
    noondwh.mxdcss_dcss.item i
ON
    tl.id_item = i.id_item
LEFT JOIN
    noondwh.mxdcss_dcss.location src_l
ON
    tl.src_key = 'location' AND CAST(tl.src_value AS STRING) =
CAST(src_l.id_location AS STRING)
LEFT JOIN
    noondwh.mxdcss_dcss.zone src_z
ON
    src_l.id_zone = src_z.id_zone
LEFT JOIN
    noondwh.mxdcss_dcss.job_container src_jc
ON
    tl.src_key = 'job_container' AND CAST(tl.src_value AS STRING) =
CAST(src_jc.id_job_container AS STRING)
LEFT JOIN
    noondwh.mxdcss_dcss.location dst_l
ON
    tl.dst_key = 'location' AND CAST(tl.dst_value AS STRING) =
CAST(dst_l.id_location AS STRING)
LEFT JOIN
    noondwh.mxdcss_dcss.zone dst_z
ON
    dst_l.id_zone = dst_z.id_zone
LEFT JOIN
    noondwh.mxdcss_dcss.job_container dst_jc
ON
    tl.dst_key = 'job_container' AND CAST(tl.dst_value AS STRING) =
CAST(dst_jc.id_job_container AS STRING)
LEFT JOIN
    noondwh.mxdcss_dcss.user u
ON
    j.id_user = u.id_user

```


PnL

Easy Guide

Purpose: Summarizes PnL-related inventory movements by day, movement type, subtype, month, and week.

Key aliases / tables: txlogs

Main fields / variables: week_no

Filter hints: BETWEEN '2026-01-01' AND CURRENT_DATE() -1

PnL UAE 2026

SQL

```
SELECT
    transaction_date,
    financial_movement_type,
    subtype,
    YEAR_MONTH,
    EXTRACT(
        WEEK
        FROM
            transaction_date
    ) AS week_no,
    SUM(qty) qty,
    SUM(qty * allocated_unit_cost_lcy) value,
FROM
    noonbinimops.fulfillment.txlogs
WHERE
    financial_movement_type IN ("adjustments", "other", "transit",
    "adjustment")
    AND id_partner = 9411
    AND transaction_date BETWEEN '2026-01-01' AND CURRENT_DATE() -1
GROUP BY
    1,
    2,
    3,
    4,
    5
```

Adj DS

Easy Guide

Purpose: Calculates dark-store inventory adjustments, enriches them with product details, and converts unit movements into value impact.

CTEs: parsed, t1, sku_info, p1, p2, t22

Key aliases / tables: stj = stock_take_job, tx = stock_tx, sl = stock_line, w = warehouse, p = product, pft = product_fulltype, f = family, pen = product_en

Main fields / variables: family, title_en, adj_type, warehouse, put_date, job_nr, put_user, price, month, wh_code

Filter hints: posted_date >= '2026-01-01'

WITH

```
    parsed AS (  
SELECT sl.*,  
stj.id_stock_take_job,  
stj.id_user,
```

```
    f.name_fr AS family,  
    pen.title AS title_en
```

```
FROM noondwh.mxfulfillment_norms.stock_take_job stj  
JOIN noondwh.wms.stock_tx tx ON (tx.tx_nr = stj.stock_take_job_nr)  
JOIN noonbicedata.wms_v3.stock_line sl USING(id_stock_tx)  
    left join noondwh.wms.warehouse w  
    on w.code = sl.warehouse_pick
```

```
LEFT JOIN  
    noondwh.instant_spanner.product p  
ON  
    (p.sku = sl.sku)  
LEFT JOIN  
    noondwh.wecat_md.product_fulltype pft  
USING  
    (id_product_fulltype)  
LEFT JOIN  
    noondwh.wecat_md.family f  
USING  
    (id_family)  
LEFT JOIN  
    noondwh.instant_spanner.product_en pen  
ON  
    (p.sku = pen.sku)  
where tx.tx_intent = 'inventory_adjustment'  
and tx.posted_date >= '2026-01-01'  
    and w.id_country = 2  
)
```

```

t1 AS (
SELECT
CASE
WHEN area_pick = 'INVADJ' AND area_put = 'STOCK' THEN 'Positive'
WHEN area_put = 'INVADJ'
AND area_pick = 'STOCK' THEN 'Negative'
END
AS adj_type,
p.*,
warehouse_pick as warehouse,
-- DATE(TIMESTAMP_ADD(updated_at, INTERVAL 4 HOUR)) AS put_date,
id_stock_take_job AS job_nr,
id_user AS put_user,
posted_date as put_date,
-- stj.id_warehouse
FROM
  parsed p ),
sku_info AS (
SELECT
  pbarcode_canonical,
  id_partner,
  category,
  brand_code,
  Product_subtype,
  product_fulltype_code
FROM
  noonbilogmis.reporting.DS_Stock_Detail
GROUP BY
  ALL),
  p1 as (
select pbarcode_canonical,
  id_partner,
  avg(price) as price
from
  noonbinimops.fulfillment.master_price_v1
group by all
)
, p2 as (
select pbarcode_canonical,
-- id_partner,
  avg(price) as price
from
  noonbinimops.fulfillment.master_price_v1
group by all

```


)

```
, t22 AS (  
  SELECT  
    adj_type,  
    put_date,  
    put_user,  
    FORMAT_DATE('%B', put_date) AS month,  
    b.id_partner,  
    b.sku,  
    title_en,  
    family,  
    si.category,  
    si.brand_code,  
    si.Product_subtype,  
    si.product_fulltype_code,  
    b.pbarcode_canonical,  
    b.barcode,  
    warehouse AS wh_code,  
    area AS warehouse_name,  
    job_nr,  
    doc_nr,  
    SUM(qty) AS units,  
  FROM  
    t1 b  
  LEFT JOIN  
    noondwh.mxfulfillment_norms.warehouse wh  
  ON  
    (wh.partner_warehouse_code = b.warehouse)
```

```
LEFT JOIN  
  sku_info si  
ON  
  si.id_partner = b.id_partner  
  AND si.pbarcode_canonical = b.pbarcode_canonical  
  LEFT JOIN  
    noonbiinstnt.darkstore.noon_catalog_with_details p  
ON  
  p.id_partner = b.id_partner  
  AND p.pbarcode_canonical = b.pbarcode_canonical
```

```

GROUP BY
ALL
),
t2 as (
select b.*,
       coalesce(p1.price, p2.price) AS value_per_item,
       coalesce(p1.price, p2.price) * units AS total_variance_value
from t22 b
left join p1
on b.pbarcode_canonical = p1.pbarcode_canonical
and b.id_partner = p1.id_partner
left join p2
on b.pbarcode_canonical = p2.pbarcode_canonical
)
,t3 AS (
SELECT
put_date,
put_user,
FORMAT_DATE('%B', put_date) AS month,
id_partner,
sku,
title_en,
family,
category,
brand_code,
Product_subtype,
product_fulltype_code,
pbarcode_canonical,
barcode,
wh_code,
warehouse_name,
job_nr,
doc_nr,
SUM(CASE
    WHEN adj_type = 'Positive' THEN total_variance_value
END
) AS positive_variance_value,
SUM(CASE
    WHEN adj_type = 'Negative' THEN total_variance_value
END
) AS negative_variance_value,
SUM(CASE
    WHEN adj_type = 'Positive' THEN units
END

```

```

    ) AS positive_variance_units,
SUM(CASE
    WHEN adj_type = 'Negative' THEN units
END
    ) AS negative_variance_units
FROM
    t2
GROUP BY
    ALL ),
t4 AS (
SELECT
    *,
    EXTRACT(month
FROM
    put_date) AS month_no,
    IFNULL(positive_variance_value,0) - IFNULL(negative_variance_value,0) AS net_value,
    IFNULL(positive_variance_units,0) - IFNULL(negative_variance_units,0) AS net_units
FROM
    t3 )
, t5 as (
SELECT
    *,
CASE
    WHEN net_units > 0 THEN 'positive_adj'
    WHEN net_units < 0 THEN 'negative_adj'
    WHEN net_units = 0 THEN 'zero'
END
    AS variance_category
FROM
    t4
WHERE
    wh_code NOT LIKE 'CAIDS%'
    and put_date >= '2026-01-01'
    )
select
    put_date,
CAST(put_user AS STRING) AS put_user,
cast(month as string) as month,
id_partner,
sku,
title_en,
family,
category,
brand_code,

```

```
Product_subtype,  
product_fulltype_code,  
pbarcode_canonical,  
barcode,  
wh_code,  
warehouse_name,  
positive_variance_units,  
negative_variance_units,  
cast(month_no as int64) as month_no,  
net_units,  
variance_category,  
job_nr,  
doc_nr,  
CAST(negative_variance_value AS NUMERIC) AS negative_variance_value,  
CAST(positive_variance_value AS NUMERIC) AS positive_variance_value,  
CAST(net_value AS NUMERIC) AS net_value  
from t5
```

Non-Live Warehouse Inventory

Easy Guide

Purpose: Builds a non-live inventory view with quantities and estimated values using fallback pricing.

CTEs: p1, p2, wms_base, wms_1, dcss_base, dcss_1

Key aliases / tables: s = stock, l = location, w = warehouse, z = zone, a = area, it = item, id = item_detail, b = wms_base

Main fields / variables: price, warehouse, wms_qty, id_partner, wms_value, pbarcode_canonical, dcss_qty, dcss_value, total_inventory, total_inventory_value

UAE

WITH

```
p1 AS (  
    SELECT  
        pbarcode_canonical,  
        id_partner,  
        avg(price) AS price  
    FROM  
        noonbinimops.fulfillment.master_price_v1  
    GROUP BY ALL  
) ,  
p2 AS (  
    SELECT  
        pbarcode_canonical,  
        -- id_partner,  
        avg(price) AS price  
    FROM  
        noonbinimops.fulfillment.master_price_v1  
    GROUP BY ALL  
)  
  
, wms_base AS (  
    SELECT  
        w.code AS warehouse,  
        id.pbarcode_canonical,
```

```

id.id_partner,
    SUM(s.qty) AS wms_qty
FROM noondwh.wms.stock s
LEFT JOIN noondwh.wms.location l
    USING (id_location)
LEFT JOIN noondwh.wms.warehouse w
    USING (id_warehouse)
LEFT JOIN noondwh.wms.zone z
    USING (id_zone)
LEFT JOIN noondwh.wms.area a
    USING (id_area)
LEFT JOIN noondwh.wms.item it
    USING (id_item)
LEFT JOIN noonbicedata.wms_v2.item_detail id
    USING (id_item)
WHERE
    (w.code LIKE '%AUHID%' OR w.code LIKE '%DXBNIM%')
    AND a.code = 'STOCK'
    AND s.qty > 0
GROUP BY all
),
wms_1 as(
    select warehouse,
    case when COALESCE(b.id_partner) in (9411,9311) then "94311" else "non_94311" end as
id_partner,
    sum(wms_qty) as wms_qty,
    sum(coalesce(p1.price, p2.price) * wms_qty) as wms_value
from wms_base b
LEFT JOIN p1
ON
    b.pbarcode_canonical = p1.pbarcode_canonical
    AND b.id_partner = p1.id_partner
LEFT JOIN p2
ON b.pbarcode_canonical = p2.pbarcode_canonical
group by all
)

```

```

,
dcss_base AS (
    SELECT
        w.partner_warehouse_code AS warehouse,
        i.barcode as pbarcode_canonical,
i.id_partner,
        SUM(li.qty - li.qty_reserved) AS dcss_qty
    FROM noondwh.mxdcss_dcsc.location_item li
    JOIN noondwh.mxdcss_dcsc.item i
        USING (id_item)
    LEFT JOIN noondwh.mxdcss_dcsc.location l
        USING (id_location)
    LEFT JOIN noondwh.mxdcss_dcsc.zone z
        USING (id_zone)
    LEFT JOIN noondwh.mxdcss_dcsc.warehouse w
        ON w.id_warehouse = z.id_warehouse
    WHERE
        id_country = 1
        AND is_saleable = 0
    GROUP BY all
)
, dcsc_1 as (
    select warehouse,
    case when COALESCE(b.id_partner) in (9411,9311) then "94311" else "non_94311" end as
id_partner,
    sum(dcss_qty) as dcsc_qty,
    sum(coalesce(p1.price, p2.price) * dcsc_qty) as dcsc_value
        from dcsc_base b
    LEFT JOIN p1
    ON
        b.pbarcode_canonical = p1.pbarcode_canonical
        AND b.id_partner = p1.id_partner
    LEFT JOIN p2
        ON b.pbarcode_canonical = p2.pbarcode_canonical
    group by all
),

```



```

final as (SELECT distinct
  COALESCE(w.id_partner, d.id_partner) AS id_partner,
  COALESCE(w.warehouse, d.warehouse) AS warehouse,
  sum(IFNULL(d.dcss_qty, 0)) AS dcss_qty,
  sum(IFNULL(w.wms_qty, 0)) AS wms_qty,
  sum(IFNULL(d.dcss_qty, 0)) + sum(IFNULL(w.wms_qty, 0)) AS total_inventory,
  sum(IFNULL(d.dcss_value, 0)) AS dcss_value,
  sum(IFNULL(w.wms_value, 0)) AS wms_value,
  sum(IFNULL(d.dcss_value, 0)) + sum(IFNULL(w.wms_value, 0)) AS total_inventory_value
FROM wms_1 w
FULL OUTER JOIN dcss_1 d
  ON w.warehouse = d.warehouse and d.id_partner=w.id_partner
group by all
ORDER BY warehouse
)

select * from final

```

KSA

```

WITH
  p1 AS (
    SELECT
      pbarcode_canonical,
      id_partner,
      avg(price) AS price
    FROM
      noonbinimops.fulfillment.master_price_ksa_v1
    GROUP BY ALL
  ),
  p2 AS (
    SELECT

```

```

        pbarcode_canonical,
        -- id_partner,
        avg(price) AS price
FROM
        noonbinimops.fulfillment.master_price_ksa_v1
GROUP BY ALL
)

, wms_base AS (
    SELECT
        w.code AS warehouse,
id.pbarcode_canonical,
id.id_partner,
        SUM(s.qty) AS wms_qty
    FROM noondwh.wms.stock s
    LEFT JOIN noondwh.wms.location l
        USING (id_location)
    LEFT JOIN noondwh.wms.warehouse w
        USING (id_warehouse)
    LEFT JOIN noondwh.wms.zone z
        USING (id_zone)
    LEFT JOIN noondwh.wms.area a
        USING (id_area)
    LEFT JOIN noondwh.wms.item it
        USING (id_item)
    LEFT JOIN noonbicedata.wms_v2.item_detail id
        USING (id_item)
    WHERE
w.code LIKE '%RUHNIM%'
        AND a.code = 'STOCK'
        AND s.qty > 0
    GROUP BY all
),
wms_1 as (
    select warehouse,

```

```

case when COALESCE(b.id_partner) in (9411,9311) then "94311" else "non_94311" end as
id_partner,
sum(wms_qty) as wms_qty,
sum(coalesce(p1.price, p2.price) * wms_qty) as wms_value
from wms_base b
LEFT JOIN p1
ON
    b.pbarcode_canonical = p1.pbarcode_canonical
    AND b.id_partner = p1.id_partner
LEFT JOIN p2
ON b.pbarcode_canonical = p2.pbarcode_canonical
group by all
)
,
dcss_base AS (
    SELECT
        w.partner_warehouse_code AS warehouse,
        i.barcode as pbarcode_canonical,
i.id_partner,
        SUM(li.qty - li.qty_reserved) AS dcss_qty
    FROM noondwh.mxdcss_dcss.location_item li
    JOIN noondwh.mxdcss_dcss.item i
        USING (id_item)
    LEFT JOIN noondwh.mxdcss_dcss.location l
        USING (id_location)
    LEFT JOIN noondwh.mxdcss_dcss.zone z
        USING (id_zone)
    LEFT JOIN noondwh.mxdcss_dcss.warehouse w
        ON w.id_warehouse = z.id_warehouse
    WHERE
        id_country = 2
        AND z.zone_code NOT IN(
            'STOCK-N-MGW-NL2',
            'STOCK-N-MGW-NL',
            'STOCK-N-RT2-NL',
            'STOCK-N-RT-NL'

```

```

        )
        AND is_saleable = 0
    GROUP BY all
)
, dcss_1 as (
    select warehouse,
    case when COALESCE(b.id_partner) in (9411,9311) then "94311" else "non_94311" end as
    id_partner,
    sum(dcss_qty) as dcss_qty,
    sum(coalesce(p1.price, p2.price) * dcss_qty) as dcss_value
    from dcss_base b
    LEFT JOIN p1
    ON
        b.pbarcode_canonical = p1.pbarcode_canonical
        AND b.id_partner = p1.id_partner
    LEFT JOIN p2
    ON b.pbarcode_canonical = p2.pbarcode_canonical
    group by all
),

final as (SELECT distinct
    COALESCE(w.id_partner, d.id_partner) AS id_partner,
    COALESCE(w.warehouse, d.warehouse) AS warehouse,
    sum(IFNULL(d.dcss_qty, 0)) AS dcss_qty,
    sum(IFNULL(w.wms_qty, 0)) AS wms_qty,
    sum(IFNULL(d.dcss_qty, 0)) + sum(IFNULL(w.wms_qty, 0)) AS total_inventory,
    sum(IFNULL(d.dcss_value, 0)) AS dcss_value,
    sum(IFNULL(w.wms_value, 0)) AS wms_value,
    sum(IFNULL(d.dcss_value, 0)) + sum(IFNULL(w.wms_value, 0)) AS total_inventory_value
FROM wms_1 w
FULL OUTER JOIN dcss_1 d
    ON w.warehouse = d.warehouse and d.id_partner=w.id_partner
group by all
ORDER BY warehouse
)

```

```
select * from final
```

ID Stock

Easy Guide

Purpose: Creates an ID/DCSS stock snapshot with gross, reserved, and net quantities at location level.

Key aliases / tables: li = location_item, i = item, l = location, z = zone, w = warehouse

Main fields / variables: snapshot_date, pbarcode_canonical, gross_qty, net_qty

```
SELECT
  current_date() as snapshot_date,
  w.id_country,
  w.partner_warehouse_code,
  z.zone_code,
  l.location_code,
  i.barcode AS pbarcode_canonical,
  i.barcode_level_code,
  i.id_partner,
  i.expiry_date,
  li.qty AS gross_qty,
  li.qty_reserved,
  li.qty - li.qty_reserved AS net_qty,
  is_saleable,
  l.is_virtual
FROM
  noondwh.mxdcss_dcsc.location_item li
JOIN
  noondwh.mxdcss_dcsc.item i
USING
  (id_item)
LEFT JOIN
  noondwh.mxdcss_dcsc.location l
USING
  (id_location)
LEFT JOIN
  noondwh.mxdcss_dcsc.zone z
USING
  (id_zone)
LEFT JOIN
  noondwh.mxdcss_dcsc.warehouse w
ON
  (w.id_warehouse = z.id_warehouse)
```

WMS Stock

Easy Guide

Purpose: Creates a current WMS stock snapshot by warehouse, zone, location, and barcode.

Key aliases / tables: s = stock, l = location, w = warehouse, z = zone, a = area, it = item, id = item_detail

Main fields / variables: warehouse, zone, location, qty, snap

```
SELECT DISTINCT
  id.id_partner,
  id.wms_barcode,
  id.pbarcode_canonical,
  w.code AS warehouse,
  z.code AS zone,
  l.code AS location,
  SUM(s.qty) AS qty,
  current_date() as snap
FROM noonndwh.wms.stock s
LEFT JOIN noonndwh.wms.location l
  USING (id_location)
LEFT JOIN noonndwh.wms.warehouse w
  USING (id_warehouse)
LEFT JOIN noonndwh.wms.zone z
  USING (id_zone)
LEFT JOIN noonndwh.wms.area a
  USING (id_area)
LEFT JOIN noonndwh.wms.item it
  USING (id_item)
LEFT JOIN noonbicedata.wms_v2.item_detail id
  USING (id_item)
WHERE
  1 = 1
  AND w.code LIKE ANY('%AUHID%', '%DXBNIM%')
  AND a.code = 'STOCK'
  AND s.qty > 0
GROUP BY ALL
```

Store <> Putaway Pendency <> Buckets

Easy Guide

Purpose: Buckets pending putaway inventory into store/temperature groups so the backlog is easy to segment.

Key aliases / tables: b = box, b_st = status, bi = box_item, pujl = putaway_job_line, dst_w = warehouse, src_w = warehouse

Main fields / variables: country, partner_warehouse_code, area, storage_condition, pending_qty_between_0_3hr, pending_qty_between_3_6hr, pending_qty_above_6hr, total_pending_qty

Filter hints: BETWEEN 0 AND 3 THEN 1; created_at >= TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 1 DAY)

SELECT

CASE WHEN dst_w.id_country = 1 THEN 'ae' ELSE 'sa' END AS country,
dst_w.partner_warehouse_code AS partner_warehouse_code,
dst_w.area AS area,

CASE

WHEN src_w.partner_warehouse_code IN (
'DXBID01','DXBID02','AUHID01','AUHID02',
'JEDID01','JEDID02','RUHID01','RUHID02'
) THEN 'pending_putaway_ambient'

WHEN src_w.partner_warehouse_code IN (
'AUHFKZ01','AUHFKZ02','AUHFKZ03','RUHID05','JEDID05'
) THEN 'pending_putaway_frozen'

WHEN src_w.partner_warehouse_code IN (
'DXBFNV01','RUHID07','JEDID07'
) THEN 'pending_putaway_fnv'

WHEN src_w.partner_warehouse_code IN (
'DXBID03','DXBID04','AUHID03','AUHID04',
'RUHID03','RUHID04','JEDID03','JEDID04',
'DXBID07','RUHID06','JEDID06'
) THEN 'pending_putaway_chiller'

ELSE 'unknown'
END AS storage_condition,

SUM(CASE
WHEN TIMESTAMP_DIFF(CURRENT_TIMESTAMP(), b.created_at, HOUR) BETWEEN 0
AND 3 THEN 1
ELSE 0
END) AS pending_qty_between_0_3hr,

SUM(CASE
WHEN TIMESTAMP_DIFF(CURRENT_TIMESTAMP(), b.created_at, HOUR) > 3
AND TIMESTAMP_DIFF(CURRENT_TIMESTAMP(), b.created_at, HOUR) <= 6 THEN 1

```

ELSE 0
END) AS pending_qty_between_3_6hr,

SUM(CASE
  WHEN TIMESTAMP_DIFF(CURRENT_TIMESTAMP(), b.created_at, HOUR) > 6 THEN 1
  ELSE 0
END) AS pending_qty_above_6hr,

COUNT(*) AS total_pending_qty

FROM noondwh.mxfulfillment_norms.box b
JOIN noondwh.mxfulfillment_norms.status b_st USING(id_status)
LEFT JOIN noondwh.mxfulfillment_norms.box_item bi USING(id_box)
LEFT JOIN noondwh.mxfulfillment_norms.putaway_job_line pujl USING(id_box_item)
LEFT JOIN noondwh.mxfulfillment_norms.warehouse dst_w ON b.id_warehouse =
dst_w.id_warehouse
LEFT JOIN noondwh.mxfulfillment_norms.warehouse src_w ON b.id_src_warehouse =
src_w.id_warehouse

WHERE b.created_at >= TIMESTAMP_SUB(CURRENT_TIMESTAMP(), INTERVAL 1 DAY)
AND b_st.code IN ('pending', 'putting')
AND (pujl.id_putaway_job_line IS NULL OR pujl.id_status = 1)
AND src_w.partner_warehouse_code IN (
  'DXBID01','DXBID02','AUHID01','AUHID02',
  'JEDID01','JEDID02','RUHID01','RUHID02',
  'AUHFKZ01','AUHFKZ02','AUHFKZ03','RUHID05','JEDID05',
  'DXBFNV01','RUHID07','JEDID07',
  'DXBID03','DXBID04','AUHID03','AUHID04',
  'RUHID03','RUHID04','JEDID03','JEDID04',
  'DXBID07','RUHID06','JEDID06'
)

GROUP BY 1,2,3,4

```

Putaway Pendency at DS - Ageging -
Raw Level

Easy Guide

Purpose: Builds dark-store putaway aging so backlog can be monitored by date, warehouse, and age bucket.

CTEs: base

Key aliases / tables: b = box, b_st = status, btm = box_tote_mapping, bi = box_item, pujl = putaway_job_line, rs = reason, dst_w = warehouse, src_w = warehouse

Main fields / variables: awb_nr, date, created_at, storage_condition, exdoc_nr, pbarcode_canonical, src_wh_code, ds_code, ds_name, country_code

WITH

base AS (

SELECT

b.awb_nr AS awb_nr,

CASE

WHEN ctry.code = 'ae' THEN DATE(b.created_at, 'Asia/Dubai')

WHEN ctry.code = 'sa' THEN DATE(b.created_at, 'Asia/Riyadh')

ELSE DATE(b.created_at)

END

AS date,

CASE

WHEN ctry.code = 'ae' THEN DATETIME(b.created_at, 'Asia/Dubai')

WHEN ctry.code = 'sa' THEN DATETIME(b.created_at, 'Asia/Riyadh')

ELSE DATETIME(b.created_at)

END

AS created_at,

CASE

WHEN src_w.partner_warehouse_code IN (
 'DXBDID01', 'DXBDID02', 'AUHID01', 'AUHID02',
 'JEDID01', 'JEDID02', 'RUHID01', 'RUHID02'
) THEN 'pending_putaway_ambient'

WHEN src_w.partner_warehouse_code IN (
 'AUHFKZ01', 'AUHFKZ02', 'AUHFKZ03', 'RUHID05', 'JEDID05'
) THEN 'pending_putaway_frozen'

WHEN src_w.partner_warehouse_code IN (
 'DXBENV01', 'RUHID07', 'JEDID07'
) THEN 'pending_putaway_fnv'

```

        WHEN src_w.partner_warehouse_code IN (
            'DXBID03', 'DXBID04', 'AUHID03', 'AUHID04',
            'RUHID03', 'RUHID04', 'JEDID03', 'JEDID04',
            'DXBID07', 'RUHID06', 'JEDID06'
        ) THEN 'pending_putaway_chiller'

        ELSE 'unknown'
    END AS storage_condition,
    b.exdoc_nr AS exdoc_nr,
    bi.wms_barcode,
    bi.pbarcode_canonical AS pbarcode_canonical,
    src_w.partner_warehouse_code AS src_wh_code,
    dst_w.partner_warehouse_code AS ds_code,
    dst_w.area AS ds_name,
    ctry.code AS country_code,
    SUM(
    IF
        (pujl.id_putaway_job_line IS NULL
         OR pujl.id_status=1, 1, 0)) AS items_pending,
FROM
    noondwh.mxfulfillment_norms.box b
LEFT JOIN
    noondwh.mxfulfillment_norms.status b_st
ON
    b.id_status = b_st.id_status
LEFT JOIN
    noondwh.mxfulfillment_norms.box_tote_mapping btm
USING
    (awb_nr)
LEFT JOIN
    noondwh.mxfulfillment_norms.box_item bi
ON
    bi.id_box=b.id_box
LEFT JOIN
    noondwh.mxfulfillment_norms.putaway_job_line pujl
ON

```

```

    pujl.id_box_item=bi.id_box_item
LEFT JOIN
    noondwh.mxfulfillment_norms.reason rs
ON
    rs.id_reason=pujl.id_reason
LEFT JOIN
    noondwh.mxfulfillment_norms.warehouse dst_w
ON
    b.id_warehouse = dst_w.id_warehouse
LEFT JOIN
    noondwh.mxfulfillment_norms.warehouse src_w
ON
    b.id_src_warehouse = src_w.id_warehouse
LEFT JOIN
    noondwh.mxfulfillment_norms.country ctry
ON
    ctry.id_country = src_w.id_country
LEFT JOIN
    noondwh.mxfulfillment_norms.exdoc e
ON
    e.exdoc_nr = b.exdoc_nr
LEFT JOIN
    noondwh.mxfulfillment_norms.exdoc_type et
ON
    et.id_exdoc_type = e.id_exdoc_type
WHERE
    DATE(b.created_at) >= CURRENT_DATE()-2
    AND b_st.code IN ('pending',
        'putting')
    AND et.code = 'goms'
GROUP BY
    ALL )
SELECT
    date,
    ds_code,
    ds_name,

```



```
src_wh_code,
storage_condition,
exdoc_nr,
awb_nr,
country_code,
wms_barcode,
pbarcode_canonical,
CASE
    WHEN country_code = 'ae' THEN TIMESTAMP_DIFF(CURRENT_DATETIME('Asia/Dubai'),
created_at, MINUTE)
    WHEN country_code = 'sa' THEN TIMESTAMP_DIFF(CURRENT_DATETIME('Asia/Riyadh'),
created_at, MINUTE)
END
AS ageing_in_mins,
SUM(items_pending) AS items_pending
FROM
base
GROUP BY
ALL
```

DS - Complains_

Easy Guide

Purpose: Lists the raw dark-store defects source tables used for UAE and KSA.

`complains_raw_all`

Source table containing all customer complaint records from the darkstore namespace. Each row is one complaint event.

Complain category:

1. `complain_category`
2. `Defect_delivery` (late / wrong / no delivery)
3. `Defect_category` (item malfunction / warranty)
4. `Defect_delivery` (damage)
5. `Defect_category` (content_mismatch)
6. `Defect_quality`
7. `Defect_fulfillment` (others)
8. `Defect_fulfillment` (missing / wrong item / order)

UAE, EGY

Select *

From ``noonbinimksa.darkstore.complains_raw_all``

KSA

Select *

From ``noonbinimksa.darkstore.complains_raw_sa``

Specific Complaints

1. complaints on expired items

```
SELECT *  
FROM `noonbinimksa.darkstore.complains_raw_all`  
where complain_reason IN ( 'expired_item' )
```

2. complaints on near expiry items

```
SELECT *  
FROM `noonbinimksa.darkstore.complains_raw_all`  
where complain_reason IN ( 'item_near_expiry', 'near_expiry', 'warranty_near_expiry' )
```

3. complaints on quality issues for dairy and milk

```
SELECT distinct *  
FROM `noonbinimksa.darkstore.complains_raw_all`  
where complain_reason IN ( 'quality_not_fresh',  
'ProductQuality_FungusorMold', 'bad_quality_item' )  
and lower(minutes_category_new) in ( 'milk', 'dairy & eggs' )
```

DS - Live Inventory

Easy Guide

Purpose: Points to the raw dark-store live inventory table.

Key aliases / tables: use country code filter

UAE, EGY, KSA: Live Qty

Below table has all the countries.

Select *

From `noonbinimksa.Stores.wh_loc_qty`

DS - Non Live Inventory

Easy Guide

Purpose: Points to the raw dark-store non-live inventory tables for UAE, KSA, and Egypt.

Key aliases / tables: KSA = DAM_non_live, EGY = DAM_non_live_ksa

UAE

Select *

From `noonbinimksa.Stores.DAM_non_live`

KSA

Select *

From `noonbinimksa.Stores.DAM_non_live_ksa`

EGY

Select *

From `noonbinimksa.Stores.DAM_non_live_all_eg`

DS - FEFO Movements

FEFO Movements :

Easy Guide

Purpose: This Gives all the skus which have breached FEFO at darkstore, date and sku level

Key Columns :

fefo_breach	exp_nl_units	ex_nl_value
-------------	--------------	-------------

```
with logs as
(select distinct a.*
from noonbinimdwh.modelling.fifo_report_logs as a
join
(select wh_code, sku, date_, max(updated_at) updated_at
from noonbinimdwh.modelling.fifo_report_logs
group by ALL) as b USING (wh_code, sku, date_, updated_at)
)
```

```
select distinct country_code, date_,hour_, wh_code, ds_code, ds_name, replen_mode,
category, sub_category, sku, brand,product_title, vendor_name,keep_life, shelf_life,
unit_cost,fefo_breach, exp_nl_units, ex_nl_value,expected_expiry_date
from logs
where date_ >= current_date - 3
and category not in ('Beverages')
order by date_ desc, ex_nl_value desc
```

FEFO Adherence

Easy Guide

Purpose: This Gives percentage of all FEFO non- adhered skus at darkstore, category at date level

```
with logs as
(select distinct a.*
from noonbinimdwh.modelling.fifo_report_logs as a
join
(select wh_code, sku, date_, max(updated_at) updated_at
from noonbinimdwh.modelling.fifo_report_logs
```

```
group by ALL) as b USING (wh_code, sku, date_, updated_at)
)
```

```
,base as (select country_code, date_,hour_, wh_code, ds_code, ds_name, replen_mode,
category, sub_category, sku, brand,product_title, vendor_name,keep_life, shelf_life,
unit_cost,fefo_breach, exp_nl_units, ex_nl_value,expected_expiry_date
from logs
where lower(sub_category) not in ('fresh juices')
and date_ >= current_date - 3),
```

```
agg_ as (select country_code, date_, category, sum(ex_nl_value) as ex_nl_value
FROM base
GROUP BY ALL),
```

```
total_nl as (
SELECT country_code, date_, category, sum(tot_nl*unit_cost) as total_nl_val
FROM noonbinimdwh.modelling.fifo_base
where date_ >= current_date - 3
and category not in ('Fruit & Vegetables','Beverages')
GROUP BY ALL
)
```

```
select a.*, total_nl_val, SAFE_DIVIDE(ex_nl_value,total_nl_val) as perc from agg_ a
LEFT JOIN total_nl b on a.date_ = b.date_ and a.country_code = b.country_code and
a.category = b.category
order by date_ DESC, ex_nl_value desc
```

DS - Non Live Migraions

Non Live Migrations:

Select *

From `noonbinimksa.darkstore.DAM\_line\_date\_new\_final`

DS - Expiry (raw and Adjusted)

Easy Guide

Purpose: Points to the raw and adjusted dark-store expiry tables by country.

Key aliases / tables: KSA = central_expiry_raw, Adjusted = central_expiry_raw_sa, KSA = central_expiry_raw_adjusted

Main fields / variables: per

Raw expiry table, system expiry:

UAE and EGY:

Select * from noonbinimksa.Stores.central_expiry_raw

KSA:

Select * from noonbinimksa.Stores.central_expiry_raw_sa

Adjusted expiry, as per KL and SL:

UAE and EGY:

Select * from noonbinimksa.Stores.central_expiry_raw_adjusted

KSA:

Select * from noonbinimksa.Stores.central_expiry_raw_adjusted_sa

DS - Picking Data

Easy Guide

Purpose: to get at date, order nr and store level picking data of non-shots order

Main fields / variables: Order Funnel Timings

Picking Data Today:

UAE : Select * from `noonbinimksa.darkstore.order\_funnel\_time`

KSA : Select * from `noonbinimksa.darkstore.order\_funnel\_time\_sa`

EGY : Select * from `noonbinimksa.darkstore.order\_funnel\_time`

Picking Data Historical (L30) :

UAE : Select * from `noonbinimops.fulfillment.order\_funnel\_time\_ae\_hist`

KSA : Select * from `noonbinimops.fulfillment.order\_funnel\_time\_sa\_hist`

EGY : Select * from `noonbinimops.fulfillment.order\_funnel\_time\_ae\_hist`

DS - Inbound and Stock Take Data

Inbound Base:

Easy Guide

Purpose: to get at date and store level inbound data

Main fields / variables: Inbound details along with user ids

UAE : Select * from noonbinimops.fulfillment.ipp

KSA : Select * from noonbinimops.fulfillment.ipp_ksa

EGY and BAH : Select * from noonbinimksa.darkstore.ipp_all

Stock Take Base:

Easy Guide

Purpose: Stock take Data at location and store level

Main fields / variables: stock_take_type (incident / full location) , expected , actual_qty, variance

Base Table:

UAE , KSA , EGY : `select distinct * from noonbinimksa.darkstore.stock_take_base`

Full Location Adherence for UAE , KSA , EGY:

```
with base as (
SELECT a.*
from noonbinimksa.darkstore.stock_take_base a
left join noonwh.instant_instant_order.warehouse b
on a.partner_warehouse_code = b.partner_wh_code
where 1=1 --LOWER(b.country_code) = 'ae'
and (status_desc IN ('completed', 'pending_approval') OR status_desc IS NULL)
)

, overall as (
select country_code,created_date, partner_warehouse_code, area_name_en,id_warehouse,
stock_take_type
, count(distinct id_stock_take_job_queue) as overall_job
from base group by all)

, completed as (
select country_code, created_date, partner_warehouse_code, stock_take_type
```

```

, coalesce(count(distinct id_stock_take_job_queue),0) as completed_job
from base
where lower(status_desc) = 'completed' group by all)

, output as (
select a.*, b.completed_job
, case
  when (b.completed_job = 0 or b.completed_job is null ) then 0
  else round(b.completed_job/a.overall_job,2)
end as fill_rate
from overall a
left join completed b using(created_date, partner_warehouse_code, stock_take_type)
where a.overall_job > 0
)

select distinct * from output

```

DS - MP - Planned Vs Actual - Biometric

Easy Guide

Purpose: Manpower planned vs actual (biometric)

Table : noonbinimksa.Stores.Biometric_base_v2_3

(Biometric is not started in KSA and EGY)

UAE

```
WITH base AS (
    SELECT
        created_date,
        employee_id,
        punch_status,
        CASE
            WHEN punch_status IN ('Punched Properly') THEN 'Present'
            WHEN punch_status = 'Week Off' THEN 'Week Off'
            WHEN punch_status = 'Annual Leave' THEN 'Annual Leave'
            WHEN LOWER(punch_status) LIKE '%leave' THEN 'Other Leave'
            WHEN punch_status = 'No Punch' THEN 'Absent'
            WHEN punch_status = 'Single Punch' THEN 'Present'
            ELSE 'Others'
        END AS final_status
    FROM noonbinimksa.Stores.Biometric_base_v2_3
    --WHERE wh_code = 'DXBDS49' AND created_date = '2026-04-20'
)

SELECT
    created_date,
    COUNT(DISTINCT employee_id) AS total_employees,

    COUNT(DISTINCT CASE WHEN final_status = 'Present' THEN employee_id END) AS present,
    COUNT(DISTINCT CASE WHEN final_status = 'Week Off' THEN employee_id END) AS
week_off,
    COUNT(DISTINCT CASE WHEN final_status like '%Leave' THEN employee_id END) AS leave_,
    COUNT(DISTINCT CASE WHEN final_status = 'Absent' THEN employee_id END) AS absent,
    COUNT(DISTINCT CASE WHEN final_status = 'Others' THEN employee_id END) AS others

FROM base
GROUP BY created_date;
```

DS - Actual Logins

Easy Guide

Purpose: Actual Job Logins across Inbound, Outbound and Stock Take

Columns : country_code, created_date, partner_warehouse_code, name, email

UAE, KSA, EGY

WITH inbound AS (

WITH ae AS (

SELECT

'AE' AS country_code,
dist_partner_code AS partner_warehouse_code,
CAST(id_user AS STRING) AS id_user,
DATE(date) AS created_date,
MIN(TIMESTAMP(start_time)) AS start_time,
MAX(TIMESTAMP(end_time)) AS end_time,
SUM(completed_count) AS num_of_items,

FROM noonbinimops.fulfillment.ipp

GROUP BY partner_warehouse_code, id_user, created_date

),

sa AS (

SELECT

'SA' AS country_code,
dist_partner_code AS partner_warehouse_code,
CAST(id_user AS STRING) AS id_user,
DATE(date) AS created_date,
MIN(TIMESTAMP(start_time)) AS start_time,
MAX(TIMESTAMP(end_time)) AS end_time,
SUM(completed_count) AS num_of_items

FROM noonbinimops.fulfillment.ipp_ksa

GROUP BY partner_warehouse_code, id_user, created_date

),

eg AS (

SELECT

'EG' AS country_code,
dist_partner_code AS partner_warehouse_code,
CAST(id_user AS STRING) AS id_user,
DATE(date) AS created_date,
MIN(TIMESTAMP(start_time)) AS start_time,

```

        MAX(TIMESTAMP(end_time)) AS end_time,
        SUM(completed_count) AS num_of_items,
    FROM noonbinimksa.darkstore.ipp_all
    WHERE country_code = 3
    GROUP BY partner_warehouse_code, id_user, created_date
)

SELECT * FROM ae
UNION ALL
SELECT * FROM sa
UNION ALL
SELECT * FROM eg
),

outbound AS (

WITH base AS (
    SELECT
        CAST(w.country_code AS STRING) AS country_code,
        partner_wh_code AS partner_warehouse_code,
        CAST(pj.id_user AS STRING) AS id_user,
        DATE(TIMESTAMP_ADD(mo.created_at, INTERVAL 4 HOUR)) AS created_date,
        MIN(TIMESTAMP_ADD(pj.created_at, INTERVAL 4 HOUR)) AS start_time,
        MAX(TIMESTAMP_ADD(pj.updated_at, INTERVAL 4 HOUR)) AS end_time,
        ANY_VALUE(mos.num_of_items) AS num_of_items
    FROM noondwh.schoms_homs_orchestration.mp_order mo
    LEFT JOIN noondwh.instant_instant_order.warehouse w
        ON mo.warehouse_code = w.wh_code
    LEFT JOIN noondwh.schoms_homs_orchestration.mp_order_history moh
        ON mo.id_mp_order = moh.id_mp_order
    LEFT JOIN noondwh.schoms_homs.status mo_s
        ON mo_s.id_status = moh.id_status
    LEFT JOIN noondwh.schoms_homs.picking_job pj
        ON mo.id_mp_order = pj.id_mp_order
    LEFT JOIN noondwh.mxfulfillment_norms.user_active_job uaj
        ON pj.id_picking_job = uaj.job_id
    LEFT JOIN noondwh.schoms_homs_orchestration.mp_order_stats mos
        ON mos.id_mp_order = mo.id_mp_order
    WHERE DATE(TIMESTAMP_ADD(mo.created_at, INTERVAL 4 HOUR))
        >= CURRENT_DATE("Asia/Dubai") - 31
    GROUP BY country_code, partner_warehouse_code, id_user, created_date

```

)

SELECT

country_code,
partner_warehouse_code,
id_user,
created_date,
MIN(TIMESTAMP(start_time)) AS start_time,
MAX(TIMESTAMP(end_time)) AS end_time,
SUM(num_of_items) AS num_of_items

FROM base

GROUP BY country_code, partner_warehouse_code, id_user, created_date

),

stock_take AS (

WITH base AS (

SELECT

UPPER(ctr.code) AS country_code,
wh.partner_warehouse_code,
CAST(uaj.id_user AS STRING) AS id_user,
DATE(TIMESTAMP_ADD(stjq.created_at, INTERVAL 4 HOUR)) AS created_date,
MIN(TIMESTAMP_ADD(stjq.created_at, INTERVAL 4 HOUR)) AS start_time,
MAX(TIMESTAMP_ADD(stjq.updated_at, INTERVAL 4 HOUR)) AS end_time,
SUM(
CASE
WHEN st.code IN ('pending_approval', 'completed')
AND stj1_st.code IN ('present', 'excess')
THEN 1 ELSE 0

END

) AS num_of_items

FROM noondwh.mxfulfillment_norms.stock_take_job_queue stj

LEFT JOIN noondwh.mxfulfillment_norms.location l USING (id_location)

LEFT JOIN noondwh.mxfulfillment_norms.warehouse wh

ON l.id_warehouse = wh.id_warehouse

LEFT JOIN noondwh.mxfulfillment_norms.stock_take_job stj

USING (id_stock_take_job_queue)

LEFT JOIN noondwh.mxfulfillment_norms.stock_take_job_line stj1

ON stj.id_stock_take_job = stj1.id_stock_take_job

LEFT JOIN noondwh.mxfulfillment_norms.user_active_job uaj

ON stj.id_stock_take_job = uaj.job_id

```

LEFT JOIN noondwh.mxfulfillment_norms.status st
  ON stj.id_status = st.id_status
LEFT JOIN noondwh.mxfulfillment_norms.status stj1_st
  ON stj1.id_status = stj1_st.id_status
LEFT JOIN noondwh.mxfulfillment_norms.country ctr USING (id_country)
GROUP BY country_code, partner_warehouse_code, id_user, created_date
)

SELECT
  country_code,
  partner_warehouse_code,
  id_user,
  created_date,
  MIN(TIMESTAMP(start_time)) AS start_time,
  MAX(TIMESTAMP(end_time)) AS end_time,
  SUM(num_of_items) AS num_of_items
FROM base
GROUP BY country_code, partner_warehouse_code, id_user, created_date
),

-- unify all
output AS (
  SELECT 'Inbound' AS type_, p.*, u.email, CONCAT(u.first_name, ' ', u.last_name) AS
name
  FROM inbound p
  LEFT JOIN noondwh.mxfulfillment_norms.user u
    ON CAST(u.id_user AS STRING) = p.id_user

  UNION ALL

  SELECT 'Outbound' AS type_, p.*, u.email, CONCAT(u.first_name, ' ', u.last_name) AS
name
  FROM outbound p
  LEFT JOIN noondwh.schoms_homs.user u
    ON CAST(u.id_user AS STRING) = p.id_user

  UNION ALL

  SELECT 'Stock Take' AS type_, p.*, u.email, CONCAT(u.first_name, ' ', u.last_name) AS
name
  FROM stock_take p

```

```

LEFT JOIN noondwh.mxfulfillment_norms.user u
  ON CAST(u.id_user AS STRING) = p.id_user
)

-- FINAL OUTPUT
SELECT
  country_code, created_date,
  partner_warehouse_code,
  id_user AS Employee_ID,
  name,
  email,

  STRING_AGG(DISTINCT id_user) AS id_users,

  SUM(CASE WHEN type_ = 'Inbound' THEN num_of_items ELSE 0 END) AS ib_qty,
  SUM(CASE WHEN type_ = 'Outbound' THEN num_of_items ELSE 0 END) AS ob_qty,
  SUM(CASE WHEN type_ = 'Stock Take' THEN num_of_items ELSE 0 END) AS st_qty,

  SUM(num_of_items) AS total_qty

FROM output
GROUP BY all
ORDER BY created_date DESC;

```

DS - Putaway Pendency Hourly Bucket

put-away delays in stores

```
SELECT
    eg.AM, eg.Assistant_manager , lower(country_code) as country_code,
    src_wh_code,
    pm.ds_code,
    pm.ds_name,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 0 AND 180 THEN items_pending
        ELSE 0
    END
    ) AS `0-3 hrs`,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 181 AND 360 THEN items_pending
        ELSE 0
    END
    ) AS `3-6 hrs`,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 361 AND 720 THEN items_pending
        ELSE 0
    END
    ) AS `6-12 hrs`,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 721 AND 1440 THEN items_pending
        ELSE 0
    END
    ) AS `12-24 hrs`,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 1441 AND 2880 THEN items_pending
        ELSE 0
    END
    ) AS `24-48 hrs`,
    SUM(CASE
        WHEN ageing_in_mins >= 2881 THEN items_pending
        ELSE 0
    END
    ) AS `48 hrs +`,
    SUM(items_pending) AS total_pending
FROM
```

```
noonbinimops.fulfillment.putaway_pendency_v2 pm
LEFT JOIN noonbinimksa.Stores.ops_logistic_fulfillment_managers_eg eg
ON
pm.ds_code = eg.partner_wh_Code
GROUP BY all
```


DS - Skips

UAE, KSA, EGY

Select *

From `noonbinimksa.darkstore.daily\_manual\_skips\_hourly\_uae\_1`

DS - Audit Scores

For UAE, KSA :

(For Egypt it is not functional)

```
select *  
from noonbinimksa.darkstore.historic_score1
```

DS - IPP

put-away delays in stores

```
SELECT
    eg.AM, eg.Assistant_manager , lower(country_code) as country_code,
    src_wh_code,
    pm.ds_code,
    pm.ds_name,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 0 AND 180 THEN items_pending
        ELSE 0
    END
    ) AS `0-3 hrs`,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 181 AND 360 THEN items_pending
        ELSE 0
    END
    ) AS `3-6 hrs`,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 361 AND 720 THEN items_pending
        ELSE 0
    END
    ) AS `6-12 hrs`,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 721 AND 1440 THEN items_pending
        ELSE 0
    END
    ) AS `12-24 hrs`,
    SUM(CASE
        WHEN ageing_in_mins BETWEEN 1441 AND 2880 THEN items_pending
        ELSE 0
    END
    ) AS `24-48 hrs`,
    SUM(CASE
        WHEN ageing_in_mins >= 2881 THEN items_pending
        ELSE 0
    END
    ) AS `48 hrs +`,
    SUM(items_pending) AS total_pending
FROM
```

```
noonbinimops.fulfillment.putaway_pendency_v2 pm
LEFT JOIN noonbinimksa.Stores.ops_logistic_fulfillment_managers_eg eg
ON
pm.ds_code = eg.partner_wh_Code
GROUP BY all
```

IPP - Inbound

UAE:

```
with ipp as ( select * from noonbinimops.fulfillment.ipp)

,base as (select
    date
    ,dist_partner_code, exdoc_nr, Box_no
    ,p.id_user,email as putter_mail,concat(u.first_name," ",u.last_name) as putter_name
    ,case when HR_Desig is null then "temp" else lower(HR_Desig) end as HR_Desig,
Employee_ID
    ,sum(total_sec) as time_taken_sec,sum(completed_count) as completed_item
from ipp p
left join noondwh.mxfulfillment_norms.user u on u.id_user = p.id_user
left join noonbinimops.fulfillment.ds_picker_details_uae pk on u.email=pk.email_final
where date >= current_date("Asia/Dubai")-31
group by all)

select date
,dist_partner_code, id_user, putter_mail,putter_name, HR_Desig, Employee_ID
, sum(time_taken_sec) as time_taken_sec,sum(completed_item) as completed_item
,case when sum(time_taken_sec)=0 then 0 else
round(sum(completed_item)*3600/sum(time_taken_sec)) end as iph
from base
group by all
```

KSA:

```
WITH ipp AS (
    SELECT *
    FROM noonbinimops.fulfillment.ipp_ksa
),
base AS (
    SELECT
        date,
        dist_partner_code,
        exdoc_nr,
```

```

Box_no,
p.id_user,
u.email AS putter_mail,
CONCAT(u.first_name," ",u.last_name) AS putter_name,
'temp' AS HR_Design,
emp_id AS Employee_ID,
SUM(total_sec) AS time_taken_sec,
SUM(completed_count) AS completed_item,
CASE
    WHEN SUM(total_sec) = 0 THEN 0
    ELSE ROUND(SAFE_DIVIDE(SUM(completed_count) * 3600, SUM(total_sec)))
END AS iph
FROM ipp p
LEFT JOIN noondwh.mxfulfillment_norms.user u
    ON u.id_user = p.id_user
LEFT JOIN noonbinimksa.Stores.ds_picker_details_ksa pk
    ON u.email = pk.email
WHERE date >= CURRENT_DATE("Asia/Riyadh") - 31
GROUP BY ALL
)
SELECT
    date,
    dist_partner_code,
    id_user,
    putter_mail,
    putter_name,
    HR_Design,
    Employee_ID,
    SUM(time_taken_sec) AS time_taken_sec,
    SUM(completed_item) AS completed_item,
    CASE
        WHEN SUM(time_taken_sec) = 0 THEN 0
        ELSE ROUND(SAFE_DIVIDE(SUM(completed_item) * 3600, SUM(time_taken_sec)))
    END AS iph
FROM base
GROUP BY ALL

```

EGY: WIP

IPP - Putaway

UAE:

```
WITH emp_list as (
    select distinct Emp_ID,a.designation, a.role, store, b.DS_code as wh_Code,
    b.New_Shif_Timing as Shift_Timing, DOJ
    from noonbinimops.fulfillment.emp_details_new a
    left join noonbinimksa.Stores.emp_functional_details_ae b on a.emp_Id = b.Employee_ID
)
, outbound as (
    SELECT
        date,
        partner_wh_code,
        id_user,
        ANY_VALUE(picker_mail) AS picker_mail,
        ANY_VALUE(picker_name) AS picker_name,
        ANY_VALUE(HR_Desig) AS HR_Desig,
        ANY_VALUE(Employee_ID) AS Employee_ID,

        SUM(total_orders) AS total_orders,
        SUM(total_time_picking) AS total_time_picking,
        SUM(total_time_packing) AS total_time_packing,
        SUM(total_time) AS total_time,
        SUM(total_item) AS total_item,

        SAFE_DIVIDE(SUM(total_time_picking), SUM(total_item)) AS picking_time_per_item,
        SAFE_DIVIDE(SUM(total_time_packing), SUM(total_item)) AS packing_time_per_item,
        SAFE_DIVIDE(SUM(total_time), SUM(total_item)) AS picking_packing_time_per_item

    FROM `noonbinimksa.Stores.speed_ae_emp`
    --WHERE partner_wh_code = 'DXBDS37' AND id_user = 9424 AND date = '2026-03-08'
    GROUP BY all
)
, final as (

    select 'Outbound' as type_, date
    , partner_wh_code as dist_partner_code, id_user
    , picker_mail as mail, picker_name as name
    , HR_Desig as desig, Employee_ID
    , total_time_picking, total_time_packing
```

```

, total_time as time_taken_sec, total_item as completed_item, '' as inbound_,
cast(picking_time_per_item as string) as outbound
, total_orders
from outbound
)

,inter2 AS (
SELECT
    type_,
    date,
    dist_partner_code AS store,
    name AS user_name,
    Employee_ID,
    completed_item,
    SAFE_CAST(inbound_ AS FLOAT64) AS inbound_iph,
    SAFE_CAST(outbound AS FLOAT64) AS outbound_time_per_item,
    time_taken_sec, total_time_picking, total_time_packing, total_orders
FROM final
),

output AS (
SELECT
    b.date,
    b.store,
    b.user_name,
    b.Employee_ID,

    SUM(IF(type_ = 'Outbound', completed_item, 0)) AS outbound_qty,
    SUM(IF(type_ = 'Outbound', time_taken_sec, 0)) AS outbound_time_,
    SUM(IF(type_ = 'Outbound', total_time_picking, 0)) AS outbound_time_picking,
    SUM(IF(type_ = 'Outbound', total_time_packing, 0)) AS outbound_time_packing,
    SUM(IF(type_ = 'Outbound', total_orders, 0)) AS total_orders

FROM inter2 b
GROUP BY
    b.date, b.store, b.user_name, b.Employee_ID
)

,output_1 AS (
SELECT

```

```

        b.date,
        b.store,
        b.user_name,
        b.Employee_ID,
        b.* EXCEPT (date, user_name, Employee_ID, store)
FROM output b
LEFT JOIN emp_list a
    ON a.Emp_ID = b.Employee_ID
)

SELECT * ,
case when outbound_qty = 0 then 0
else safe_divide(outbound_qty*3600, outbound_time_) end as outbound_iph,

FROM output_1

```

KSA:

```

WITH outbound AS (
    WITH base AS (
        SELECT
            mo.id_mp_order,
            mo.order_nr,
            partner_wh_code AS partner_warehouse_code,
            area_name_en,
            pj.id_user,
            ANY_VALUE(mo.warehouse_code) AS warehouse_code,
            DATE(TIMESTAMP_ADD(mo.created_at, INTERVAL 3 HOUR)) AS created_date,
            MIN(TIMESTAMP_ADD(pj.ready_to_start_at , INTERVAL 3 HOUR)) AS
fulfillemnt_ready_to_start_at,
            MIN(
                GREATEST(
                    TIMESTAMP_ADD(pj.ready_to_start_at , INTERVAL 3 HOUR),
                    COALESCE(
                        LEAST(
                            COALESCE(TIMESTAMP_ADD(prj.preparing_ended_at , INTERVAL 3
HOUR), TIMESTAMP('2100-01-01')),

```

```

        TIMESTAMP_ADD(TIMESTAMP_ADD(prj.preparing_started_at ,
INTERVAL 3 MINUTE), INTERVAL 3 MINUTE)
    ),
    TIMESTAMP('1980-01-01')
)
)
) AS picking_ready_to_start_at,
MIN(CASE WHEN mo_s.code = 'created' THEN TIMESTAMP_ADD(moh.modified_at,
INTERVAL 4 HOUR) END) AS created,
MIN(CASE WHEN mo_s.code = 'fulfilling' THEN TIMESTAMP_ADD(moh.modified_at,
INTERVAL 4 HOUR) END) AS fulfilling,
MIN(CASE WHEN mo_s.code = 'fulfilled' THEN TIMESTAMP_ADD(moh.modified_at,
INTERVAL 4 HOUR) END) AS fulfilled,
MIN(CASE WHEN mo_s.code = 'delivered' THEN TIMESTAMP_ADD(moh.modified_at,
INTERVAL 4 HOUR) END) AS delivered,
MIN(CASE WHEN mo_s.code = 'to_scan' THEN TIMESTAMP_ADD(moh.modified_at,
INTERVAL 4 HOUR) END) AS to_scan,
MIN(CASE WHEN mo_s.code = 'scanned' THEN TIMESTAMP_ADD(moh.modified_at,
INTERVAL 4 HOUR) END) AS scanned,
MIN(CASE WHEN mo_s.code = 'picked_up' THEN TIMESTAMP_ADD(moh.modified_at,
INTERVAL 4 HOUR) END) AS picked_up,
MIN(CASE WHEN pj_s.code = 'pending' THEN TIMESTAMP_ADD(pjh.modified_at,
INTERVAL 4 HOUR) END) AS pending,
MIN(CASE WHEN pj_s.code = 'picking' THEN TIMESTAMP_ADD(pjh.modified_at,
INTERVAL 4 HOUR) END) AS picking,
MIN(CASE WHEN pj_s.code = 'packing' THEN TIMESTAMP_ADD(pjh.modified_at,
INTERVAL 4 HOUR) END) AS packing,
MIN(CASE WHEN pj_s.code = 'completed' THEN TIMESTAMP_ADD(pjh.modified_at,
INTERVAL 4 HOUR) END) AS completed,
ANY_VALUE(mos.num_of_items) AS num_of_items
FROM noondwh.schoms_homs_orchestration.mp_order mo
LEFT JOIN noondwh.schoms_homs.country ctry USING(id_country)
LEFT JOIN noondwh.schoms_homs_orchestration.mp_order_history moh ON
mo.id_mp_order = moh.id_mp_order
LEFT JOIN noondwh.schoms_homs.status mo_s ON mo_s.id_status = moh.id_status
LEFT JOIN noondwh.schoms_homs.picking_job pj ON mo.id_mp_order = pj.id_mp_order
LEFT JOIN noondwh.schoms_homs.picking_job_history pjh ON pj.id_picking_job =
pjh.id_picking_job
LEFT JOIN noondwh.schoms_homs.status pj_s ON pjh.id_status = pj_s.id_status
LEFT JOIN noondwh.schoms_homs.prepping_job prj ON pj.id_picking_job =
prj.id_picking_job

```



```

        LEFT JOIN noondwh.schoms_homs_orchestration.mp_order_stats mos ON
mos.id_mp_order = mo.id_mp_order
        LEFT JOIN noondwh.instant_instant_order.warehouse w ON mo.warehouse_code =
w.wh_code
        LEFT JOIN noondwh.instant_instant_order.sales_order so ON mo.order_nr =
so.order_nr
        WHERE DATE(TIMESTAMP_ADD(mo.created_at, INTERVAL 4 HOUR)) >=
CURRENT_DATE("Asia/Dubai") - 31
        AND w.country_code = 'SA'
        GROUP BY ALL
    ),
    DeliveryTimes AS (
        SELECT
            DISTINCT created_date AS date,
            partner_warehouse_code AS partner_wh_code,
            order_nr,
            id_user,
            TIMESTAMP_DIFF(packing, fulfilling, SECOND) AS picking_time,
            TIMESTAMP_DIFF(fulfilled, packing, SECOND) AS packing_time,
            TIMESTAMP_DIFF(fulfilled, fulfilling, SECOND) AS picking_packing_time
        FROM base
        WHERE DATE(created) >= CURRENT_DATE('Asia/Riyadh') - 31
    ),
    item_count AS (
        SELECT
            EXTRACT(DATE FROM TIMESTAMP_ADD(a.created_at, INTERVAL 3 HOUR)) AS date,
            order_nr,
            a.wh_Code,
            a.country_code,
            a.id_sales_order,
            w.area_name_en AS name,
            w.partner_wh_code,
            COUNT(DISTINCT item_nr) AS item_count
        FROM noondwh.instant_instant_order.sales_order a
        LEFT JOIN noondwh.instant_instant_order.sales_order_item z ON a.id_sales_order =
z.id_sales_order
        LEFT JOIN noondwh.instant_instant_order.warehouse w ON a.wh_code = w.wh_code
        WHERE status_code_order = "delivered" AND cancel_reason_code IS NULL
        AND EXTRACT(DATE FROM TIMESTAMP_ADD(a.created_at, INTERVAL 4 HOUR)) >=
CURRENT_DATE("Asia/Dubai") - 31
        AND a.country_code = "SA"
    )

```

```

        GROUP BY ALL
    ),
    picker_base AS (
        SELECT
            DISTINCT a.date,
            a.partner_wh_code,
            id_user,
            a.order_nr,
            SUM(a.picking_time) AS time_taken_picking,
            SUM(a.packing_time) AS time_taken_packing,
            SUM(a.picking_packing_time) AS time_taken_picking_packing,
            SUM(i.item_count) AS picked_items
        FROM DeliveryTimes a
        INNER JOIN item_count i ON a.order_nr = i.order_nr
        GROUP BY ALL
    ),
    final AS (
        SELECT
            a.date,
            a.partner_wh_code,
            a.id_user,
            u.email AS picker_mail,
            CONCAT(u.first_name, " ", u.last_name) AS picker_name,
            a.order_nr,
            time_taken_picking,
            time_taken_packing,
            time_taken_picking_packing,
            picked_items,
            "temp" AS HR_Design,
            Emp_id AS Employee_ID
        FROM picker_base a
        LEFT JOIN noonwh.schoms_homs.user u ON a.id_user = u.id_user
        LEFT JOIN noonbinimksa.Stores.ds_picker_details_ksa pk ON u.email = pk.email
    )
    SELECT
        date,
        partner_wh_code,
        id_user,
        picker_mail,
        picker_name,
        HR_Design,

```

```

Employee_ID,
COUNT(DISTINCT order_nr) AS total_orders,
SUM(time_taken_picking) AS total_time_picking,
SUM(time_taken_packing) AS total_time_packing,
SUM(time_taken_picking_packing) AS total_time,
SUM(picked_items) AS total_item,
ROUND(SAFE_DIVIDE(SUM(time_taken_picking), SUM(picked_items)), 2) AS
picking_time_per_item,
ROUND(SAFE_DIVIDE(SUM(time_taken_packing), SUM(picked_items)), 2) AS
packing_time_per_item,
ROUND(SAFE_DIVIDE(SUM(time_taken_picking_packing), SUM(picked_items)), 2) AS
picking_packing_time_per_item
FROM final
GROUP BY ALL
),

final AS (
SELECT
'Outbound' AS type_,
date,
partner_wh_code AS dist_partner_code,
STRING_AGG(DISTINCT CAST(id_user AS STRING)) AS id_user,
picker_mail AS mail,
picker_name AS name,
HR_Design AS design,
Employee_ID,
total_time_picking,
total_time_packing,
total_time AS time_taken_sec,
total_item AS completed_item,
'' AS inbound_,
CAST(picking_time_per_item AS STRING) AS outbound
FROM outbound
GROUP BY ALL
),

inter1 AS (
SELECT *
FROM final
),

```

```

inter2 AS (
    SELECT
        type_,
        date,
        dist_partner_code AS store,
        name AS user_name,
        STRING_AGG(DISTINCT id_user) AS id_user,
        STRING_AGG(DISTINCT Employee_ID) AS Employee_ID,
        completed_item,
        SAFE_CAST(inbound_ AS FLOAT64) AS inbound_iph,
        SAFE_CAST(outbound AS FLOAT64) AS outbound_time_per_item,
        time_taken_sec,
        total_time_picking,
        total_time_packing
    FROM inter1
    GROUP BY ALL
),

```

```

output AS (
    SELECT
        b.date,
        b.store,
        b.user_name,
        b.id_user,
        SUM(IF(type_ = 'Outbound', completed_item, 0)) AS outbound_qty,
        SUM(IF(type_ = 'Outbound', time_taken_sec, 0)) AS outbound_time_,
        SUM(IF(type_ = 'Outbound', total_time_picking, 0)) AS outbound_time_picking,
        SUM(IF(type_ = 'Outbound', total_time_packing, 0)) AS outbound_time_packing,
        Employee_ID
    FROM inter2 b
    GROUP BY b.date, b.store, b.user_name, b.id_user, Employee_ID
),

```

```

output_1 AS (
    SELECT
        b.date,
        b.store,
        b.user_name,
        b.Employee_ID,
        b.* EXCEPT(date, user_name, store, id_user, Employee_ID)
    FROM output b

```

)

```
SELECT *,  
case when outbound_qty = 0 then 0  
else safe_divide(outbound_qty*3600, outbound_time_) end as outbound_iph  
FROM output_1
```

DS - Adjustments

Adjustments :

UAE:

```
SELECT * FROM `noonbinimops.fulfillment.adjustments_master_uae`
```

KSA:

```
SELECT * FROM `noonbinimops.fulfillment.adjustments_master_ksa`
```

EGY:

```
SELECT * FROM `noonbinimops.fulfillment.adjustments_master_eg`
```

DS - Missing

DS-MISSING

Easy Guide

Purpose: for each inbound timestamp, it gives bifurcation of qtys (Expected, Pending, Completed, Missing, Expired, Damaged, Excess) at country_code level

UAE, KSA, EGY

*SELECT DISTINCT **

FROM noonbinimops.fulfillment.missing_base_v3

GROUP BY ALL

Easy Guide

Purpose: For each inbound date, it gives bifurcation of qtys (Expected, Pending, Completed, Missing, Expired, Damaged, Excess) at country_code level

UAE, KSA, EGY

SELECT DISTINCT

```
country_code, ds_code, ds_name, DATE,
psku_code, wms_barcode, pbarcode_canonical, id_partner,
SUM(qty_expected) AS qty_expected,
SUM(qty_pending) AS qty_pending,
SUM(qty_completed) AS qty_completed,
SUM(qty_missing) AS qty_missing,
SUM(qty_expired) AS qty_expired,
SUM(qty_damaged) AS qty_damaged,
SUM(qty_excess) AS qty_excess,
AVG(cost_price) AS cost_price,
STRING_AGG(DISTINCT category) AS category,
STRING_AGG(DISTINCT src_wh_code) AS source_warehouse,
STRING_AGG(DISTINCT CAST(id_box AS STRING)) AS id_box,
STRING_AGG(DISTINCT CAST(awb_nr AS STRING)) AS awb_nr,
STRING_AGG(DISTINCT CAST(exdoc_nr AS STRING)) AS exdoc_nr,
STRING_AGG(DISTINCT CAST(tote_code AS STRING)) AS tote_code,
STRING_AGG(DISTINCT CAST(ts AS STRING)) AS ts
FROM noonbinimops.fulfillment.missing_base_v3
GROUP BY ALL
```


DS noonbinimksa table <> DAG
mapping <> Raw tables

DS noonbinimksa tables <> Raw Table/Dag mapping

Heading / Tab	Dag Link	Derived table name	Main / primary raw table	If primary raw is derived: raw DAG	If primary raw is derived: raw source
DS - Complains	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/minutes_fulfillment_uae/Defects/complain_details_uae_auto.toml?ref_type=heads	noonbinimksa.darkstore.complains_raw_all	noonbinimksa.darkstore.complains_raw_ae	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/daily_ops/UAE/complains_raw_ae.toml?ref_type=heads	noondwh.instant_instant_order.cs_order_adjustment noondwh.instant_instant_order.sales_order noondwh.instant_instant_order.warehouse noonbiinstnt.mktg.sales_complete
DS - Complains	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/daily_ops/KSA/complaints/complaints_raw_sa.toml?ref_type=heads	noonbinimksa.darkstore.complains_raw_sa	noonbiinstnt.minutes.minutes_commercial_category_mapping		
Specific Complains	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/minutes_fulfillment_uae/Defects/complain_details_uae_auto.toml?ref_type=heads	noonbinimksa.darkstore.complains_raw_all	noonbinimksa.darkstore.complains_raw_ae	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/daily_ops/UAE/complains_raw_ae.toml?ref_type=heads	noondwh.instant_instant_order.cs_order_adjustment noondwh.instant_instant_order.sales_order noondwh.instant_instant_order.warehouse noonbiinstnt.mktg.sales_complete
DS - Live Inventory	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Hourly_Tracking/wh_loc_qt	noonbinimksa.Stores.wh_loc_qty	noondwh.wms.stock		

	y.toml?ref_type=heads				
DS - Non Live Inventory	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Disposal/NonLive/DAM.toml?ref_type=heads	noonbinimksa.Stores.DAM_n on_live	noonbinimksa.Stores.wh_loc_qty	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Hourly_Tracking/wh_loc_qty.toml?ref_type=heads	noondwh.wms.stock noondwh.wms.location noondwh.wms.warehouse noondwh.wms.zone
DS - Non Live Inventory	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Disposal/NonLive/ksa_DAM.toml?ref_type=heads	noonbinimksa.Stores.DAM_n on_live_ksa	noonbinimksa.Stores.wh_loc_qty	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Hourly_Tracking/wh_loc_qty.toml?ref_type=heads	noondwh.wms.stock noondwh.wms.location noondwh.wms.warehouse noondwh.wms.zone
DS - Non Live Inventory	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Disposal/NonLive/DAM_non_live_all_eg.toml?ref_type=heads	noonbinimksa.Stores.DAM_n on_live_all_eg	noonbinimksa.Stores.wh_loc_qty	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Hourly_Tracking/wh_loc_qty.toml?ref_type=heads	noondwh.wms.stock noondwh.wms.location noondwh.wms.warehouse noondwh.wms.zone
DS - Non Live Migrations	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/Non_Live/Today/NL_DB.toml?ref_type=heads	noonbinimksa.darkstore.DAM_line_date_new_final	noondwh.mxfulfillment_norms.stock_migration_job		
DS - Expiry (raw and adjusted)	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Expiry/CentralExpiryTable/raw_data.toml?ref_type=heads	noonbinimksa.Stores.central_expiry_raw	noondwh.mxfulfillment_norms.expirable_item		

DS - Expiry (raw and adjusted)	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Expiry/CentralExpiryTable/central_expiry_raw_sa.toml?ref_type=heads	noonbinimksa.Stores.central_expiry_raw_sa	noondwh.mxfulfillment_norms.expirable_item		
DS - Expiry (raw and adjusted)	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Expiry/CentralExpiryTable/central_expiry_raw_adjusted.toml?ref_type=heads	noonbinimksa.Stores.central_expiry_raw_adjusted	noondwh.instant_spanner.product_ops_attributes		
DS - Expiry (raw and adjusted)	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Expiry/CentralExpiryTable/central_expiry_raw_adjusted_sa.toml?ref_type=heads	noonbinimksa.Stores.central_expiry_raw_adjusted_sa	noondwh.instant_spanner.product_ops_attributes		
DS - Picking Data	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/order_funnel_time.toml?ref_type=heads	noonbinimksa.darkstore.order_funnel_time	noondwh.schoms_homs_orchestration.mp_order		
DS - Picking Data	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/order_funnel_time_sa.toml?ref_type=heads	noonbinimksa.darkstore.order_funnel_time_sa	noondwh.schoms_homs_orchestration.mp_order		

DS - Inbound and Stock Take Data	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/minutes_fulfillment_uae/ipp%20txw/ipp_txw.toml?ref_type=heads	noonbinimksa.darkstore.ipp_a ll	noondwh.mxfulfillment_norms.putaway_job_line		
DS - Inbound and Stock Take Data	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/stock_take/stock_take_base.toml?ref_type=heads	noonbinimksa.darkstore.stock_take_base	noondwh.mxfulfillment_norms.stock_take_job_queue		
DS - MP - Planned Vs Actual - Biometric	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/ManPower/Biometric/Biometric_base_v2_3.toml?ref_type=heads	noonbinimksa.Stores.Biometric_base_v2_3	noonbinimksa.Stores.biometric_roaster_3	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/ManPower/Biometric/Biometric_roaster_historical.toml?ref_type=heads	noonbinimksa.Stores.table_mp_ae noonbinimksa.Stores.Biometric_base_v2_3
DS - Actual Logins	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/minutes_fulfillment_uae/ipp%20txw/ipp_txw.toml?ref_type=heads	noonbinimksa.darkstore.ipp_a ll	noondwh.mxfulfillment_norms.putaway_job_line		
DS - Putaway Pendency Hourly Bucket	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/together_ops_manager.toml?ref_type=heads	noonbinimksa.Stores.ops_logistic_fulfillment_managers_eg	noonbinimops.fulfillment.putaway_pendency_v2		

DS - Skips	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/daily_ops/UAE/daily_manual_skips_hourly_uae.toml?ref_type=heads	noonbinimksa.darkstore.daily_manual_skips_hourly_uae_1	noondwh.scho.ms_homs.picking_incident		
DS - Audit Scores	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Audit/table5.toml?ref_type=heads	noonbinimksa.darkstore.historic_score1	noonbinimksa.darkstore.audit_staff_qus_ans	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/Audit/staff_answers.toml?ref_type=heads	noondwh.instant_instant_ds_audit.question_group noondwh.instant_instant_ds_audit.question noondwh.instant_instant_ds_audit.audit_submission noondwh.instant_instant_order.warehouse
DS - IPP	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/together_ops_manager.toml?ref_type=heads	noonbinimksa.Stores.ops_logistic_fulfillment_managers_eg	noonbinimops.fulfillment.putaway_pendency_v2		
IPP - Inbound	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/ManPower/ds_picker_details_ksa.toml?ref_type=heads	noonbinimksa.Stores.ds_picker_details_ksa	noondwh.scho.ms_homs.user		
IPP - Putaway	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/KSA/Chiller/Excel/emp_functional_details_ae.toml?ref_type=heads	noonbinimksa.Stores.emp_functional_details_ae	noonbinimops.fulfillment.emp_details_new		

IPP - Putaway	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/central_ops/minutes_fulfillment_uae/Speed/speed_ae_emp.toml?ref_type=heads	noonbinimksa.Stores.speed_ae_emp	noondwh.schoms_homs_orchestration.mp_order		
IPP - Putaway	https://dp-gitlab.noon.team/dagman-group/dp-dagman-nimksa/-/blob/master/project/ops/ManPower/ds_picker_details_ksa.toml?ref_type=heads	noonbinimksa.Stores.ds_picker_details_ksa	noondwh.schoms_homs.user		